Multiverse analysis

First Author[1] & Ernst-August Doelle[1,2]

[1] Wilhelm-Wundt-University

[2] Konstanz Business School

Author Note

Add complete departmental affiliations for each author here. Each new line herein must be indented, like this line.

Enter author note here.

The authors made the following contributions. First Author: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; Ernst-August Doelle: Writing - Review & Editing, Supervision.

Correspondence concerning this article should be addressed to First Author, Postal address. E-mail: my@email.com

Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Keywords: keywords

Word count: X

Multiverse analysis

Results

Study characteristics

Table 1 describes each study in the primary analytic sample, and Table 2 shows how each study's measure of socioeconomic status was harmonised.

Model diagnostics

The model diagnostics for the main analysis, and for all models presented in this document are in Table 3

Fixed Effect Model

In this document we show the results using another analytic technique. Namely treating study ID as a fixed effect. This is what we wrote in the protocol originally.

The association between physical activity volume and sleep. We estimated the associations between physical activity and sleep (RQ1) using fixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 4 and Figure 1.

The association between physical activity intensity and sleep. We estimated how physical activity intensity is associated with sleep across different age groups. We present the results controlling for SES, sex, BMI z-score, and the fixed effects of study ID, in Table 4 and Figure 2.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by age.

Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 3.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 4.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 5.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 6.

Log Transforming PA Volume

Based on preliminary modelling on a subset of the data, we chose to leave physical activity volume in its natural form to enhance interpretability. The log scale meant that the variable became unintepretable towards the edge of its range. Below, we present the results had we not made this decision. These results are all moderated by age, as this was our primary analysis. The associated figures are back-transformed to the scale range.

The association between physical activity volume and sleep. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 6 and Figure 7.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 8.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 9.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 10.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 11.

All Eligible Participants Regardless of Valid Wear Days

The primary analysis required at least 4 days of valid wear time. Here we refit the primary models to every eligible participant, regardless of how many valid days they contributed. Both analyses share the same imputations and standardisation constants. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 8 and 9 and Figures 12 to 17.

Participants With at Least Seven Valid Days

Some recommend at least seven nights to estimate sleep regularity. Here we refit the primary models to the 14,981 participants with at least seven days of valid wear time. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 10 and 11 and Figures 18 to 23.

Moderation

In the following section we present models for a range of moderators which were pre-defined in our protocol.

BMI z-score

The association between physical activity volume and sleep by BMI z-score. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by BMI z-score, and the results are presented in Table 12 and Figure 24

The association between physical activity intensity and sleep by BMI z-score. We estimated how physical activity intensity is associated with sleep across BMI z-score. We present the results in Table 12 and Figure 25.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 26.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 27.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 28.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by BMI z-score. Results are presented in Table 13 and Figure 29.

SES

The association between physical activity volume and sleep by SES. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by SES, and the results are presented in Table 14 and Figure 30

The association between physical activity intensity and sleep by SES. We estimated how physical activity intensity is associated with sleep across SES. We present the results in Table 14 and Figure 31.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 32.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 33.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 34.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by SES. Results are presented in Table 15 and Figure 35.

sex

The association between physical activity volume and sleep by sex. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by sex, and the results are presented in Table 16 and Figure 36

The association between physical activity intensity and sleep by sex. We estimated how physical activity intensity is associated with sleep across sex. We present the results in Table 16 and Figure 37.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 38.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 39.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 40.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by sex. Results are presented in Table 17 and Figure 41.

weekday

The association between physical activity volume and sleep by weekday. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by weekday, and the results are presented in Table 18 and Figure 42

The association between physical activity intensity and sleep by weekday. We estimated how physical activity intensity is associated with sleep across weekday. We present the results in Table 18 and Figure 43.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 44.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 45.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 46.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by weekday. Results are presented in Table 19 and Figure 47.

season

The association between physical activity volume and sleep by season. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by season, and the results are presented in Table 20 and Figure 48

The association between physical activity intensity and sleep by season. We estimated how physical activity intensity is associated with sleep across season. We present the results in Table 20 and Figure 49.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 50.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 51.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 52.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by season. Results are presented in Table 21 and Figure 53.

region

The association between physical activity volume and sleep by region. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by region, and the results are presented in Table 22 and Figure 54

The association between physical activity intensity and sleep by region. We estimated how physical activity intensity is associated with sleep across region. We present the results in Table 22 and Figure 55.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 56.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 57.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 58.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by region. Results are presented in Table 23 and Figure 59.

daylight hours

The association between physical activity volume and sleep by daylight hours. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by daylight hours, and the results are presented in Table 24 and Figure 60

The association between physical activity intensity and sleep by daylight hours. We estimated how physical activity intensity is associated with sleep across daylight hours. We present the results in Table 24 and Figure 61.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 62.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 63.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 64.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by daylight hours. Results are presented in Table 25 and Figure 65.

wear location

The association between physical activity volume and sleep by wear location. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by wear location, and the results are presented in Table 26 and Figure 66

The association between physical activity intensity and sleep by wear location. We estimated how physical activity intensity is associated with sleep across wear location. We present the results in Table 26 and Figure 67.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 68.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 69.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 70.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by wear location. Results are presented in Table 27 and Figure 71.

most active hour

The association between physical activity volume and sleep by most active hour. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by most active hour, and the results are presented in Table 28 and Figure 72

The association between physical activity intensity and sleep by most active hour. We estimated how physical activity intensity is associated with sleep across most active hour. We present the results in Table 28 and Figure 73.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 74.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 75.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 76.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by most active hour. Results are presented in Table 29 and Figure 77.

ethnicity

The association between physical activity volume and sleep by ethnicity. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by ethnicity, and the results are presented in Table 30 and Figure 78

The association between physical activity intensity and sleep by ethnicity. We estimated how physical activity intensity is associated with sleep across ethnicity. We present the results in Table 30 and Figure 79.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 80.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 81.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 82.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by ethnicity. Results are presented in Table 31 and Figure 83.

Table 1

Characteristics of the included studies (primary analytic sample)

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range
100			12 countries	GT3X	Hip	6,360	10.4 (9-11)
101			Switzerland	GENEActiv	Wrist	2,928	61.9 (45-86)
102			Denmark	GT3X	Hip	691	10.0 (8-11)
103			New Zealand	Actigraph	Wrist	651	19.3 (8-58)
104			Brazil	GT3X+	Wrist	737	16.4 (14-22)
105			Czech Republic	GT3X+, GT9X Link, wGT3X-BT, wGT3X+	Wrist	1,170	14.0 (8-19)
106			Australia	GENEActiv	Wrist	301	42.0 (18-65)
107			Spain	GT3X+	Wrist	2,423	72.6 (64-94)
108			Australia	GENEActiv	Wrist	1,038	8.8 (7-11)
110			Finland	Actisleep	Wrist	801	62.6 (58-64)
111			United Kingdom	Axivity AX3	Wrist	692	44.7 (20-70)
112			Chile	GENEActiv	Wrist	144	9.5 (8-12)
113			Spain	Actisleep	Wrist	353	8.4 (8-9)
114			United States	WGT3X-BT	Wrist	154	3.5 (2-4)
115			Ecuador	wGT3X-BT	Hip	1,019	8.8 (7-13)
117			Spain	GENEActiv	Wrist	175	8.8 (4-11)
118			Australia	Actigraph	Wrist	304	16.6 (15-18)
212			Spain	WGT3X	Hip	285	4.7 (2-6)
221			United Kingdom	GENEActiv	Wrist	4,071	69.4 (59-82)
222			Brazil	GT3X	Wrist	2,939	3.8 (3-4)

Table 2

Harmonisation of socioeconomic status

Study ID	Original category
100	1
100	2
100	4
100	3
101	High school
101	Apprenticeship
101	Mandatory education
101	University education
102	Vocational basic training/vocational internship and main course
102	Short higher education
102	Long higher education/ Research degree
102	Regular secondary education and vocational secondary education
102	Medium higher education
102	Primary school
102	Unanswered
104	Incomplete elementary school
104	Complete Bachelor degree
104	Incomplete bachelor degree
104	Complete high school
104	Complete elementary school
104	Unknown
104	Has not studied
105	vocational education
105	high school education
105	master's degree or higher
105	bachelor's degree or equivalent
105	10th grade or less
105	short tertiary education
105	26-100 books
105	11-25 books
105	>200 books
105	101-200 books
105	0-10 books
106	High school or less

Table 2 continued

Study ID	Original category
106	Technical or further education institutions (e.g., TAFE)
106	University degree
107	Secondary education 2nd stage (i.e., High shool graduate, complete 12th grade or equivalent
107	Incomplete primary education (i.e., less than 6th grade or equivalent)
107	Complete primary education (i.e., complete 6th grade or equivalent)
107	Medium university studies (i.e., university diplomature or equivalent)
107	Higher university studies (i.e., Bachelor degree, Master degree, Doctoral degree or equivalen
107	Secondary education 1st stage (i.e., complete 10th grade or equivalent)
107	Illiterate (i.e., not able to read or write)
108	Enough to fill two bookcases (101-200 Books)
108	Enough to fill one shelf (11-25 Books)
108	Enough to fill 3 or more bookcases (more than 200 Books)
108	Enough to fill one bookcases (26-100 books)
108	None or very few (0-10 Books)
110	lower-level non-manual occupation
110	higher-level non-manual occupation
110	manual occupation
110	service occupation
111	10
113	primary school
113	university degree
113	bachelor's degree
113	high school
113	master's degree or higher
113	vocational training
114	poor
114	O level
114	A/S level
114	ba/bsc
114	higher degree
114	No academic quals
115	Medium
115	High
115	Low
117	secondary education

Table 2 continued

Study ID	Original category
117	higher education
117	primary education
118	2
118	7
118	5
118	3
118	10
118	8
118	9
118	6
118	4
118	1
221	O level
221	A/S level
221	ba/bsc
221	higher degree
221	No academic quals
222	q5-wealthiest
222	q4
222	q3
222	q2
222	q1-poorest

Table 3

Model diagnostics

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.66	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.38	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	-0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age fixedef				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.66	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.38	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	-0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age log				
Scale sleep duration by log PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by log PA volume	-0.91	3.67	100.00%	0.00%
Scale sleep onset by log PA volume	0.16	9.40	100.00%	0.00%
Scale sleep regularity by log PA volume	-0.72	2.13	100.00%	0.00%
Scale sleep duration by PA intensity	-0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.09	100.00%	0.00%
Log PA volume by sleep duration(lagged)	-0.46	2.16	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Log PA volume by sleep efficiency(lagged)	-0.46	2.18	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Log PA volume by sleep onset(lagged)	-0.46	2.18	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Log PA volume by sleep regularity(lagged)	-0.33	1.71	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age allwear				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.92	3.73	100.00%	0.00%
Scale sleep onset by PA volume	0.15	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	-0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.92	3.73	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age 7days				

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA volume	0.02	2.79	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.83	3.00	100.00%	0.00%
Scale sleep onset by PA volume	0.08	12.15	100.00%	0.00%
Scale sleep regularity by PA volume	-0.80	2.66	100.00%	0.00%
Scale sleep duration by PA intensity	0.02	2.79	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.83	2.99	100.00%	0.00%
Scale sleep onset by PA intensity	0.10	12.21	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.80	2.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.53	5.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.54	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.53	5.16	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.52	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.53	5.11	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.52	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.60	5.81	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.33	100.00%	0.00%
Models moderated by bmi				
Scale sleep duration by PA volume	-0.12	2.48	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.92	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.13	11.82	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.12	100.00%	0.00%
Scale sleep duration by PA intensity	-0.12	2.47	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.92	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.15	11.85	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.71	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.00	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.17	1.52	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.02	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.16	1.52	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.01	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.16	1.51	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.54	100.00%	0.00%
Models moderated by ses				
Scale sleep duration by PA volume	-0.03	2.15	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA volume	-0.90	3.55	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.60	100.00%	0.00%
Scale sleep regularity by PA volume	-0.72	2.20	100.00%	0.00%
Scale sleep duration by PA intensity	-0.03	2.12	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.90	3.55	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.61	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.16	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.26	1.67	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.02	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.26	1.66	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.02	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.26	1.66	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.22	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.23	1.57	100.00%	0.00%
Models moderated by weekday				
Scale sleep duration by PA volume	-0.08	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.09	9.83	100.00%	0.00%
Scale sleep regularity by PA volume	-0.79	2.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.08	2.16	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.90	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.13	9.73	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.80	2.69	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.73	5.07	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.29	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.73	5.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.29	1.69	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.73	5.09	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.31	1.72	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.79	5.16	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.63	100.00%	0.00%
Models moderated by season				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.69	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep onset by PA volume	0.17	9.34	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.71	2.06	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by region				
Scale sleep duration by PA volume	-0.05	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.71	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.39	100.00%	0.00%
Scale sleep regularity by PA volume	-0.72	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.70	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.01	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	4.95	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.12	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by daylight				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.34	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep regularity by PA volume	-0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.71	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by wear location				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.70	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	-0.72	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.68	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.23	1.57	100.00%	0.00%
Models moderated by PA mostactivehr				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.18	9.34	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.10	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.20	9.38	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.71	2.06	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.98	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.66	5.01	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.75	5.12	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by sex				
Scale sleep duration by PA volume	-0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.68	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.36	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.71	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by ethnicity				
Scale sleep duration by PA volume	-0.05	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	-0.05	2.14	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA intensity	-0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.71	2.05	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.98	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%

Table 4

Physical activity predicting sleep controlling for SES, sex, BMI z-score, and the fixed effects of study IDs.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.72 [0.68, 0.76]	< .001	0.61 [0.57, 0.65]	< .001
Physical activity	0.10 [0.09, 0.11]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.02 [-0.03, 0.00]	.024	-0.02 [-0.03, 0.00]	.013
Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.00 [0.00, 0.01]	.591
Physical activity × Age	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [-0.01, 0.00]	.001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.51 [-0.55, -0.46]	< .001	-0.58 [-0.62, -0.54]	< .001
Physical activity	0.03 [0.03, 0.04]	< .001	0.02 [0.02, 0.03]	< .001
Age	-0.01 [-0.03, 0.00]	.146	-0.02 [-0.03, 0.00]	.024
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.02 [-0.02, -0.01]	< .001
Physical activity × Age	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.094	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	-1.14 [-1.18, -1.11]	< .001	-1.04 [-1.07, -1.01]	< .001
Physical activity	-0.12 [-0.12, -0.11]	< .001	-0.06 [-0.06, -0.05]	< .001
Age	-0.01 [-0.03, 0.00]	.025	0.00 [-0.02, 0.01]	.436
Physical activity^2	0.02 [0.02, 0.03]	< .001	-0.01 [-0.02, -0.01]	< .001
Physical activity × Age	-0.01 [-0.01, -0.01]	< .001	0.01 [0.00, 0.01]	< .001
Age × Physical activity^2	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	-0.06 [-0.11, 0.00]	.045	-0.49 [-0.54, -0.43]	< .001
Physical activity	0.29 [0.27, 0.30]	< .001	0.14 [0.13, 0.15]	< .001

Table 4 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.00 [-0.02, 0.02]	.939	-0.09 [-0.11, -0.07]	$< .001$
Physical activity^2	-0.15 [-0.16, -0.14]	$< .001$	-0.09 [-0.09, -0.08]	$< .001$
Physical activity $\times$ Age	0.00 [-0.01, 0.00]	.044	-0.05 [-0.05, -0.05]	$< .001$
Age $\times$ Physical activity^2	-0.03 [-0.04, -0.03]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 5

Sleep predicting physical activity controlling for SES, sex, BMI z-score, and the fixed effects of study IDs

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-1.09 [-1.12, -1.05]	< .001	-0.41 [-0.44, -0.37]	< .001
Sleep duration	0.02 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	-0.23 [-0.24, -0.22]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.050
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.330
Sleep efficiency (z)				
(Intercept)	-1.07 [-1.10, -1.03]	< .001	-0.38 [-0.41, -0.35]	< .001
Sleep efficiency	0.00 [0.00, 0.01]	.283	0.01 [0.00, 0.01]	.010
Age	-0.23 [-0.24, -0.21]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.204
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	-1.10 [-1.14, -1.07]	< .001	-0.44 [-0.47, -0.40]	< .001
Sleep onset	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age	-0.22 [-0.24, -0.21]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.001	0.00 [0.00, 0.00]	.796
Sleep onset × Age	0.00 [0.00, 0.00]	.011	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	-1.11 [-1.15, -1.06]	< .001	-0.52 [-0.56, -0.48]	< .001
Sleep regularity	0.12 [0.11, 0.12]	< .001	0.06 [0.06, 0.07]	< .001

Table 5 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.24 [-0.26, -0.22]	$< .001$	-0.33 [-0.34, -0.31]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.445	0.00 [0.00, 0.00]	.408
Sleep regularity $\times$ Age	-0.01 [-0.01, -0.01]	$< .001$	-0.01 [-0.01, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.146	0.00 [0.00, 0.00]	.024

Note. Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 6

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.17 [-0.08, 0.42]	.185	0.00 [-0.15, 0.14]	.995
Log pa volume	-0.27 [-0.38, -0.16]	< .001	0.05 [0.05, 0.06]	< .001
Age	-0.06 [-0.13, 0.01]	.081	-0.02 [-0.03, -0.01]	.003
Log pa volume^2	0.06 [0.04, 0.08]	< .001	0.00 [0.00, 0.01]	.612
Log pa volume × Age	0.05 [0.01, 0.09]	.016	-0.01 [-0.01, -0.01]	< .001
Age × Log pa volume^2	-0.01 [-0.02, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-1.12 [-1.37, -0.87]	< .001	0.15 [-0.02, 0.32]	.086
Log pa volume	0.67 [0.56, 0.78]	< .001	0.02 [0.02, 0.03]	< .001
Age	0.22 [0.15, 0.29]	< .001	-0.01 [-0.03, 0.00]	.067
Log pa volume^2	-0.09 [-0.10, -0.07]	< .001	-0.02 [-0.02, -0.01]	< .001
Log pa volume × Age	-0.09 [-0.13, -0.06]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Log pa volume^2	0.01 [0.00, 0.01]	.003	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	-0.37 [-0.68, -0.06]	.018	0.12 [-0.14, 0.37]	.379
Log pa volume	0.47 [0.39, 0.56]	< .001	-0.06 [-0.06, -0.05]	< .001
Age	0.31 [0.26, 0.37]	< .001	0.00 [-0.02, 0.01]	.524
Log pa volume^2	-0.09 [-0.10, -0.08]	< .001	-0.01 [-0.02, -0.01]	< .001
Log pa volume × Age	-0.17 [-0.20, -0.14]	< .001	0.01 [0.00, 0.01]	< .001
Age × Log pa volume^2	0.02 [0.02, 0.03]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	-4.20 [-4.48, -3.93]	< .001	0.06 [-0.14, 0.25]	.569
Log pa volume	1.98 [1.84, 2.12]	< .001	0.14 [0.13, 0.15]	< .001

Table 6 continued

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.39 [-0.47, -0.30]	$< .001$	-0.08 [-0.10, -0.06]	$< .001$
Log pa volume^2	-0.22 [-0.24, -0.20]	$< .001$	-0.09 [-0.09, -0.08]	$< .001$
Log pa volume $\times$ Age	0.22 [0.17, 0.27]	$< .001$	-0.05 [-0.05, -0.05]	$< .001$
Age $\times$ Log pa volume^2	-0.03 [-0.04, -0.02]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised and (ln) natural-log transformed; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 7

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	-0.07 [-0.19, 0.05]	.254
Sleep duration	0.01 [0.00, 0.01]	< .001	0.03 [0.02, 0.03]	< .001
Age	-0.13 [-0.14, -0.12]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep duration^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.051
Age × Sleep duration^2	0.00 [0.00, 0.00]	.992	0.00 [0.00, 0.00]	.326
Sleep efficiency (z)				
(Intercept)	3.65 [3.51, 3.79]	< .001	-0.07 [-0.20, 0.05]	.242
Sleep efficiency	0.01 [0.00, 0.01]	< .001	0.01 [0.00, 0.01]	.009
Age	-0.13 [-0.14, -0.12]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.010	0.00 [0.00, 0.00]	.200
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	-0.08 [-0.21, 0.04]	.202
Sleep onset	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Age	-0.13 [-0.14, -0.12]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep onset^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.811
Sleep onset × Age	0.00 [0.00, 0.00]	.001	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	3.61 [3.47, 3.75]	< .001	-0.18 [-0.30, -0.05]	.006
Sleep regularity	0.08 [0.08, 0.09]	< .001	0.06 [0.06, 0.07]	< .001

Table 7 continued

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.13 [-0.14, -0.13]	$< .001$	-0.32 [-0.34, -0.31]	$< .001$
Sleep regularity^2	-0.01 [-0.01, -0.01]	$< .001$	0.00 [0.00, 0.00]	.414
Sleep regularity $\times$ Age	0.00 [0.00, 0.00]	$< .001$	-0.01 [-0.01, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.481	0.00 [0.00, 0.00]	.023

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised and (ln) natural-log transformed; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 8

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [-0.16, 0.15]	.960	-0.01 [-0.15, 0.14]	.944
Physical activity	0.10 [0.09, 0.11]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.02 [-0.03, -0.01]	.004	-0.02 [-0.03, -0.01]	.002
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [0.00, 0.01]	.691
Physical activity × Age	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [-0.01, 0.00]	.002	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [-0.01, 0.31]	.067	0.15 [-0.02, 0.31]	.090
Physical activity	0.04 [0.03, 0.04]	< .001	0.02 [0.02, 0.03]	< .001
Age	-0.01 [-0.02, 0.01]	.466	-0.01 [-0.03, 0.00]	.100
Physical activity^2	-0.02 [-0.03, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Age	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.149	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.09 [-0.18, 0.36]	.515	0.11 [-0.14, 0.37]	.386
Physical activity	-0.12 [-0.12, -0.11]	< .001	-0.06 [-0.06, -0.05]	< .001
Age	-0.01 [-0.02, 0.00]	.073	0.00 [-0.01, 0.01]	.744
Physical activity^2	0.02 [0.02, 0.03]	< .001	-0.01 [-0.02, -0.01]	< .001
Physical activity × Age	-0.01 [-0.01, -0.01]	< .001	0.01 [0.00, 0.01]	< .001
Age × Physical activity^2	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.19 [0.04, 0.33]	.012	0.06 [-0.13, 0.25]	.548
Physical activity	0.29 [0.27, 0.30]	< .001	0.14 [0.13, 0.15]	< .001

Table 8 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.01, 0.02]	.571	-0.08 [-0.10, -0.06]	$< .001$
Physical activity^2	-0.15 [-0.16, -0.14]	$< .001$	-0.09 [-0.09, -0.08]	$< .001$
Physical activity $\times$ Age	0.00 [-0.01, 0.00]	.063	-0.05 [-0.05, -0.05]	$< .001$
Age $\times$ Physical activity^2	-0.03 [-0.04, -0.03]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 9

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.13 [-0.17, 0.44]	.382	-0.07 [-0.19, 0.05]	.248
Sleep duration	0.02 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	-0.23 [-0.24, -0.22]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.045
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.374
Sleep efficiency (z)				
(Intercept)	0.12 [-0.18, 0.42]	.416	-0.07 [-0.20, 0.05]	.236
Sleep efficiency	0.00 [0.00, 0.01]	.292	0.01 [0.00, 0.01]	.010
Age	-0.23 [-0.24, -0.21]	< .001	-0.27 [-0.29, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.209
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.224
Sleep onset (z)				
(Intercept)	0.13 [-0.18, 0.43]	.414	-0.08 [-0.21, 0.04]	.197
Sleep onset	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age	-0.22 [-0.24, -0.21]	< .001	-0.27 [-0.29, -0.26]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.892
Sleep onset × Age	0.00 [0.00, 0.00]	.011	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.016
Sleep regularity (z)				
(Intercept)	0.04 [-0.27, 0.34]	.819	-0.18 [-0.30, -0.05]	.006
Sleep regularity	0.12 [0.11, 0.12]	< .001	0.06 [0.06, 0.07]	< .001

Table 9 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.24 [-0.26, -0.22]	$< .001$	-0.32 [-0.34, -0.31]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.386	0.00 [0.00, 0.00]	.364
Sleep regularity $\times$ Age	-0.01 [-0.01, -0.01]	$< .001$	-0.01 [-0.01, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.151	0.00 [0.00, 0.00]	.015

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 10

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.03 [-0.28, 0.23]	.834	-0.04 [-0.28, 0.20]	.757
Physical activity	0.09 [0.08, 0.10]	< .001	0.05 [0.04, 0.06]	< .001
Age	0.03 [0.01, 0.04]	< .001	0.03 [0.01, 0.04]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.403
Physical activity × Age	-0.01 [-0.01, 0.00]	.004	-0.01 [-0.01, 0.00]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.303	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.12 [-0.18, 0.43]	.425	0.11 [-0.21, 0.44]	.499
Physical activity	0.06 [0.05, 0.07]	< .001	0.04 [0.03, 0.05]	< .001
Age	-0.05 [-0.06, -0.03]	< .001	-0.05 [-0.07, -0.03]	< .001
Physical activity^2	-0.02 [-0.03, -0.02]	< .001	-0.02 [-0.03, -0.02]	< .001
Physical activity × Age	-0.03 [-0.03, -0.03]	< .001	-0.02 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.01]	.021	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.00 [-0.40, 0.39]	.984	0.02 [-0.36, 0.39]	.937
Physical activity	-0.09 [-0.10, -0.08]	< .001	-0.05 [-0.05, -0.04]	< .001
Age	-0.07 [-0.08, -0.06]	< .001	-0.06 [-0.07, -0.04]	< .001
Physical activity^2	0.01 [0.00, 0.01]	< .001	-0.01 [-0.01, 0.00]	.001
Physical activity × Age	-0.03 [-0.03, -0.02]	< .001	0.00 [-0.01, 0.00]	.034
Age × Physical activity^2	0.01 [0.00, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.21 [0.06, 0.36]	.006	0.05 [-0.23, 0.33]	.723
Physical activity	0.28 [0.27, 0.30]	< .001	0.16 [0.14, 0.17]	< .001

Table 10 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.02, 0.04]	.381	-0.12 [-0.15, -0.09]	$< .001$
Physical activity^2	-0.17 [-0.18, -0.15]	$< .001$	-0.11 [-0.12, -0.10]	$< .001$
Physical activity $\times$ Age	-0.01 [-0.02, -0.01]	$< .001$	-0.05 [-0.06, -0.05]	$< .001$
Age $\times$ Physical activity^2	-0.04 [-0.05, -0.04]	$< .001$	0.02 [0.02, 0.02]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 11

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.01 [-0.37, 0.34]	.939	-0.10 [-0.25, 0.04]	.167
Sleep duration	0.02 [0.02, 0.03]	< .001	0.04 [0.04, 0.05]	< .001
Age	-0.19 [-0.20, -0.18]	< .001	-0.24 [-0.25, -0.22]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.02 [-0.37, 0.33]	.903	-0.11 [-0.25, 0.04]	.160
Sleep efficiency	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.001
Age	-0.18 [-0.20, -0.17]	< .001	-0.23 [-0.24, -0.22]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.852	0.00 [0.00, 0.00]	.138
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	.037	0.00 [0.00, 0.00]	.016
Sleep onset (z)				
(Intercept)	-0.05 [-0.42, 0.33]	.799	-0.10 [-0.25, 0.06]	.217
Sleep onset	-0.05 [-0.06, -0.04]	< .001	-0.02 [-0.03, -0.01]	< .001
Age	-0.19 [-0.21, -0.18]	< .001	-0.23 [-0.25, -0.22]	< .001
Sleep onset^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.065
Sleep onset × Age	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.116
Sleep regularity (z)				
(Intercept)	-0.12 [-0.47, 0.24]	.524	-0.22 [-0.38, -0.05]	.009
Sleep regularity	0.11 [0.10, 0.11]	< .001	0.07 [0.06, 0.08]	< .001

Table 11 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.18 [-0.20, -0.16]	$< .001$	-0.30 [-0.32, -0.28]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.232	0.00 [-0.01, 0.00]	.443
Sleep regularity $\times$ Age	-0.01 [-0.01, 0.00]	$< .001$	-0.02 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.269	0.00 [0.00, 0.00]	.915

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 12

Physical activity predicting sleep controlling for SES, age, and sex.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.04 [-0.19, 0.11]	.603	-0.05 [-0.18, 0.09]	.524
Physical activity	0.11 [0.10, 0.12]	< .001	0.06 [0.05, 0.07]	< .001
BMI z-score	0.00 [-0.01, 0.00]	.249	-0.02 [-0.02, -0.01]	< .001
Physical activity^2	0.00 [-0.01, 0.00]	.037	0.01 [0.01, 0.02]	< .001
Physical activity × BMI z-score	0.01 [0.00, 0.02]	.003	0.01 [0.01, 0.02]	< .001
BMI z-score × Physical activity^2	0.00 [0.00, 0.00]	.154	0.00 [0.00, 0.01]	.016
Sleep efficiency (z)				
(Intercept)	0.21 [0.07, 0.36]	.005	0.20 [0.05, 0.36]	.011
Physical activity	0.06 [0.05, 0.07]	< .001	0.01 [0.00, 0.01]	.072
BMI z-score	-0.01 [-0.02, 0.00]	.012	-0.02 [-0.03, -0.01]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [0.00, 0.00]	.947
Physical activity × BMI z-score	0.02 [0.02, 0.03]	< .001	0.01 [0.01, 0.02]	< .001
BMI z-score × Physical activity^2	0.00 [-0.01, 0.00]	.001	0.00 [-0.01, 0.00]	.576
Sleep onset (z)				
(Intercept)	0.15 [-0.13, 0.43]	.286	0.15 [-0.11, 0.41]	.260
Physical activity	-0.10 [-0.11, -0.09]	< .001	-0.08 [-0.08, -0.07]	< .001
BMI z-score	0.00 [-0.01, 0.00]	.218	0.00 [0.00, 0.01]	.547
Physical activity^2	0.00 [0.00, 0.00]	.594	-0.01 [-0.01, -0.01]	< .001
Physical activity × BMI z-score	-0.01 [-0.01, 0.00]	.006	-0.01 [-0.02, -0.01]	< .001
BMI z-score × Physical activity^2	0.00 [0.00, 0.00]	.920	0.00 [-0.01, 0.00]	.010
Sleep regularity (z)				
(Intercept)	0.14 [0.00, 0.29]	.058	0.14 [-0.04, 0.31]	.119
Physical activity	0.31 [0.30, 0.32]	< .001	0.10 [0.09, 0.11]	< .001

Table 12 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	-0.05 [-0.06, -0.05]	$< .001$	-0.06 [-0.07, -0.05]	$< .001$
Physical activity^2	-0.06 [-0.06, -0.05]	$< .001$	-0.02 [-0.02, -0.02]	$< .001$
Physical activity $\times$ BMI z-score	0.04 [0.03, 0.05]	$< .001$	0.03 [0.02, 0.03]	$< .001$
BMI z-score $\times$ Physical activity^2	-0.01 [-0.01, 0.00]	$< .001$	-0.01 [-0.01, 0.00]	.004

Note. Adjusted for SES, age, and sex. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 13

Sleep predicting physical activity controlling for SES, age, and sex

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.07 [-0.25, 0.39]	.661	-0.16 [-0.29, -0.02]	.025
Sleep duration	0.02 [0.02, 0.03]	< .001	0.02 [0.02, 0.03]	< .001
BMI z-score	-0.05 [-0.06, -0.04]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep duration^2	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × BMI z-score	0.00 [0.00, 0.01]	.026	0.00 [0.00, 0.01]	.041
BMI z-score × Sleep duration^2	0.00 [0.00, 0.00]	.783	0.00 [0.00, 0.01]	.019
Sleep efficiency (z)				
(Intercept)	0.06 [-0.26, 0.38]	.723	-0.16 [-0.30, -0.03]	.020
Sleep efficiency	0.00 [0.00, 0.01]	.552	0.01 [0.00, 0.01]	.002
BMI z-score	-0.05 [-0.06, -0.04]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × BMI z-score	-0.01 [-0.01, 0.00]	.017	0.00 [-0.01, 0.00]	.245
BMI z-score × Sleep efficiency^2	0.00 [0.00, 0.00]	.783	0.00 [0.00, 0.00]	.854
Sleep onset (z)				
(Intercept)	0.06 [-0.27, 0.38]	.730	-0.17 [-0.30, -0.03]	.019
Sleep onset	-0.02 [-0.03, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
BMI z-score	-0.05 [-0.06, -0.05]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.020	0.00 [0.00, 0.01]	.212
Sleep onset × BMI z-score	0.00 [-0.01, 0.00]	.191	0.00 [0.00, 0.01]	.682
BMI z-score × Sleep onset^2	0.00 [0.00, 0.01]	.026	0.00 [0.00, 0.00]	.838
Sleep regularity (z)				
(Intercept)	0.04 [-0.26, 0.35]	.784	-0.17 [-0.30, -0.04]	.009
Sleep regularity	0.12 [0.12, 0.13]	< .001	0.07 [0.07, 0.08]	< .001

Table 13 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	-0.04 [-0.05, -0.04]	$< .001$	-0.05 [-0.06, -0.05]	$< .001$
Sleep regularity^2	0.00 [-0.01, 0.00]	.083	0.00 [-0.01, 0.00]	.045
Sleep regularity $\times$ BMI z-score	0.00 [-0.01, 0.00]	.272	0.00 [0.00, 0.01]	.324
BMI z-score $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.685	0.00 [0.00, 0.00]	.479

Note. Adjusted for SES, age, and sex. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 14

Physical activity predicting sleep controlling for age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [-0.19, 0.20]	.946	0.01 [-0.17, 0.20]	.884
Physical activity	0.09 [0.08, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Medium	0.01 [-0.01, 0.03]	.324	0.01 [-0.01, 0.04]	.240
High	0.03 [0.00, 0.05]	.022	0.01 [-0.01, 0.03]	.412
Physical activity^2	0.00 [-0.01, 0.00]	.522	0.01 [0.00, 0.02]	.035
Physical activity × Medium	0.01 [-0.01, 0.03]	.300	0.00 [-0.01, 0.02]	.631
Physical activity × High	0.02 [0.00, 0.04]	.094	0.00 [-0.02, 0.01]	.761
Medium × Physical activity^2	0.00 [0.00, 0.01]	.439	-0.01 [-0.02, 0.00]	.239
High × Physical activity^2	-0.01 [-0.01, 0.00]	.151	0.00 [-0.01, 0.01]	.719
Sleep efficiency (z)				
(Intercept)	0.15 [-0.04, 0.33]	.129	0.13 [-0.06, 0.33]	.191
Physical activity	0.03 [0.01, 0.04]	< .001	-0.01 [-0.02, 0.01]	.253
Medium	0.03 [0.00, 0.05]	.021	0.04 [0.01, 0.06]	.002
High	0.04 [0.01, 0.06]	.002	0.03 [0.01, 0.06]	.014
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.01 [0.00, 0.01]	.193
Physical activity × Medium	0.01 [-0.01, 0.02]	.620	0.01 [0.00, 0.03]	.096
Physical activity × High	0.04 [0.02, 0.06]	< .001	0.03 [0.01, 0.04]	.001
Medium × Physical activity^2	0.00 [0.00, 0.01]	.252	-0.01 [-0.02, 0.00]	.128
High × Physical activity^2	-0.01 [-0.01, 0.00]	.051	-0.01 [-0.02, 0.00]	.299
Sleep onset (z)				
(Intercept)	0.06 [-0.26, 0.39]	.701	0.05 [-0.25, 0.36]	.746
Physical activity	-0.14 [-0.16, -0.13]	< .001	-0.09 [-0.10, -0.08]	< .001
Medium	0.04 [0.02, 0.05]	< .001	0.04 [0.02, 0.06]	< .001

Table 14 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High	0.02 [0.00, 0.04]	.045	0.03 [0.01, 0.05]	.006
Physical activity^2	0.01 [0.00, 0.01]	.010	0.00 [-0.01, 0.00]	.221
Physical activity × Medium	0.03 [0.01, 0.04]	.001	0.01 [0.00, 0.02]	.093
Physical activity × High	0.06 [0.04, 0.08]	$<$ .001	0.03 [0.02, 0.04]	$<$ .001
Medium × Physical activity^2	0.00 [-0.01, 0.00]	.431	0.00 [-0.01, 0.01]	.826
High × Physical activity^2	0.00 [-0.01, 0.00]	.084	0.00 [-0.01, 0.00]	.243
Sleep regularity (z)				
(Intercept)	0.15 [-0.01, 0.32]	.068	0.15 [-0.03, 0.34]	.104
Physical activity	0.34 [0.32, 0.36]	$<$ .001	0.08 [0.07, 0.10]	$<$ .001
Medium	0.01 [-0.02, 0.04]	.385	0.00 [-0.03, 0.02]	.743
High	0.04 [0.01, 0.07]	.003	0.03 [0.00, 0.06]	.027
Physical activity^2	-0.06 [-0.07, -0.06]	$<$ .001	-0.03 [-0.04, -0.02]	$<$ .001
Physical activity × Medium	-0.03 [-0.06, -0.01]	.008	0.00 [-0.02, 0.01]	.677
Physical activity × High	-0.03 [-0.06, -0.01]	.009	0.03 [0.01, 0.05]	.001
Medium × Physical activity^2	0.01 [0.00, 0.01]	.097	0.01 [0.00, 0.02]	.114
High × Physical activity^2	0.01 [0.00, 0.02]	.056	0.00 [-0.02, 0.01]	.487

Note. Adjusted for age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 15

Sleep predicting physical activity controlling for age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.16 [-0.14, 0.47]	.295	-0.06 [-0.19, 0.06]	.312
Sleep duration	-0.01 [-0.01, 0.00]	.087	0.02 [0.01, 0.03]	< .001
Medium	-0.06 [-0.08, -0.04]	< .001	0.01 [-0.01, 0.02]	.453
High	-0.08 [-0.10, -0.06]	< .001	0.02 [0.00, 0.03]	.088
Sleep duration^2	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Medium	0.02 [0.01, 0.03]	< .001	0.00 [-0.01, 0.01]	.666
Sleep duration × High	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.029
Medium × Sleep duration^2	0.00 [0.00, 0.01]	.527	0.00 [0.00, 0.01]	.360
High × Sleep duration^2	0.00 [0.00, 0.01]	.417	0.00 [0.00, 0.01]	.658
Sleep efficiency (z)				
(Intercept)	0.15 [-0.16, 0.45]	.347	-0.07 [-0.20, 0.05]	.265
Sleep efficiency	0.01 [0.00, 0.02]	.029	0.00 [-0.01, 0.01]	.463
Medium	-0.06 [-0.07, -0.04]	< .001	0.01 [-0.01, 0.03]	.177
High	-0.07 [-0.09, -0.05]	< .001	0.03 [0.01, 0.05]	.003
Sleep efficiency^2	0.00 [-0.01, 0.00]	.096	0.00 [-0.01, 0.00]	.053
Sleep efficiency × Medium	-0.01 [-0.02, 0.01]	.290	0.01 [0.00, 0.02]	.229
Sleep efficiency × High	0.00 [-0.01, 0.01]	.712	0.00 [-0.01, 0.02]	.578
Medium × Sleep efficiency^2	0.00 [-0.01, 0.00]	.304	0.00 [-0.01, 0.00]	.580
High × Sleep efficiency^2	-0.01 [-0.01, 0.00]	.025	-0.01 [-0.01, 0.00]	.003
Sleep onset (z)				
(Intercept)	0.16 [-0.15, 0.47]	.312	-0.07 [-0.20, 0.05]	.263
Sleep onset	-0.02 [-0.03, -0.01]	< .001	0.00 [-0.01, 0.01]	.836
Medium	-0.07 [-0.09, -0.05]	< .001	0.01 [-0.01, 0.03]	.302

Table 15 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High	-0.09 [-0.11, -0.07]	< .001	0.01 [-0.01, 0.03]	.154
Sleep onset^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.986
Sleep onset × Medium	0.01 [0.00, 0.02]	.051	-0.01 [-0.02, 0.01]	.394
Sleep onset × High	0.00 [-0.01, 0.02]	.672	-0.02 [-0.03, -0.01]	.003
Medium × Sleep onset^2	0.01 [0.00, 0.02]	.002	0.00 [-0.01, 0.01]	.815
High × Sleep onset^2	0.01 [0.00, 0.02]	.022	0.00 [0.00, 0.01]	.226
Sleep regularity (z)				
(Intercept)	0.09 [-0.23, 0.40]	.598	-0.15 [-0.27, -0.04]	.010
Sleep regularity	0.13 [0.12, 0.14]	< .001	0.08 [0.07, 0.09]	< .001
Medium	-0.04 [-0.06, -0.02]	< .001	0.02 [0.00, 0.04]	.022
High	-0.06 [-0.08, -0.04]	< .001	0.03 [0.01, 0.05]	.001
Sleep regularity^2	0.00 [0.00, 0.01]	.714	0.00 [0.00, 0.01]	.513
Sleep regularity × Medium	-0.01 [-0.03, 0.00]	.050	-0.02 [-0.04, -0.01]	< .001
Sleep regularity × High	-0.01 [-0.02, 0.00]	.104	0.00 [-0.01, 0.02]	.469
Medium × Sleep regularity^2	-0.01 [-0.01, 0.00]	.124	0.00 [-0.01, 0.00]	.236
High × Sleep regularity^2	-0.01 [-0.01, 0.00]	.143	-0.01 [-0.02, 0.00]	.041

Note. Adjusted for age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 16

Physical activity predicting sleep controlling for SES, age, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [-0.15, 0.17]	.865	0.01 [-0.13, 0.16]	.857
Physical activity	0.12 [0.11, 0.13]	< .001	0.07 [0.06, 0.08]	< .001
Male	-0.13 [-0.15, -0.12]	< .001	-0.13 [-0.15, -0.11]	< .001
Physical activity^2	0.00 [-0.01, 0.00]	.089	0.01 [0.00, 0.02]	< .001
Physical activity × Male	-0.02 [-0.04, -0.01]	.004	0.00 [-0.01, 0.01]	.996
Male × Physical activity^2	0.00 [0.00, 0.01]	.444	0.00 [-0.01, 0.01]	.486
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.34]	.020	0.17 [0.00, 0.33]	.044
Physical activity	0.05 [0.04, 0.06]	< .001	-0.01 [-0.02, 0.00]	.058
Male	-0.18 [-0.19, -0.16]	< .001	-0.16 [-0.18, -0.15]	< .001
Physical activity^2	-0.02 [-0.02, -0.02]	< .001	0.00 [-0.01, 0.01]	.991
Physical activity × Male	-0.01 [-0.02, 0.01]	.392	0.03 [0.02, 0.04]	< .001
Male × Physical activity^2	0.01 [0.00, 0.02]	< .001	0.00 [-0.01, 0.01]	.517
Sleep onset (z)				
(Intercept)	0.12 [-0.16, 0.39]	.397	0.11 [-0.15, 0.36]	.415
Physical activity	-0.12 [-0.13, -0.11]	< .001	-0.09 [-0.10, -0.08]	< .001
Male	0.01 [0.00, 0.02]	.125	0.02 [0.01, 0.03]	.004
Physical activity^2	0.00 [0.00, 0.00]	.637	0.00 [-0.01, 0.00]	.054
Physical activity × Male	0.01 [0.00, 0.02]	.026	0.02 [0.01, 0.03]	< .001
Male × Physical activity^2	0.00 [0.00, 0.01]	.476	-0.01 [-0.01, 0.00]	.013
Sleep regularity (z)				
(Intercept)	0.16 [0.01, 0.30]	.034	0.14 [-0.03, 0.31]	.108
Physical activity	0.33 [0.32, 0.35]	< .001	0.09 [0.08, 0.10]	< .001

Table 16 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male	-0.18 [-0.19, -0.16]	$< .001$	-0.15 [-0.17, -0.13]	$< .001$
Physical activity^2	-0.07 [-0.08, -0.07]	$< .001$	-0.03 [-0.03, -0.02]	$< .001$
Physical activity $\times$ Male	-0.03 [-0.04, -0.01]	.003	0.03 [0.02, 0.04]	$< .001$
Male $\times$ Physical activity^2	0.02 [0.01, 0.03]	$< .001$	0.01 [0.00, 0.02]	.012

Note. Adjusted for SES, age, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 17

Sleep predicting physical activity controlling for SES, age, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [-0.16, 0.44]	.366	-0.07 [-0.19, 0.05]	.253
Sleep duration	0.01 [0.00, 0.01]	.027	0.02 [0.02, 0.03]	< .001
Male	0.16 [0.15, 0.17]	< .001	0.22 [0.21, 0.24]	< .001
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep duration × Male	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.459
Male × Sleep duration^2	0.00 [-0.01, 0.00]	.832	0.00 [-0.01, 0.00]	.782
Sleep efficiency (z)				
(Intercept)	0.12 [-0.18, 0.42]	.419	-0.07 [-0.20, 0.05]	.243
Sleep efficiency	0.02 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	< .001
Male	0.16 [0.15, 0.17]	< .001	0.22 [0.21, 0.23]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Male	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, 0.00]	.068
Male × Sleep efficiency^2	0.00 [-0.01, 0.00]	.100	0.00 [0.00, 0.01]	.612
Sleep onset (z)				
(Intercept)	0.13 [-0.18, 0.43]	.410	-0.08 [-0.20, 0.05]	.215
Sleep onset	0.00 [-0.01, 0.01]	.550	-0.01 [-0.02, 0.00]	.006
Male	0.15 [0.14, 0.17]	< .001	0.22 [0.21, 0.23]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.011	0.00 [0.00, 0.01]	.256
Sleep onset × Male	-0.05 [-0.06, -0.04]	< .001	-0.01 [-0.02, 0.00]	.073
Male × Sleep onset^2	0.00 [0.00, 0.01]	.336	0.00 [0.00, 0.01]	.575
Sleep regularity (z)				
(Intercept)	0.04 [-0.27, 0.34]	.814	-0.18 [-0.30, -0.05]	.007
Sleep regularity	0.11 [0.10, 0.11]	< .001	0.07 [0.06, 0.08]	< .001

Table 17 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male	0.20 [0.19, 0.22]	$< .001$	0.24 [0.23, 0.26]	$< .001$
Sleep regularity^2	0.00 [-0.01, 0.00]	.022	0.00 [-0.01, 0.00]	.137
Sleep regularity $\times$ Male	0.04 [0.03, 0.05]	$< .001$	0.00 [-0.01, 0.01]	.501
Male $\times$ Sleep regularity^2	0.01 [0.00, 0.01]	.015	0.00 [0.00, 0.01]	.584

Note. Adjusted for SES, age, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 18

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.10 [-0.06, 0.26]	.215	0.11 [-0.03, 0.26]	.133
Physical activity	0.10 [0.09, 0.12]	< .001	0.03 [0.02, 0.04]	< .001
Monday	-0.11 [-0.13, -0.10]	< .001	-0.12 [-0.14, -0.10]	< .001
Saturday	-0.01 [-0.03, 0.00]	.160	-0.03 [-0.05, -0.01]	< .001
Sunday	-0.14 [-0.15, -0.12]	< .001	-0.14 [-0.16, -0.12]	< .001
Thursday	-0.12 [-0.14, -0.11]	< .001	-0.13 [-0.15, -0.11]	< .001
Tuesday	-0.13 [-0.15, -0.12]	< .001	-0.15 [-0.17, -0.13]	< .001
Wednesday	-0.14 [-0.15, -0.12]	< .001	-0.15 [-0.17, -0.13]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.02, 0.00]	.038
Physical activity × Monday	0.01 [-0.01, 0.03]	.504	0.07 [0.05, 0.08]	< .001
Physical activity × Saturday	0.03 [0.02, 0.05]	< .001	0.00 [-0.01, 0.02]	.753
Physical activity × Sunday	-0.01 [-0.03, 0.01]	.500	0.04 [0.03, 0.05]	< .001
Physical activity × Thursday	-0.01 [-0.03, 0.01]	.144	0.06 [0.05, 0.07]	< .001
Physical activity × Tuesday	0.00 [-0.02, 0.02]	.698	0.08 [0.06, 0.09]	< .001
Physical activity × Wednesday	-0.02 [-0.04, 0.00]	.119	0.06 [0.04, 0.07]	< .001
Monday × Physical activity^2	0.02 [0.01, 0.02]	< .001	0.02 [0.00, 0.03]	.007
Saturday × Physical activity^2	-0.01 [-0.02, 0.00]	.033	0.01 [-0.01, 0.02]	.337
Sunday × Physical activity^2	0.03 [0.02, 0.03]	< .001	0.02 [0.01, 0.03]	< .001
Thursday × Physical activity^2	0.02 [0.01, 0.03]	< .001	0.02 [0.01, 0.04]	< .001
Tuesday × Physical activity^2	0.02 [0.01, 0.03]	< .001	0.03 [0.02, 0.04]	< .001
Wednesday × Physical activity^2	0.02 [0.02, 0.03]	< .001	0.03 [0.02, 0.04]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [0.00, 0.31]	.057	0.15 [-0.02, 0.31]	.078

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity	0.04 [0.03, 0.06]	< .001	-0.02 [-0.03, -0.01]	.002
Monday	0.06 [0.04, 0.07]	< .001	0.07 [0.05, 0.09]	< .001
Saturday	-0.03 [-0.04, -0.01]	< .001	-0.06 [-0.08, -0.05]	< .001
Sunday	0.00 [-0.02, 0.01]	.938	0.00 [-0.02, 0.01]	.733
Thursday	0.05 [0.04, 0.07]	< .001	0.06 [0.05, 0.08]	< .001
Tuesday	0.05 [0.03, 0.06]	< .001	0.06 [0.04, 0.08]	< .001
Wednesday	0.06 [0.04, 0.07]	< .001	0.07 [0.05, 0.09]	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	.015	0.00 [0.00, 0.01]	.349
Physical activity × Monday	-0.03 [-0.05, -0.01]	.009	0.03 [0.01, 0.04]	< .001
Physical activity × Saturday	0.06 [0.04, 0.08]	< .001	0.01 [0.00, 0.03]	.028
Physical activity × Sunday	-0.01 [-0.03, 0.01]	.440	0.01 [-0.01, 0.02]	.411
Physical activity × Thursday	-0.03 [-0.05, -0.01]	.002	0.02 [0.00, 0.03]	.018
Physical activity × Tuesday	-0.02 [-0.04, 0.00]	.036	0.02 [0.01, 0.04]	< .001
Physical activity × Wednesday	-0.01 [-0.03, 0.01]	.206	0.02 [0.01, 0.04]	.001
Monday × Physical activity^2	0.00 [-0.01, 0.00]	.345	-0.02 [-0.03, -0.01]	.003
Saturday × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.01 [0.00, 0.02]	.024
Sunday × Physical activity^2	0.00 [-0.01, 0.01]	.794	-0.01 [-0.02, 0.01]	.322
Thursday × Physical activity^2	0.00 [-0.01, 0.01]	.790	-0.01 [-0.02, 0.00]	.035
Tuesday × Physical activity^2	0.00 [-0.01, 0.01]	.542	-0.01 [-0.03, 0.00]	.014
Wednesday × Physical activity^2	0.00 [-0.01, 0.01]	.486	-0.02 [-0.03, 0.00]	.008
Sleep onset (z)				
(Intercept)	0.20 [-0.07, 0.47]	.147	0.19 [-0.07, 0.44]	.153
Physical activity	-0.07 [-0.08, -0.06]	< .001	-0.06 [-0.07, -0.05]	< .001
Monday	-0.16 [-0.17, -0.15]	< .001	-0.14 [-0.15, -0.13]	< .001

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday	0.09 [0.07, 0.10]	< .001	0.07 [0.05, 0.08]	< .001
Sunday	-0.15 [-0.16, -0.14]	< .001	-0.14 [-0.15, -0.13]	< .001
Thursday	-0.13 [-0.14, -0.12]	< .001	-0.11 [-0.13, -0.10]	< .001
Tuesday	-0.14 [-0.16, -0.13]	< .001	-0.12 [-0.14, -0.11]	< .001
Wednesday	-0.14 [-0.15, -0.13]	< .001	-0.12 [-0.14, -0.11]	< .001
Physical activity^2	0.00 [0.00, 0.00]	.890	0.00 [-0.01, 0.00]	.572
Physical activity × Monday	-0.09 [-0.11, -0.08]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Saturday	0.05 [0.03, 0.06]	< .001	0.01 [0.00, 0.02]	.021
Physical activity × Sunday	-0.06 [-0.07, -0.05]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Thursday	-0.07 [-0.09, -0.06]	< .001	-0.03 [-0.04, -0.02]	< .001
Physical activity × Tuesday	-0.08 [-0.10, -0.07]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Wednesday	-0.08 [-0.10, -0.07]	< .001	-0.02 [-0.03, -0.01]	< .001
Monday × Physical activity^2	0.01 [0.00, 0.02]	< .001	0.00 [-0.01, 0.00]	.272
Saturday × Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.811
Sunday × Physical activity^2	0.00 [0.00, 0.01]	.163	0.00 [0.00, 0.01]	.430
Thursday × Physical activity^2	0.01 [0.00, 0.01]	.021	-0.01 [-0.02, 0.00]	.046
Tuesday × Physical activity^2	0.01 [0.00, 0.01]	.003	-0.01 [-0.02, 0.00]	.009
Wednesday × Physical activity^2	0.01 [0.00, 0.01]	.003	-0.01 [-0.02, 0.00]	.062
Sleep regularity (z)				
(Intercept)	-0.07 [-0.22, 0.08]	.341	-0.07 [-0.25, 0.11]	.453
Physical activity	0.28 [0.26, 0.29]	< .001	0.00 [-0.02, 0.01]	.484
Monday	0.43 [0.41, 0.45]	< .001	0.46 [0.44, 0.48]	< .001
Saturday	-0.03 [-0.05, -0.01]	< .001	-0.06 [-0.08, -0.04]	< .001
Sunday	-0.01 [-0.03, 0.00]	.152	-0.07 [-0.09, -0.05]	< .001

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	0.34 [0.32, 0.35]	< .001	0.36 [0.35, 0.38]	< .001
Tuesday	0.42 [0.41, 0.44]	< .001	0.45 [0.43, 0.47]	< .001
Wednesday	0.41 [0.39, 0.42]	< .001	0.45 [0.43, 0.47]	< .001
Physical activity^2	-0.04 [-0.05, -0.03]	< .001	0.00 [-0.01, 0.01]	.770
Physical activity × Monday	-0.03 [-0.05, 0.00]	.018	0.11 [0.10, 0.13]	< .001
Physical activity × Saturday	-0.01 [-0.03, 0.01]	.463	0.02 [0.01, 0.04]	.001
Physical activity × Sunday	0.03 [0.01, 0.05]	.004	0.06 [0.05, 0.08]	< .001
Physical activity × Thursday	-0.01 [-0.03, 0.01]	.213	0.08 [0.06, 0.09]	< .001
Physical activity × Tuesday	-0.02 [-0.04, 0.00]	.092	0.11 [0.10, 0.13]	< .001
Physical activity × Wednesday	-0.03 [-0.05, -0.01]	.014	0.08 [0.07, 0.10]	< .001
Monday × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	-0.07 [-0.08, -0.05]	< .001
Saturday × Physical activity^2	0.00 [-0.01, 0.01]	.820	0.00 [-0.01, 0.01]	.922
Sunday × Physical activity^2	-0.01 [-0.02, 0.00]	.055	0.01 [-0.01, 0.02]	.286
Thursday × Physical activity^2	-0.01 [-0.02, 0.00]	.040	-0.05 [-0.06, -0.03]	< .001
Tuesday × Physical activity^2	-0.01 [-0.02, 0.00]	.086	-0.05 [-0.06, -0.04]	< .001
Wednesday × Physical activity^2	-0.01 [-0.02, 0.00]	.004	-0.07 [-0.08, -0.05]	< .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 19

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.21 [-0.10, 0.51]	.182	-0.04 [-0.16, 0.09]	.545
Sleep duration	0.03 [0.02, 0.04]	< .001	0.06 [0.05, 0.07]	< .001
Monday	-0.07 [-0.08, -0.06]	< .001	-0.01 [-0.03, 0.00]	.032
Saturday	-0.12 [-0.13, -0.11]	< .001	-0.12 [-0.13, -0.11]	< .001
Sunday	-0.18 [-0.20, -0.17]	< .001	-0.13 [-0.14, -0.11]	< .001
Thursday	0.00 [-0.01, 0.01]	.883	0.03 [0.01, 0.04]	< .001
Tuesday	-0.02 [-0.03, -0.01]	.003	0.02 [0.01, 0.03]	.004
Wednesday	-0.04 [-0.05, -0.03]	< .001	0.02 [0.01, 0.04]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	0.00 [-0.01, 0.00]	.667
Sleep duration × Monday	-0.01 [-0.02, 0.00]	.023	-0.02 [-0.03, -0.01]	< .001
Sleep duration × Saturday	-0.05 [-0.06, -0.04]	< .001	-0.08 [-0.09, -0.06]	< .001
Sleep duration × Sunday	-0.03 [-0.04, -0.02]	< .001	-0.07 [-0.08, -0.06]	< .001
Sleep duration × Thursday	0.01 [0.00, 0.02]	.033	0.01 [0.00, 0.02]	.170
Sleep duration × Tuesday	0.01 [-0.01, 0.02]	.291	-0.01 [-0.02, 0.00]	.102
Sleep duration × Wednesday	0.01 [-0.01, 0.02]	.383	0.01 [0.00, 0.02]	.143
Monday × Sleep duration^2	0.01 [0.00, 0.02]	.003	0.00 [-0.01, 0.01]	.855
Saturday × Sleep duration^2	0.00 [0.00, 0.01]	.360	-0.01 [-0.01, 0.00]	.043
Sunday × Sleep duration^2	0.01 [0.00, 0.01]	.032	0.00 [-0.01, 0.00]	.369
Thursday × Sleep duration^2	0.00 [-0.01, 0.01]	.940	-0.01 [-0.02, 0.00]	.028
Tuesday × Sleep duration^2	0.00 [-0.01, 0.01]	.969	-0.01 [-0.02, 0.00]	.032
Wednesday × Sleep duration^2	0.01 [0.00, 0.02]	.070	0.00 [-0.01, 0.00]	.354
Sleep efficiency (z)				
(Intercept)	0.19 [-0.11, 0.49]	.223	-0.03 [-0.16, 0.09]	.585

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency	-0.01 [-0.02, 0.00]	.070	-0.02 [-0.03, -0.01]	< .001
Monday	-0.07 [-0.08, -0.05]	< .001	-0.02 [-0.03, 0.00]	.015
Saturday	-0.12 [-0.13, -0.11]	< .001	-0.14 [-0.15, -0.12]	< .001
Sunday	-0.17 [-0.18, -0.16]	< .001	-0.13 [-0.14, -0.12]	< .001
Thursday	0.01 [-0.01, 0.02]	.439	0.02 [0.00, 0.03]	.014
Tuesday	-0.01 [-0.02, 0.00]	.126	0.02 [0.00, 0.03]	.021
Wednesday	-0.03 [-0.04, -0.02]	< .001	0.02 [0.01, 0.03]	.004
Sleep efficiency^2	0.00 [-0.01, 0.00]	.130	-0.01 [-0.01, 0.00]	.003
Sleep efficiency × Monday	0.02 [0.01, 0.03]	.001	0.02 [0.01, 0.03]	< .001
Sleep efficiency × Saturday	0.02 [0.01, 0.04]	< .001	0.06 [0.05, 0.07]	< .001
Sleep efficiency × Sunday	0.00 [-0.01, 0.02]	.555	0.06 [0.05, 0.08]	< .001
Sleep efficiency × Thursday	0.00 [-0.01, 0.02]	.657	0.00 [-0.01, 0.01]	.963
Sleep efficiency × Tuesday	0.00 [-0.01, 0.02]	.556	0.01 [0.00, 0.03]	.032
Sleep efficiency × Wednesday	0.01 [0.00, 0.02]	.185	0.01 [0.00, 0.02]	.165
Monday × Sleep efficiency^2	0.00 [0.00, 0.01]	.244	0.00 [-0.01, 0.01]	.762
Saturday × Sleep efficiency^2	0.00 [0.00, 0.01]	.203	0.01 [0.00, 0.01]	.025
Sunday × Sleep efficiency^2	-0.01 [-0.01, 0.00]	.017	0.00 [0.00, 0.01]	.570
Thursday × Sleep efficiency^2	0.00 [-0.01, 0.00]	.201	0.00 [-0.01, 0.01]	.840
Tuesday × Sleep efficiency^2	-0.01 [-0.02, 0.00]	.013	0.00 [-0.01, 0.00]	.407
Wednesday × Sleep efficiency^2	0.00 [-0.01, 0.01]	.948	0.00 [-0.01, 0.01]	.567
Sleep onset (z)				
(Intercept)	0.18 [-0.12, 0.49]	.232	-0.06 [-0.18, 0.07]	.380
Sleep onset	-0.04 [-0.05, -0.03]	< .001	-0.05 [-0.06, -0.04]	< .001
Monday	-0.06 [-0.08, -0.05]	< .001	-0.01 [-0.03, 0.00]	.049

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday	-0.11 [-0.12, -0.10]	< .001	-0.10 [-0.11, -0.09]	< .001
Sunday	-0.17 [-0.18, -0.16]	< .001	-0.11 [-0.12, -0.10]	< .001
Thursday	0.00 [-0.02, 0.01]	.734	0.01 [0.00, 0.03]	.045
Tuesday	-0.01 [-0.03, 0.00]	.025	0.02 [0.01, 0.03]	.002
Wednesday	-0.03 [-0.05, -0.02]	< .001	0.02 [0.01, 0.03]	.003
Sleep onset^2	0.00 [-0.01, 0.00]	.316	0.01 [0.00, 0.02]	< .001
Sleep onset × Monday	0.04 [0.02, 0.05]	< .001	0.04 [0.03, 0.05]	< .001
Sleep onset × Saturday	0.07 [0.06, 0.08]	< .001	0.12 [0.11, 0.13]	< .001
Sleep onset × Sunday	0.05 [0.04, 0.06]	< .001	0.13 [0.12, 0.14]	< .001
Sleep onset × Thursday	0.00 [-0.02, 0.01]	.453	-0.01 [-0.02, 0.00]	.228
Sleep onset × Tuesday	0.01 [0.00, 0.02]	.106	0.02 [0.01, 0.03]	< .001
Sleep onset × Wednesday	0.01 [0.00, 0.02]	.117	0.00 [-0.01, 0.02]	.417
Monday × Sleep onset^2	0.01 [0.00, 0.01]	.163	0.00 [-0.01, 0.01]	.865
Saturday × Sleep onset^2	-0.01 [-0.01, 0.00]	.051	-0.03 [-0.04, -0.02]	< .001
Sunday × Sleep onset^2	0.00 [-0.01, 0.00]	.261	-0.03 [-0.03, -0.02]	< .001
Thursday × Sleep onset^2	0.00 [-0.01, 0.01]	.639	0.00 [-0.01, 0.01]	.674
Tuesday × Sleep onset^2	-0.01 [-0.01, 0.00]	.180	-0.01 [-0.02, 0.00]	.014
Wednesday × Sleep onset^2	0.00 [-0.01, 0.01]	.701	0.00 [-0.01, 0.01]	.640
Sleep regularity (z)				
(Intercept)	0.08 [-0.23, 0.39]	.603	-0.13 [-0.26, 0.00]	.049
Sleep regularity	0.13 [0.12, 0.14]	< .001	0.04 [0.03, 0.05]	< .001
Monday	-0.03 [-0.05, -0.02]	< .001	0.00 [-0.01, 0.01]	.986
Saturday	-0.08 [-0.09, -0.06]	< .001	-0.13 [-0.15, -0.12]	< .001
Sunday	-0.11 [-0.13, -0.10]	< .001	-0.14 [-0.16, -0.13]	< .001

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	-0.01 [-0.02, 0.01]	.352	0.03 [0.01, 0.04]	$< .001$
Tuesday	-0.03 [-0.05, -0.02]	$< .001$	0.00 [-0.01, 0.02]	.607
Wednesday	-0.04 [-0.05, -0.02]	$< .001$	0.02 [0.00, 0.04]	.014
Sleep regularity^2	0.00 [-0.01, 0.01]	.685	-0.01 [-0.01, 0.00]	.099
Sleep regularity × Monday	-0.04 [-0.05, -0.03]	$< .001$	-0.02 [-0.03, 0.00]	.013
Sleep regularity × Saturday	-0.01 [-0.03, 0.00]	.039	0.02 [0.01, 0.03]	.005
Sleep regularity × Sunday	-0.03 [-0.04, -0.01]	$< .001$	0.01 [0.00, 0.03]	.049
Sleep regularity × Thursday	-0.02 [-0.03, 0.00]	.014	0.00 [-0.02, 0.01]	.681
Sleep regularity × Tuesday	-0.01 [-0.02, 0.01]	.248	0.01 [0.00, 0.03]	.051
Sleep regularity × Wednesday	-0.02 [-0.04, -0.01]	.001	0.01 [0.00, 0.02]	.209
Monday × Sleep regularity^2	0.01 [0.00, 0.01]	.257	0.00 [-0.01, 0.01]	.972
Saturday × Sleep regularity^2	-0.01 [-0.02, 0.00]	.003	0.00 [-0.01, 0.01]	.697
Sunday × Sleep regularity^2	-0.02 [-0.03, -0.01]	$< .001$	0.00 [-0.01, 0.01]	.978
Thursday × Sleep regularity^2	0.00 [-0.01, 0.01]	.610	-0.01 [-0.02, 0.00]	.185
Tuesday × Sleep regularity^2	0.01 [0.00, 0.02]	.188	0.01 [0.00, 0.02]	.290
Wednesday × Sleep regularity^2	-0.01 [-0.02, 0.00]	.238	0.00 [-0.01, 0.01]	.913

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 20

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [-0.14, 0.18]	.768	0.02 [-0.13, 0.17]	.792
Physical activity	0.12 [0.10, 0.13]	< .001	0.06 [0.05, 0.07]	< .001
Spring	-0.02 [-0.05, 0.00]	.026	-0.01 [-0.03, 0.01]	.289
Summer	-0.10 [-0.12, -0.08]	< .001	-0.08 [-0.11, -0.06]	< .001
Winter	0.08 [0.06, 0.10]	< .001	0.07 [0.05, 0.09]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.781	0.02 [0.01, 0.02]	< .001
Physical activity × Spring	-0.02 [-0.04, 0.00]	.020	0.00 [-0.02, 0.01]	.638
Physical activity × Summer	-0.02 [-0.04, 0.01]	.132	0.00 [-0.02, 0.02]	.870
Physical activity × Winter	0.00 [-0.02, 0.02]	.754	0.01 [0.00, 0.03]	.065
Spring × Physical activity^2	0.00 [-0.01, 0.00]	.284	-0.01 [-0.02, 0.00]	.014
Summer × Physical activity^2	0.00 [-0.01, 0.01]	.894	-0.01 [-0.02, 0.00]	.162
Winter × Physical activity^2	-0.01 [-0.02, 0.00]	.004	-0.01 [-0.02, 0.00]	.099
Sleep efficiency (z)				
(Intercept)	0.17 [0.02, 0.33]	.029	0.16 [0.00, 0.32]	.056
Physical activity	0.06 [0.04, 0.07]	< .001	0.01 [0.00, 0.02]	.018
Spring	0.00 [-0.02, 0.03]	.712	0.01 [-0.02, 0.03]	.488
Summer	-0.02 [-0.04, 0.00]	.080	-0.01 [-0.03, 0.02]	.555
Winter	0.04 [0.01, 0.06]	.001	0.04 [0.02, 0.07]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.694
Physical activity × Spring	-0.03 [-0.05, -0.01]	.009	-0.02 [-0.04, -0.01]	.006
Physical activity × Summer	-0.02 [-0.04, 0.01]	.156	-0.01 [-0.03, 0.01]	.211
Physical activity × Winter	0.00 [-0.02, 0.02]	.995	0.00 [-0.02, 0.01]	.907
Spring × Physical activity^2	0.01 [0.00, 0.01]	.118	0.00 [-0.01, 0.01]	.837

Table 20 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Physical activity^2	0.01 [0.00, 0.01]	.123	-0.01 [-0.02, 0.01]	.398
Winter × Physical activity^2	0.00 [0.00, 0.01]	.569	-0.01 [-0.02, 0.00]	.122
Sleep onset (z)				
(Intercept)	0.10 [-0.17, 0.37]	.473	0.09 [-0.16, 0.35]	.471
Physical activity	-0.12 [-0.13, -0.11]	< .001	-0.08 [-0.09, -0.07]	< .001
Spring	0.02 [0.00, 0.04]	.052	0.02 [0.00, 0.04]	.064
Summer	0.07 [0.05, 0.09]	< .001	0.06 [0.04, 0.08]	< .001
Winter	-0.01 [-0.03, 0.01]	.210	0.00 [-0.01, 0.02]	.768
Physical activity^2	-0.01 [-0.01, 0.00]	.001	-0.01 [-0.02, -0.01]	< .001
Physical activity × Spring	0.01 [-0.01, 0.02]	.499	0.00 [-0.01, 0.01]	.997
Physical activity × Summer	0.04 [0.02, 0.06]	< .001	0.02 [0.01, 0.04]	< .001
Physical activity × Winter	0.00 [-0.02, 0.01]	.915	-0.01 [-0.02, 0.00]	.100
Spring × Physical activity^2	0.02 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.007
Summer × Physical activity^2	0.01 [0.00, 0.01]	.012	0.02 [0.01, 0.03]	< .001
Winter × Physical activity^2	0.01 [0.01, 0.02]	< .001	0.00 [-0.01, 0.01]	.679
Sleep regularity (z)				
(Intercept)	0.18 [0.03, 0.32]	.016	0.19 [0.01, 0.37]	.034
Physical activity	0.28 [0.27, 0.30]	< .001	0.12 [0.11, 0.13]	< .001
Spring	-0.08 [-0.10, -0.05]	< .001	-0.12 [-0.15, -0.09]	< .001
Summer	0.03 [0.00, 0.06]	.024	0.01 [-0.02, 0.04]	.586
Winter	-0.03 [-0.05, 0.00]	.029	-0.04 [-0.07, -0.02]	< .001
Physical activity^2	-0.05 [-0.05, -0.04]	< .001	-0.04 [-0.04, -0.03]	< .001
Physical activity × Spring	0.05 [0.03, 0.07]	< .001	-0.04 [-0.06, -0.02]	< .001
Physical activity × Summer	0.07 [0.04, 0.09]	< .001	-0.01 [-0.03, 0.01]	.359

Table 20 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Winter	0.04 [0.01, 0.06]	.001	0.00 [-0.02, 0.01]	.614
Spring × Physical activity^2	-0.02 [-0.03, -0.01]	$<$ .001	0.03 [0.01, 0.04]	$<$ .001
Summer × Physical activity^2	-0.01 [-0.02, 0.00]	.008	0.03 [0.02, 0.04]	$<$ .001
Winter × Physical activity^2	-0.01 [-0.02, 0.00]	.030	0.00 [-0.01, 0.01]	.861

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 21

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [-0.16, 0.44]	.368	-0.06 [-0.18, 0.07]	.385
Sleep duration	0.04 [0.03, 0.05]	< .001	0.04 [0.03, 0.05]	< .001
Spring	0.00 [-0.01, 0.02]	.627	-0.01 [-0.03, 0.01]	.174
Summer	0.02 [0.00, 0.04]	.035	-0.04 [-0.05, -0.02]	< .001
Winter	-0.04 [-0.06, -0.02]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep duration^2	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.01, 0.00]	.004
Sleep duration × Spring	-0.05 [-0.06, -0.04]	< .001	-0.03 [-0.04, -0.02]	< .001
Sleep duration × Summer	-0.05 [-0.06, -0.03]	< .001	-0.03 [-0.04, -0.02]	< .001
Sleep duration × Winter	0.00 [-0.01, 0.01]	.408	0.00 [-0.01, 0.01]	.543
Spring × Sleep duration^2	0.00 [-0.01, 0.01]	.843	0.00 [-0.01, 0.00]	.165
Summer × Sleep duration^2	0.00 [-0.01, 0.01]	.718	-0.01 [-0.02, 0.00]	.021
Winter × Sleep duration^2	0.00 [0.00, 0.01]	.456	0.00 [0.00, 0.01]	.583
Sleep efficiency (z)				
(Intercept)	0.13 [-0.17, 0.43]	.391	-0.05 [-0.17, 0.08]	.441
Sleep efficiency	0.01 [0.00, 0.02]	.113	0.01 [0.00, 0.02]	.018
Spring	0.00 [-0.02, 0.02]	.853	-0.02 [-0.04, -0.01]	.008
Summer	0.02 [0.00, 0.04]	.101	-0.05 [-0.07, -0.03]	< .001
Winter	-0.05 [-0.06, -0.03]	< .001	-0.04 [-0.05, -0.02]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency × Spring	0.00 [-0.02, 0.01]	.591	-0.01 [-0.02, 0.00]	.245
Sleep efficiency × Summer	-0.01 [-0.02, 0.01]	.445	0.00 [-0.02, 0.01]	.701
Sleep efficiency × Winter	0.00 [-0.01, 0.01]	.709	-0.01 [-0.02, 0.01]	.272
Spring × Sleep efficiency^2	0.01 [0.00, 0.01]	.050	0.01 [0.00, 0.01]	.059

Table 21 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Sleep efficiency^2	0.00 [0.00, 0.01]	.172	0.01 [0.00, 0.01]	.067
Winter × Sleep efficiency^2	0.01 [0.00, 0.01]	.037	0.00 [0.00, 0.01]	.093
Sleep onset (z)				
(Intercept)	0.13 [-0.17, 0.44]	.396	-0.06 [-0.19, 0.07]	.352
Sleep onset	-0.02 [-0.02, -0.01]	.001	-0.01 [-0.02, -0.01]	.002
Spring	0.01 [-0.01, 0.03]	.366	-0.02 [-0.03, 0.00]	.102
Summer	0.01 [-0.01, 0.03]	.223	-0.05 [-0.06, -0.03]	< .001
Winter	-0.04 [-0.06, -0.02]	< .001	-0.03 [-0.05, -0.01]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.255	0.00 [0.00, 0.01]	.173
Sleep onset × Spring	-0.02 [-0.03, -0.01]	< .001	0.00 [-0.01, 0.01]	.844
Sleep onset × Summer	0.03 [0.01, 0.04]	< .001	0.02 [0.01, 0.04]	.002
Sleep onset × Winter	-0.03 [-0.05, -0.02]	< .001	-0.01 [-0.02, 0.00]	.104
Spring × Sleep onset^2	0.00 [-0.01, 0.00]	.220	0.00 [-0.01, 0.00]	.561
Summer × Sleep onset^2	0.00 [-0.01, 0.01]	.485	0.00 [-0.01, 0.01]	.616
Winter × Sleep onset^2	0.00 [-0.01, 0.01]	.972	0.00 [-0.01, 0.01]	.896
Sleep regularity (z)				
(Intercept)	0.03 [-0.28, 0.34]	.850	-0.16 [-0.29, -0.03]	.014
Sleep regularity	0.12 [0.11, 0.13]	< .001	0.08 [0.08, 0.09]	< .001
Spring	0.01 [-0.01, 0.03]	.193	-0.01 [-0.03, 0.01]	.416
Summer	0.03 [0.00, 0.05]	.025	-0.04 [-0.06, -0.02]	< .001
Winter	-0.05 [-0.07, -0.03]	< .001	-0.03 [-0.04, -0.01]	.006
Sleep regularity^2	-0.01 [-0.01, 0.00]	.040	0.00 [-0.01, 0.00]	.632
Sleep regularity × Spring	0.01 [-0.01, 0.02]	.356	-0.02 [-0.04, -0.01]	< .001
Sleep regularity × Summer	-0.02 [-0.03, 0.00]	.008	-0.03 [-0.05, -0.02]	< .001

Table 21 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity × Winter	0.02 [0.01, 0.03]	.005	0.00 [-0.01, 0.01]	.703
Spring × Sleep regularity^2	0.01 [0.00, 0.01]	.128	0.00 [-0.01, 0.00]	.421
Summer × Sleep regularity^2	0.00 [-0.01, 0.01]	.899	-0.01 [-0.02, 0.00]	.026
Winter × Sleep regularity^2	0.01 [0.00, 0.02]	.012	0.01 [0.00, 0.01]	.085

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 22

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.19 [0.02, 0.36]	.031	0.23 [0.06, 0.39]	.006
Physical activity	0.03 [0.00, 0.05]	.037	0.02 [-0.01, 0.04]	.252
Europe	-0.18 [-0.24, -0.12]	< .001	-0.20 [-0.26, -0.14]	< .001
Africa	-0.19 [-0.26, -0.12]	< .001	-0.22 [-0.29, -0.15]	< .001
Asia	-0.44 [-0.52, -0.36]	< .001	-0.50 [-0.57, -0.43]	< .001
North America	-0.07 [-0.14, 0.00]	.047	-0.09 [-0.16, -0.03]	.005
South America	-0.36 [-0.43, -0.30]	< .001	-0.46 [-0.52, -0.40]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.105	0.04 [0.02, 0.05]	< .001
Physical activity × Europe	0.06 [0.03, 0.09]	< .001	0.04 [0.01, 0.07]	.008
Physical activity × Africa	0.06 [0.00, 0.11]	.039	0.01 [-0.04, 0.07]	.599
Physical activity × Asia	0.07 [-0.02, 0.16]	.135	0.06 [0.01, 0.11]	.023
Physical activity × North America	0.07 [0.02, 0.12]	.007	0.02 [-0.02, 0.07]	.317
Physical activity × South America	0.16 [0.13, 0.19]	< .001	0.06 [0.02, 0.09]	.001
Europe × Physical activity^2	-0.01 [-0.02, 0.00]	.020	-0.04 [-0.05, -0.02]	< .001
Africa × Physical activity^2	-0.04 [-0.09, 0.02]	.189	-0.02 [-0.05, 0.02]	.355
Asia × Physical activity^2	-0.06 [-0.14, 0.01]	.102	-0.04 [-0.09, 0.00]	.060
North America × Physical activity^2	0.00 [-0.02, 0.02]	.856	-0.02 [-0.06, 0.01]	.151
South America × Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.02, 0.02]	.797
Sleep efficiency (z)				
(Intercept)	0.29 [0.12, 0.46]	< .001	0.33 [0.16, 0.50]	< .001
Physical activity	-0.01 [-0.04, 0.01]	.336	0.00 [-0.03, 0.03]	.937
Europe	-0.15 [-0.21, -0.08]	< .001	-0.18 [-0.24, -0.11]	< .001
Africa	0.02 [-0.05, 0.09]	.624	-0.03 [-0.11, 0.04]	.337

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia	-0.20 [-0.29, -0.12]	< .001	-0.34 [-0.41, -0.26]	< .001
North America	0.03 [-0.04, 0.11]	.361	-0.09 [-0.16, -0.02]	.012
South America	-0.21 [-0.27, -0.14]	< .001	-0.30 [-0.37, -0.24]	< .001
Physical activity^2	0.00 [-0.01, 0.01]	.981	-0.01 [-0.03, 0.00]	.054
Physical activity × Europe	0.03 [0.00, 0.05]	.060	0.00 [-0.03, 0.03]	.946
Physical activity × Africa	0.05 [-0.01, 0.10]	.097	-0.01 [-0.06, 0.04]	.723
Physical activity × Asia	0.01 [-0.08, 0.10]	.866	0.07 [0.01, 0.12]	.013
Physical activity × North America	0.15 [0.10, 0.20]	< .001	0.06 [0.02, 0.11]	.010
Physical activity × South America	0.14 [0.11, 0.17]	< .001	0.07 [0.04, 0.11]	< .001
Europe × Physical activity^2	-0.01 [-0.02, 0.00]	.014	0.02 [0.00, 0.03]	.019
Africa × Physical activity^2	-0.06 [-0.11, -0.01]	.030	0.02 [-0.02, 0.05]	.358
Asia × Physical activity^2	-0.21 [-0.28, -0.13]	< .001	-0.04 [-0.09, 0.00]	.061
North America × Physical activity^2	-0.04 [-0.06, -0.02]	< .001	-0.02 [-0.05, 0.02]	.316
South America × Physical activity^2	-0.03 [-0.04, -0.02]	< .001	-0.03 [-0.06, -0.01]	.002
Sleep onset (z)				
(Intercept)	0.06 [-0.22, 0.34]	.675	0.07 [-0.19, 0.33]	.575
Physical activity	-0.10 [-0.11, -0.08]	< .001	-0.06 [-0.08, -0.04]	< .001
Europe	0.07 [0.02, 0.12]	.010	0.05 [0.00, 0.10]	.067
Africa	0.08 [0.02, 0.14]	.007	0.00 [-0.05, 0.06]	.882
Asia	0.23 [0.17, 0.30]	< .001	0.16 [0.10, 0.22]	< .001
North America	0.07 [0.01, 0.13]	.021	-0.04 [-0.10, 0.01]	.150
South America	0.04 [-0.01, 0.10]	.105	0.08 [0.02, 0.13]	.005
Physical activity^2	0.00 [0.00, 0.01]	.162	-0.05 [-0.06, -0.04]	< .001
Physical activity × Europe	-0.02 [-0.04, 0.00]	.084	0.00 [-0.03, 0.02]	.697

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Africa	0.07 [0.02, 0.11]	.002	0.02 [-0.02, 0.05]	.455
Physical activity × Asia	0.02 [-0.05, 0.09]	.573	0.01 [-0.03, 0.05]	.565
Physical activity × North America	0.11 [0.07, 0.15]	< .001	0.06 [0.03, 0.10]	< .001
Physical activity × South America	-0.06 [-0.09, -0.04]	< .001	-0.02 [-0.04, 0.01]	.250
Europe × Physical activity^2	0.01 [0.01, 0.02]	< .001	0.05 [0.04, 0.07]	< .001
Africa × Physical activity^2	-0.02 [-0.07, 0.02]	.249	0.04 [0.01, 0.07]	.002
Asia × Physical activity^2	-0.06 [-0.12, 0.00]	.041	0.02 [-0.01, 0.05]	.249
North America × Physical activity^2	-0.02 [-0.04, -0.01]	.006	0.02 [-0.01, 0.04]	.236
South America × Physical activity^2	0.00 [-0.01, 0.00]	.467	-0.02 [-0.03, 0.00]	.039
Sleep regularity (z)				
(Intercept)	0.00 [-0.15, 0.15]	.966	0.05 [-0.13, 0.23]	.561
Physical activity	0.32 [0.29, 0.35]	< .001	0.21 [0.18, 0.24]	< .001
Europe	0.27 [0.21, 0.34]	< .001	0.16 [0.09, 0.23]	< .001
Africa	0.43 [0.36, 0.51]	< .001	0.38 [0.30, 0.46]	< .001
Asia	0.46 [0.37, 0.55]	< .001	0.42 [0.34, 0.50]	< .001
North America	0.15 [0.07, 0.22]	< .001	0.03 [-0.04, 0.11]	.421
South America	0.05 [-0.02, 0.11]	.184	-0.05 [-0.12, 0.02]	.169
Physical activity^2	-0.05 [-0.06, -0.05]	< .001	-0.12 [-0.14, -0.11]	< .001
Physical activity × Europe	0.01 [-0.02, 0.04]	.391	-0.13 [-0.16, -0.10]	< .001
Physical activity × Africa	-0.34 [-0.40, -0.28]	< .001	0.10 [0.04, 0.16]	< .001
Physical activity × Asia	-0.36 [-0.45, -0.26]	< .001	0.03 [-0.03, 0.08]	.339
Physical activity × North America	0.02 [-0.04, 0.07]	.508	0.02 [-0.04, 0.07]	.551
Physical activity × South America	0.00 [-0.03, 0.03]	.998	0.05 [0.02, 0.09]	.006
Europe × Physical activity^2	-0.03 [-0.04, -0.02]	< .001	0.12 [0.10, 0.14]	< .001

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Physical activity^2	-0.26 [-0.32, -0.20]	< .001	-0.06 [-0.09, -0.02]	.005
Asia × Physical activity^2	-0.26 [-0.33, -0.18]	< .001	0.02 [-0.03, 0.07]	.439
North America × Physical activity^2	-0.03 [-0.05, -0.01]	.003	-0.02 [-0.05, 0.02]	.350
South America × Physical activity^2	0.00 [-0.01, 0.01]	.579	0.01 [-0.01, 0.03]	.399

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 23

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.28 [-0.01, 0.58]	.062	0.08 [-0.05, 0.20]	.224
Sleep duration	0.10 [0.09, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Europe	-0.15 [-0.20, -0.09]	< .001	-0.15 [-0.20, -0.10]	< .001
Africa	-0.12 [-0.18, -0.06]	< .001	-0.02 [-0.08, 0.04]	.491
Asia	-0.36 [-0.43, -0.30]	< .001	-0.42 [-0.48, -0.36]	< .001
North America	-0.29 [-0.35, -0.22]	< .001	-0.24 [-0.30, -0.19]	< .001
South America	-0.19 [-0.25, -0.14]	< .001	-0.25 [-0.31, -0.20]	< .001
Sleep duration^2	-0.04 [-0.04, -0.03]	< .001	-0.01 [-0.02, 0.00]	.002
Sleep duration × Europe	-0.11 [-0.12, -0.09]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep duration × Africa	-0.06 [-0.10, -0.03]	< .001	-0.06 [-0.09, -0.02]	< .001
Sleep duration × Asia	-0.09 [-0.13, -0.06]	< .001	0.01 [-0.03, 0.05]	.614
Sleep duration × North America	-0.04 [-0.07, -0.01]	.003	-0.02 [-0.05, 0.00]	.104
Sleep duration × South America	-0.08 [-0.10, -0.07]	< .001	-0.07 [-0.09, -0.06]	< .001
Europe × Sleep duration^2	0.02 [0.02, 0.03]	< .001	0.00 [0.00, 0.01]	.269
Africa × Sleep duration^2	0.02 [0.00, 0.04]	.088	-0.03 [-0.05, -0.01]	.004
Asia × Sleep duration^2	0.03 [0.00, 0.06]	.021	-0.05 [-0.08, -0.02]	< .001
North America × Sleep duration^2	0.02 [0.01, 0.04]	.009	0.00 [-0.01, 0.02]	.693
South America × Sleep duration^2	0.00 [-0.01, 0.01]	.452	-0.01 [-0.02, 0.00]	.047
Sleep efficiency (z)				
(Intercept)	0.31 [0.02, 0.61]	.038	0.11 [-0.02, 0.23]	.096
Sleep efficiency	-0.02 [-0.04, 0.00]	.019	0.01 [-0.01, 0.02]	.522
Europe	-0.20 [-0.26, -0.15]	< .001	-0.17 [-0.22, -0.12]	< .001
Africa	-0.17 [-0.23, -0.11]	< .001	-0.12 [-0.18, -0.07]	< .001

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia	-0.42 [-0.49, -0.36]	< .001	-0.47 [-0.53, -0.42]	< .001
North America	-0.31 [-0.37, -0.25]	< .001	-0.28 [-0.33, -0.23]	< .001
South America	-0.28 [-0.34, -0.22]	< .001	-0.33 [-0.38, -0.28]	< .001
Sleep efficiency^2	-0.03 [-0.03, -0.02]	< .001	-0.02 [-0.03, -0.01]	< .001
Sleep efficiency × Europe	0.04 [0.02, 0.06]	< .001	0.00 [-0.01, 0.02]	.587
Sleep efficiency × Africa	0.02 [-0.01, 0.05]	.178	-0.02 [-0.05, 0.01]	.144
Sleep efficiency × Asia	0.04 [0.00, 0.07]	.038	0.00 [-0.03, 0.04]	.843
Sleep efficiency × North America	0.02 [-0.01, 0.05]	.192	0.00 [-0.03, 0.03]	.929
Sleep efficiency × South America	-0.01 [-0.03, 0.01]	.432	-0.01 [-0.03, 0.01]	.448
Europe × Sleep efficiency^2	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.01]	.253
Africa × Sleep efficiency^2	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.03]	.031
Asia × Sleep efficiency^2	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.187
North America × Sleep efficiency^2	0.02 [0.01, 0.03]	.001	0.01 [0.00, 0.03]	.014
South America × Sleep efficiency^2	0.02 [0.01, 0.02]	.002	0.01 [0.00, 0.02]	.003
Sleep onset (z)				
(Intercept)	0.14 [-0.16, 0.44]	.350	0.01 [-0.12, 0.13]	.925
Sleep onset	-0.16 [-0.18, -0.15]	< .001	-0.10 [-0.11, -0.08]	< .001
Europe	-0.02 [-0.08, 0.04]	.523	-0.06 [-0.12, -0.01]	.020
Africa	0.07 [-0.02, 0.15]	.142	-0.04 [-0.12, 0.05]	.382
Asia	-0.20 [-0.28, -0.12]	< .001	-0.35 [-0.43, -0.28]	< .001
North America	-0.07 [-0.15, 0.00]	.060	-0.17 [-0.24, -0.10]	< .001
South America	0.01 [-0.06, 0.07]	.861	-0.18 [-0.24, -0.13]	< .001
Sleep onset^2	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.01]	.012
Sleep onset × Europe	0.14 [0.12, 0.15]	< .001	0.09 [0.07, 0.11]	< .001

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset × Africa	0.16 [0.03, 0.30]	.019	-0.01 [-0.15, 0.13]	.876
Sleep onset × Asia	0.14 [0.03, 0.25]	.015	0.02 [-0.10, 0.13]	.765
Sleep onset × North America	0.21 [0.15, 0.27]	< .001	0.08 [0.03, 0.14]	.004
Sleep onset × South America	0.21 [0.19, 0.23]	< .001	0.11 [0.09, 0.13]	< .001
Europe × Sleep onset^2	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.01, 0.00]	.114
Africa × Sleep onset^2	-0.02 [-0.10, 0.05]	.517	-0.05 [-0.13, 0.02]	.169
Asia × Sleep onset^2	-0.03 [-0.10, 0.04]	.377	-0.07 [-0.15, 0.00]	.048
North America × Sleep onset^2	0.01 [-0.02, 0.04]	.511	0.00 [-0.03, 0.03]	.983
South America × Sleep onset^2	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, 0.00]	.014
Sleep regularity (z)				
(Intercept)	0.23 [-0.07, 0.53]	.135	0.00 [-0.13, 0.12]	.948
Sleep regularity	0.15 [0.14, 0.17]	< .001	0.07 [0.05, 0.08]	< .001
Europe	-0.24 [-0.29, -0.18]	< .001	-0.18 [-0.23, -0.13]	< .001
Africa	-0.20 [-0.26, -0.14]	< .001	-0.13 [-0.18, -0.07]	< .001
Asia	-0.46 [-0.52, -0.40]	< .001	-0.49 [-0.55, -0.43]	< .001
North America	-0.32 [-0.38, -0.26]	< .001	-0.24 [-0.29, -0.18]	< .001
South America	-0.27 [-0.33, -0.22]	< .001	-0.31 [-0.36, -0.26]	< .001
Sleep regularity^2	0.00 [0.00, 0.01]	.139	0.00 [0.00, 0.01]	.182
Sleep regularity × Europe	-0.03 [-0.05, -0.02]	< .001	-0.01 [-0.02, 0.01]	.458
Sleep regularity × Africa	-0.09 [-0.12, -0.07]	< .001	0.03 [0.01, 0.06]	.008
Sleep regularity × Asia	-0.10 [-0.13, -0.07]	< .001	0.06 [0.04, 0.09]	< .001
Sleep regularity × North America	-0.05 [-0.08, -0.02]	< .001	0.03 [0.00, 0.05]	.048
Sleep regularity × South America	-0.01 [-0.03, 0.00]	.077	0.01 [0.00, 0.03]	.156
Europe × Sleep regularity^2	-0.01 [-0.02, 0.00]	.016	-0.01 [-0.02, 0.00]	.003

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Sleep regularity^2	-0.03 [-0.04, -0.01]	< .001	0.00 [-0.02, 0.01]	.797
Asia × Sleep regularity^2	-0.02 [-0.04, 0.00]	.044	-0.01 [-0.03, 0.01]	.349
North America × Sleep regularity^2	-0.01 [-0.02, 0.00]	.178	-0.02 [-0.03, 0.00]	.030
South America × Sleep regularity^2	0.00 [-0.01, 0.01]	.804	0.00 [-0.01, 0.01]	.802

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 24

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.38 [0.22, 0.54]	< .001	0.34 [0.19, 0.49]	< .001
Physical activity	0.20 [0.15, 0.24]	< .001	0.13 [0.10, 0.16]	< .001
Daylight hours	-0.03 [-0.03, -0.03]	< .001	-0.03 [-0.03, -0.02]	< .001
Physical activity^2	-0.03 [-0.04, -0.01]	.002	0.01 [-0.01, 0.03]	.238
Physical activity × Daylight hours	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.009	0.00 [0.00, 0.00]	.698
Sleep efficiency (z)				
(Intercept)	0.27 [0.10, 0.43]	.001	0.24 [0.07, 0.41]	.006
Physical activity	0.13 [0.09, 0.18]	< .001	0.04 [0.01, 0.07]	.006
Daylight hours	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	.004
Physical activity^2	-0.03 [-0.05, -0.02]	< .001	-0.01 [-0.03, 0.01]	.369
Physical activity × Daylight hours	-0.01 [-0.01, 0.00]	< .001	0.00 [-0.01, 0.00]	.015
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.019	0.00 [0.00, 0.00]	.425
Sleep onset (z)				
(Intercept)	-0.06 [-0.33, 0.22]	.693	-0.02 [-0.27, 0.24]	.904
Physical activity	-0.20 [-0.23, -0.17]	< .001	-0.14 [-0.16, -0.12]	< .001
Daylight hours	0.01 [0.01, 0.02]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity^2	-0.01 [-0.03, 0.00]	.044	-0.04 [-0.06, -0.03]	< .001
Physical activity × Daylight hours	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.012	0.00 [0.00, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	0.04 [-0.11, 0.20]	.571	0.05 [-0.14, 0.23]	.629
Physical activity	0.32 [0.27, 0.37]	< .001	0.19 [0.16, 0.22]	< .001

Table 24 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.01 [0.00, 0.01]	$< .001$	0.01 [0.00, 0.01]	.003
Physical activity^2	-0.04 [-0.06, -0.02]	$< .001$	-0.06 [-0.08, -0.04]	$< .001$
Physical activity $\times$ Daylight hours	0.00 [0.00, 0.00]	.933	-0.01 [-0.01, 0.00]	$< .001$
Daylight hours $\times$ Physical activity^2	0.00 [0.00, 0.00]	.047	0.00 [0.00, 0.01]	$< .001$

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 25

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.07 [-0.37, 0.24]	.658	-0.10 [-0.23, 0.03]	.116
Sleep duration	0.12 [0.10, 0.14]	< .001	0.09 [0.07, 0.11]	< .001
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.094
Sleep duration^2	-0.01 [-0.02, 0.00]	.160	0.02 [0.00, 0.03]	.011
Sleep duration × Daylight hours	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Daylight hours × Sleep duration^2	0.00 [0.00, 0.00]	.088	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.08 [-0.38, 0.23]	.619	-0.06 [-0.19, 0.06]	.320
Sleep efficiency	0.04 [0.01, 0.06]	.003	0.02 [-0.01, 0.04]	.171
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.586
Sleep efficiency^2	-0.02 [-0.03, 0.00]	.006	-0.03 [-0.04, -0.01]	< .001
Sleep efficiency × Daylight hours	0.00 [0.00, 0.00]	.008	0.00 [0.00, 0.00]	.379
Daylight hours × Sleep efficiency^2	0.00 [0.00, 0.00]	.086	0.00 [0.00, 0.00]	.002
Sleep onset (z)				
(Intercept)	-0.08 [-0.38, 0.23]	.619	-0.09 [-0.22, 0.04]	.181
Sleep onset	-0.05 [-0.08, -0.02]	< .001	-0.04 [-0.06, -0.01]	.008
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.662
Sleep onset^2	-0.01 [-0.02, 0.01]	.466	0.00 [-0.01, 0.02]	.556
Sleep onset × Daylight hours	0.00 [0.00, 0.00]	.078	0.00 [0.00, 0.00]	.105
Daylight hours × Sleep onset^2	0.00 [0.00, 0.00]	.775	0.00 [0.00, 0.00]	.806
Sleep regularity (z)				
(Intercept)	-0.21 [-0.51, 0.10]	.191	-0.19 [-0.32, -0.05]	.006
Sleep regularity	0.17 [0.15, 0.20]	< .001	0.15 [0.13, 0.18]	< .001

Table 25 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.02 [0.02, 0.02]	$< .001$	0.00 [0.00, 0.00]	.649
Sleep regularity^2	0.01 [-0.01, 0.02]	.243	0.01 [0.00, 0.03]	.156
Sleep regularity $\times$ Daylight hours	0.00 [-0.01, 0.00]	$< .001$	-0.01 [-0.01, 0.00]	$< .001$
Daylight hours $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.165	0.00 [0.00, 0.00]	.092

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 26

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.56 [0.33, 0.80]	< .001	0.46 [0.22, 0.70]	< .001
Physical activity	0.11 [0.09, 0.13]	< .001	0.08 [0.06, 0.09]	< .001
Wrist	-0.69 [-0.95, -0.43]	< .001	-0.56 [-0.82, -0.29]	< .001
Physical activity^2	-0.03 [-0.05, -0.02]	< .001	0.00 [-0.01, 0.01]	.594
Physical activity × Wrist	-0.01 [-0.03, 0.01]	.332	-0.01 [-0.03, 0.00]	.081
Wrist × Physical activity^2	0.03 [0.02, 0.05]	< .001	0.01 [-0.01, 0.02]	.312
Sleep efficiency (z)				
(Intercept)	-0.35 [-0.55, -0.15]	< .001	-0.47 [-0.67, -0.27]	< .001
Physical activity	0.12 [0.10, 0.13]	< .001	0.09 [0.07, 0.10]	< .001
Wrist	0.68 [0.46, 0.90]	< .001	0.79 [0.57, 1.01]	< .001
Physical activity^2	-0.12 [-0.14, -0.10]	< .001	-0.04 [-0.06, -0.03]	< .001
Physical activity × Wrist	-0.14 [-0.16, -0.12]	< .001	-0.11 [-0.12, -0.09]	< .001
Wrist × Physical activity^2	0.12 [0.11, 0.14]	< .001	0.04 [0.03, 0.05]	< .001
Sleep onset (z)				
(Intercept)	-0.85 [-1.21, -0.49]	< .001	-0.81 [-1.18, -0.45]	< .001
Physical activity	-0.02 [-0.03, -0.01]	.001	-0.03 [-0.04, -0.02]	< .001
Wrist	1.22 [0.81, 1.62]	< .001	1.15 [0.75, 1.56]	< .001
Physical activity^2	-0.03 [-0.04, -0.02]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Wrist	-0.15 [-0.16, -0.13]	< .001	-0.08 [-0.09, -0.06]	< .001
Wrist × Physical activity^2	0.05 [0.03, 0.06]	< .001	0.01 [0.00, 0.02]	.057
Sleep regularity (z)				
(Intercept)	-0.01 [-0.32, 0.29]	.936	-0.37 [-0.69, -0.06]	.020
Physical activity	0.19 [0.17, 0.21]	< .001	0.30 [0.28, 0.31]	< .001

Table 26 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist	0.20 [-0.14, 0.54]	.243	0.62 [0.26, 0.97]	$< .001$
Physical activity^2	-0.25 [-0.27, -0.23]	$< .001$	-0.14 [-0.15, -0.13]	$< .001$
Physical activity $\times$ Wrist	0.13 [0.11, 0.15]	$< .001$	-0.25 [-0.27, -0.24]	$< .001$
Wrist $\times$ Physical activity^2	0.19 [0.17, 0.21]	$< .001$	0.13 [0.11, 0.14]	$< .001$

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 27

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.98 [-1.38, -0.59]	< .001	-0.40 [-0.63, -0.17]	< .001
Sleep duration	0.06 [0.05, 0.07]	< .001	0.07 [0.06, 0.08]	< .001
Wrist	1.40 [0.96, 1.84]	< .001	0.41 [0.15, 0.66]	.002
Sleep duration^2	-0.02 [-0.03, -0.01]	< .001	-0.03 [-0.03, -0.02]	< .001
Sleep duration × Wrist	-0.06 [-0.07, -0.05]	< .001	-0.05 [-0.06, -0.04]	< .001
Wrist × Sleep duration^2	0.00 [-0.01, 0.01]	.699	0.02 [0.01, 0.03]	< .001
Sleep efficiency (z)				
(Intercept)	-0.95 [-1.35, -0.56]	< .001	-0.37 [-0.60, -0.14]	.002
Sleep efficiency	0.02 [0.01, 0.03]	< .001	0.01 [0.01, 0.02]	.001
Wrist	1.35 [0.90, 1.79]	< .001	0.37 [0.11, 0.63]	.005
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Wrist	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, 0.00]	.010
Wrist × Sleep efficiency^2	0.00 [0.00, 0.01]	.039	0.01 [0.00, 0.01]	.015
Sleep onset (z)				
(Intercept)	-0.97 [-1.37, -0.58]	< .001	-0.41 [-0.64, -0.18]	< .001
Sleep onset	-0.02 [-0.05, 0.01]	.188	-0.06 [-0.09, -0.03]	< .001
Wrist	1.38 [0.93, 1.82]	< .001	0.40 [0.15, 0.66]	.002
Sleep onset^2	-0.01 [-0.02, 0.01]	.298	-0.02 [-0.04, -0.01]	.003
Sleep onset × Wrist	-0.01 [-0.04, 0.02]	.418	0.05 [0.02, 0.07]	.001
Wrist × Sleep onset^2	0.01 [-0.01, 0.02]	.381	0.03 [0.01, 0.04]	.001
Sleep regularity (z)				
(Intercept)	-1.03 [-1.41, -0.65]	< .001	-0.53 [-0.75, -0.31]	< .001
Sleep regularity	0.11 [0.11, 0.12]	< .001	0.13 [0.13, 0.14]	< .001

Table 27 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist	1.34 [0.91, 1.76]	$< .001$	0.44 [0.20, 0.69]	$< .001$
Sleep regularity^2	-0.01 [-0.01, 0.00]	.019	0.00 [0.00, 0.01]	.474
Sleep regularity $\times$ Wrist	0.01 [0.00, 0.02]	.020	-0.10 [-0.11, -0.09]	$< .001$
Wrist $\times$ Sleep regularity^2	0.01 [0.00, 0.01]	.053	0.00 [0.00, 0.01]	.579

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 28

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.09 [-0.07, 0.25]	.261	0.11 [-0.03, 0.26]	.130
Physical activity	0.07 [0.05, 0.09]	< .001	0.06 [0.05, 0.08]	< .001
Most active hour	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Physical activity^2	0.00 [-0.01, 0.01]	.832	-0.01 [-0.02, 0.00]	.046
Physical activity × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.818
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	.264	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.34]	.025	0.20 [0.04, 0.37]	.016
Physical activity	-0.04 [-0.06, -0.02]	< .001	-0.02 [-0.04, -0.01]	.009
Most active hour	0.00 [0.00, 0.00]	.447	0.00 [0.00, 0.00]	< .001
Physical activity^2	0.00 [-0.01, 0.01]	.852	-0.02 [-0.03, 0.00]	.010
Physical activity × Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.014
Sleep onset (z)				
(Intercept)	-0.06 [-0.34, 0.21]	.647	-0.06 [-0.31, 0.20]	.651
Physical activity	-0.20 [-0.21, -0.18]	< .001	-0.11 [-0.12, -0.09]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity^2	0.01 [0.00, 0.02]	.007	0.00 [-0.01, 0.01]	.477
Physical activity × Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	.009	0.00 [0.00, 0.00]	.006
Sleep regularity (z)				
(Intercept)	0.23 [0.08, 0.37]	.002	0.22 [0.04, 0.39]	.014
Physical activity	0.27 [0.24, 0.29]	< .001	0.06 [0.04, 0.08]	< .001

Table 28 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	-0.01 [-0.01, 0.00]	$< .001$	-0.01 [-0.01, 0.00]	$< .001$
Physical activity^2	-0.06 [-0.07, -0.05]	$< .001$	-0.04 [-0.05, -0.02]	$< .001$
Physical activity $\times$ Most active hour	0.00 [0.00, 0.01]	$< .001$	0.00 [0.00, 0.00]	$< .001$
Most active hour $\times$ Physical activity^2	0.00 [0.00, 0.00]	.528	0.00 [0.00, 0.00]	.051

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor.

Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 29

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [-0.27, 0.32]	.871	-0.09 [-0.22, 0.03]	.135
Sleep duration	0.00 [-0.01, 0.01]	.641	0.06 [0.05, 0.07]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep duration^2	0.00 [-0.01, 0.01]	.726	0.00 [-0.01, 0.01]	.942
Sleep duration × Most active hour	0.00 [0.00, 0.00]	.005	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.019
Sleep efficiency (z)				
(Intercept)	0.03 [-0.27, 0.32]	.866	-0.10 [-0.22, 0.03]	.120
Sleep efficiency	-0.01 [-0.02, 0.00]	.092	-0.04 [-0.05, -0.03]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency^2	0.00 [-0.01, 0.01]	.718	-0.01 [-0.01, 0.00]	.110
Sleep efficiency × Most active hour	0.00 [0.00, 0.00]	.010	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep efficiency^2	0.00 [0.00, 0.00]	.022	0.00 [0.00, 0.00]	.654
Sleep onset (z)				
(Intercept)	0.02 [-0.28, 0.33]	.877	-0.12 [-0.24, 0.01]	.068
Sleep onset	-0.04 [-0.05, -0.03]	< .001	-0.06 [-0.08, -0.05]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep onset^2	0.00 [0.00, 0.01]	.161	0.02 [0.01, 0.02]	< .001
Sleep onset × Most active hour	0.00 [0.00, 0.00]	.009	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep onset^2	0.00 [0.00, 0.00]	.004	0.00 [0.00, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	-0.06 [-0.36, 0.24]	.703	-0.19 [-0.32, -0.06]	.005
Sleep regularity	0.06 [0.05, 0.07]	< .001	0.03 [0.01, 0.04]	< .001

Table 29 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.01 [0.01, 0.01]	$< .001$	0.00 [0.00, 0.00]	.341
Sleep regularity^2	0.00 [-0.01, 0.01]	.630	0.00 [-0.01, 0.01]	.771
Sleep regularity $\times$ Most active hour	0.00 [0.00, 0.01]	$< .001$	0.00 [0.00, 0.00]	$< .001$
Most active hour $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.467	0.00 [0.00, 0.00]	.899

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 30

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.08 [-0.09, 0.24]	.365	0.08 [-0.07, 0.23]	.316
Physical activity	0.11 [0.09, 0.12]	< .001	0.07 [0.06, 0.08]	< .001
Non-white	-0.14 [-0.17, -0.11]	< .001	-0.16 [-0.19, -0.14]	< .001
Unclear	-0.11 [-0.14, -0.08]	< .001	-0.10 [-0.13, -0.07]	< .001
Physical activity^2	0.00 [0.00, 0.00]	.993	0.00 [0.00, 0.01]	.118
Physical activity × Non-white	0.06 [0.03, 0.08]	< .001	0.03 [0.01, 0.05]	.002
Physical activity × Unclear	-0.02 [-0.04, 0.00]	.045	-0.02 [-0.04, -0.01]	.002
Non-white × Physical activity^2	0.00 [-0.01, 0.01]	.512	0.01 [-0.01, 0.02]	.374
Unclear × Physical activity^2	0.00 [-0.01, 0.00]	.162	0.01 [0.00, 0.02]	.060
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.35]	.019	0.18 [0.02, 0.34]	.031
Physical activity	0.04 [0.03, 0.05]	< .001	0.01 [0.00, 0.01]	.246
Non-white	-0.07 [-0.10, -0.04]	< .001	-0.11 [-0.14, -0.08]	< .001
Unclear	-0.01 [-0.04, 0.02]	.604	0.00 [-0.03, 0.03]	.938
Physical activity^2	-0.01 [-0.01, -0.01]	< .001	0.00 [-0.01, 0.00]	.408
Physical activity × Non-white	0.07 [0.05, 0.10]	< .001	0.06 [0.04, 0.08]	< .001
Physical activity × Unclear	0.00 [-0.02, 0.02]	.802	-0.02 [-0.03, 0.00]	.025
Non-white × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.03, 0.00]	.149
Unclear × Physical activity^2	0.00 [-0.01, 0.01]	.871	0.00 [0.00, 0.01]	.429
Sleep onset (z)				
(Intercept)	0.09 [-0.19, 0.37]	.520	0.09 [-0.17, 0.34]	.505
Physical activity	-0.13 [-0.14, -0.12]	< .001	-0.08 [-0.09, -0.07]	< .001
Non-white	0.07 [0.05, 0.09]	< .001	0.06 [0.03, 0.08]	< .001

Table 30 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear	0.04 [0.02, 0.07]	< .001	0.04 [0.02, 0.07]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.078	-0.01 [-0.01, 0.00]	.006
Physical activity × Non-white	0.03 [0.01, 0.04]	.007	0.00 [-0.02, 0.02]	.999
Physical activity × Unclear	0.04 [0.02, 0.05]	< .001	-0.01 [-0.02, 0.00]	.276
Non-white × Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.02, 0.00]	.205
Unclear × Physical activity^2	0.00 [0.00, 0.01]	.746	0.00 [-0.01, 0.00]	.262
Sleep regularity (z)				
(Intercept)	0.21 [0.05, 0.36]	.008	0.17 [-0.01, 0.35]	.062
Physical activity	0.33 [0.32, 0.35]	< .001	0.11 [0.10, 0.12]	< .001
Non-white	0.00 [-0.03, 0.03]	.851	0.05 [0.02, 0.09]	.001
Unclear	-0.12 [-0.15, -0.09]	< .001	-0.07 [-0.10, -0.03]	< .001
Physical activity^2	-0.06 [-0.06, -0.06]	< .001	-0.01 [-0.01, 0.00]	.082
Physical activity × Non-white	-0.09 [-0.11, -0.06]	< .001	0.08 [0.06, 0.11]	< .001
Physical activity × Unclear	-0.02 [-0.04, 0.00]	.091	-0.04 [-0.06, -0.03]	< .001
Non-white × Physical activity^2	0.02 [0.01, 0.03]	< .001	-0.08 [-0.09, -0.06]	< .001
Unclear × Physical activity^2	0.00 [-0.01, 0.01]	.770	-0.03 [-0.03, -0.02]	< .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 31

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.13 [-0.18, 0.43]	.414	-0.05 [-0.18, 0.07]	.381
Sleep duration	0.02 [0.01, 0.02]	< .001	0.04 [0.03, 0.04]	< .001
Non-white	-0.03 [-0.05, 0.00]	.025	-0.03 [-0.05, -0.01]	.012
Unclear	0.03 [0.01, 0.06]	.016	-0.03 [-0.05, -0.01]	.013
Sleep duration^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.095
Sleep duration × Non-white	-0.01 [-0.02, 0.00]	.150	-0.04 [-0.05, -0.02]	< .001
Sleep duration × Unclear	-0.01 [-0.02, 0.00]	.002	-0.02 [-0.03, -0.01]	< .001
Non-white × Sleep duration^2	-0.01 [-0.02, 0.00]	.071	-0.01 [-0.02, -0.01]	< .001
Unclear × Sleep duration^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.12 [-0.18, 0.42]	.427	-0.04 [-0.16, 0.08]	.506
Sleep efficiency	0.01 [0.00, 0.01]	.017	0.01 [0.01, 0.02]	< .001
Non-white	-0.05 [-0.07, -0.02]	< .001	-0.06 [-0.08, -0.04]	< .001
Unclear	0.01 [-0.01, 0.04]	.266	-0.06 [-0.08, -0.03]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Non-white	-0.02 [-0.03, 0.00]	.045	-0.02 [-0.03, 0.00]	.030
Sleep efficiency × Unclear	0.00 [-0.01, 0.01]	.529	-0.01 [-0.02, 0.00]	.015
Non-white × Sleep efficiency^2	0.00 [0.00, 0.01]	.270	0.00 [0.00, 0.01]	.312
Unclear × Sleep efficiency^2	0.00 [0.00, 0.01]	.490	0.00 [0.00, 0.01]	.045
Sleep onset (z)				
(Intercept)	0.13 [-0.17, 0.43]	.399	-0.05 [-0.17, 0.07]	.413
Sleep onset	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Non-white	-0.03 [-0.06, 0.00]	.020	-0.04 [-0.06, -0.01]	.002

Table 31 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear	-0.01 [-0.03, 0.02]	.683	-0.05 [-0.08, -0.03]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.009	0.01 [0.00, 0.01]	< .001
Sleep onset × Non-white	0.05 [0.04, 0.07]	< .001	0.03 [0.01, 0.04]	< .001
Sleep onset × Unclear	-0.03 [-0.04, -0.01]	< .001	-0.02 [-0.03, -0.01]	.004
Non-white × Sleep onset^2	0.01 [0.00, 0.02]	.011	0.00 [-0.01, 0.01]	.446
Unclear × Sleep onset^2	0.00 [0.00, 0.01]	.345	-0.01 [-0.01, 0.00]	.017
Sleep regularity (z)				
(Intercept)	0.01 [-0.29, 0.32]	.934	-0.15 [-0.28, -0.02]	.022
Sleep regularity	0.13 [0.13, 0.14]	< .001	0.08 [0.07, 0.08]	< .001
Non-white	-0.03 [-0.06, -0.01]	.005	-0.05 [-0.07, -0.02]	< .001
Unclear	0.04 [0.01, 0.06]	.009	-0.05 [-0.07, -0.02]	< .001
Sleep regularity^2	0.00 [-0.01, 0.00]	.223	0.00 [-0.01, 0.00]	.199
Sleep regularity × Non-white	-0.04 [-0.06, -0.03]	< .001	0.01 [0.00, 0.03]	.130
Sleep regularity × Unclear	-0.01 [-0.02, 0.00]	.079	-0.02 [-0.03, -0.01]	.001
Non-white × Sleep regularity^2	-0.01 [-0.02, 0.00]	.035	-0.01 [-0.01, 0.00]	.211
Unclear × Sleep regularity^2	0.00 [0.00, 0.01]	.161	0.00 [0.00, 0.01]	.365

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

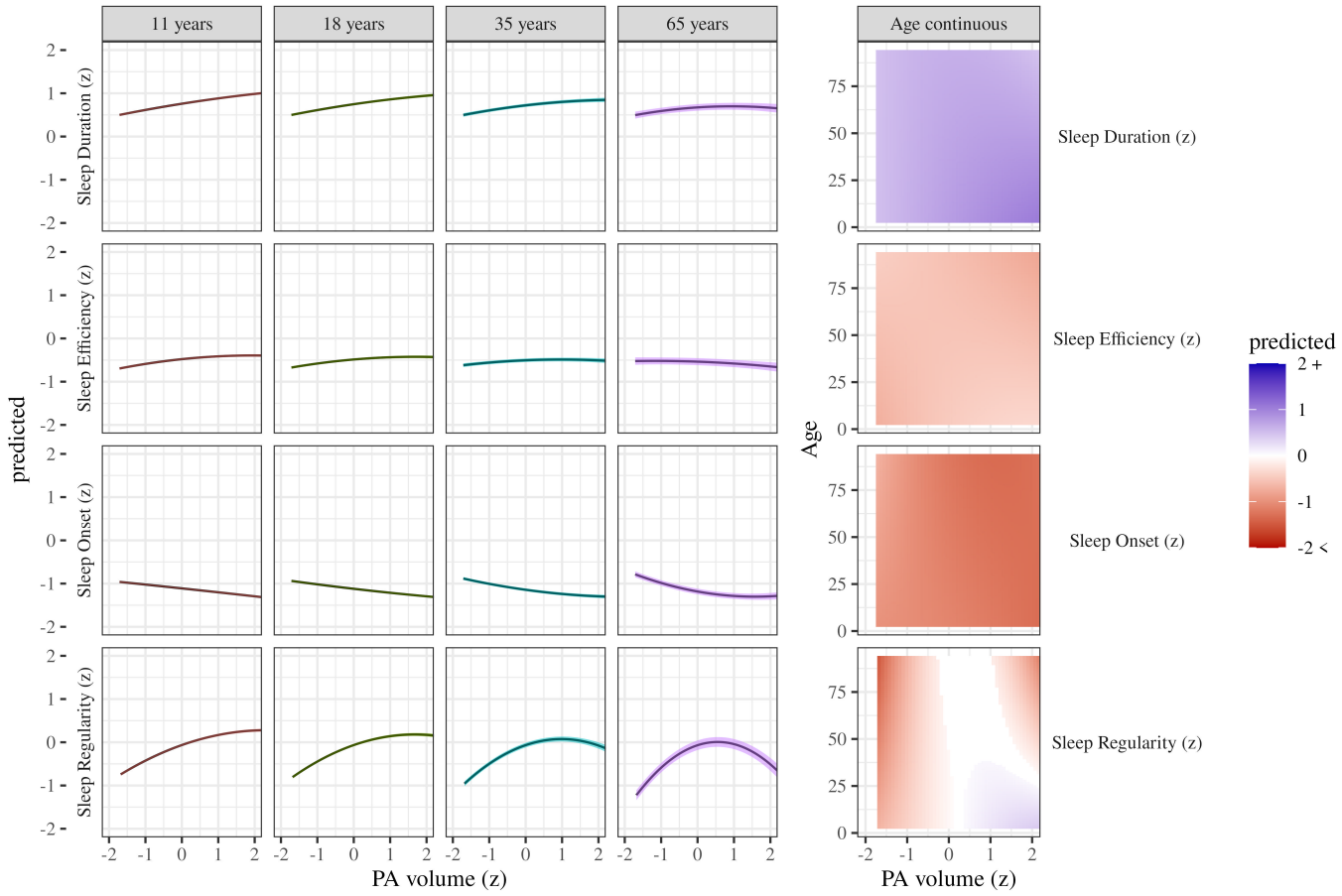

Figure 1. Sleep metrics on Physical activity volume

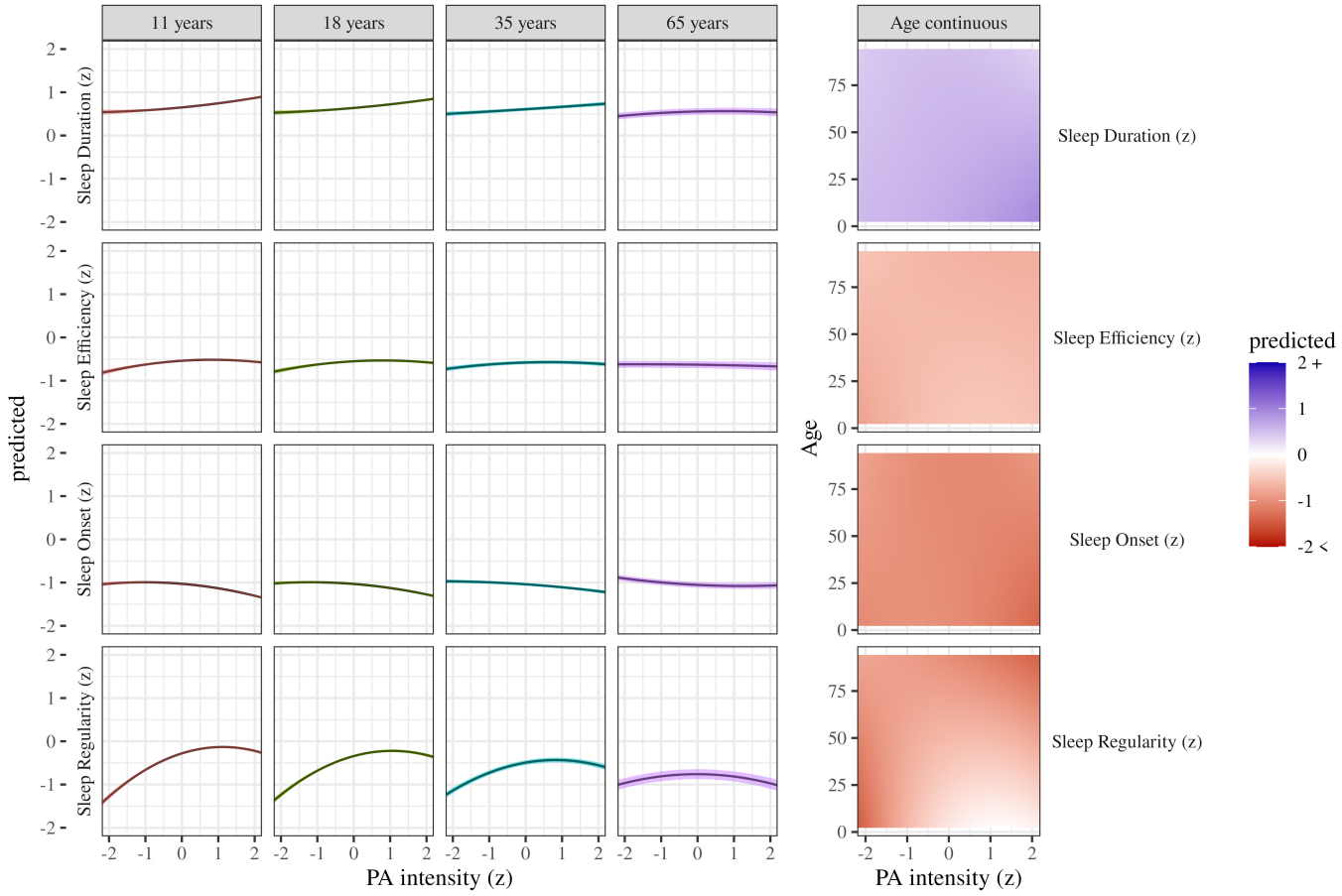

Figure 2. Sleep metrics on Physical activity intensity

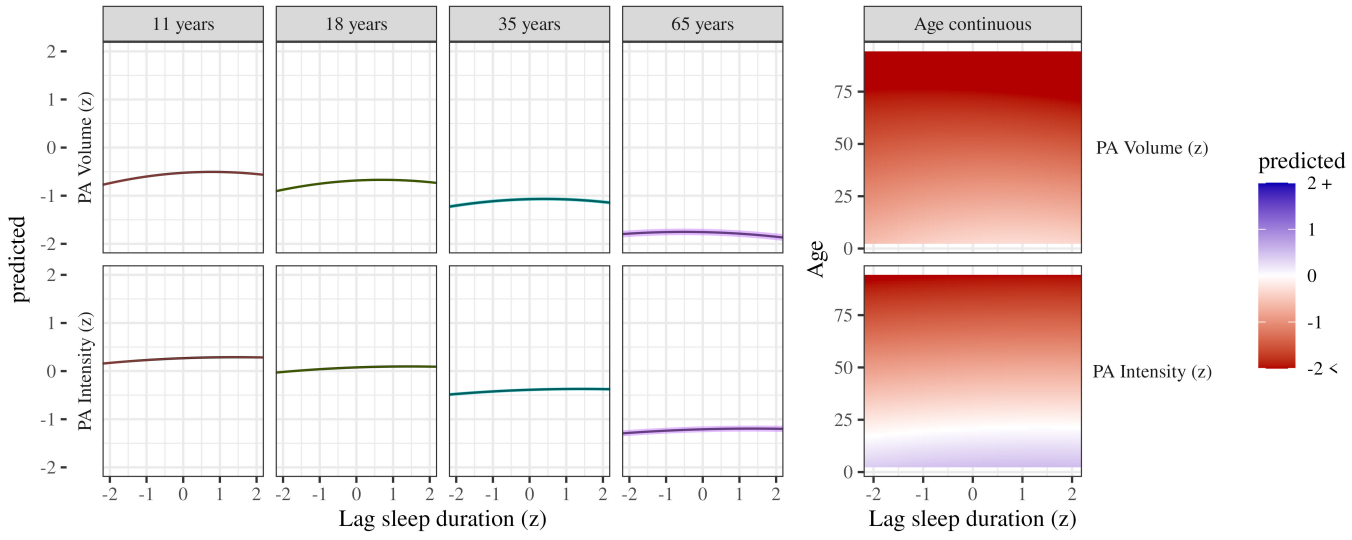

Figure 3. Physical activity by sleep duration

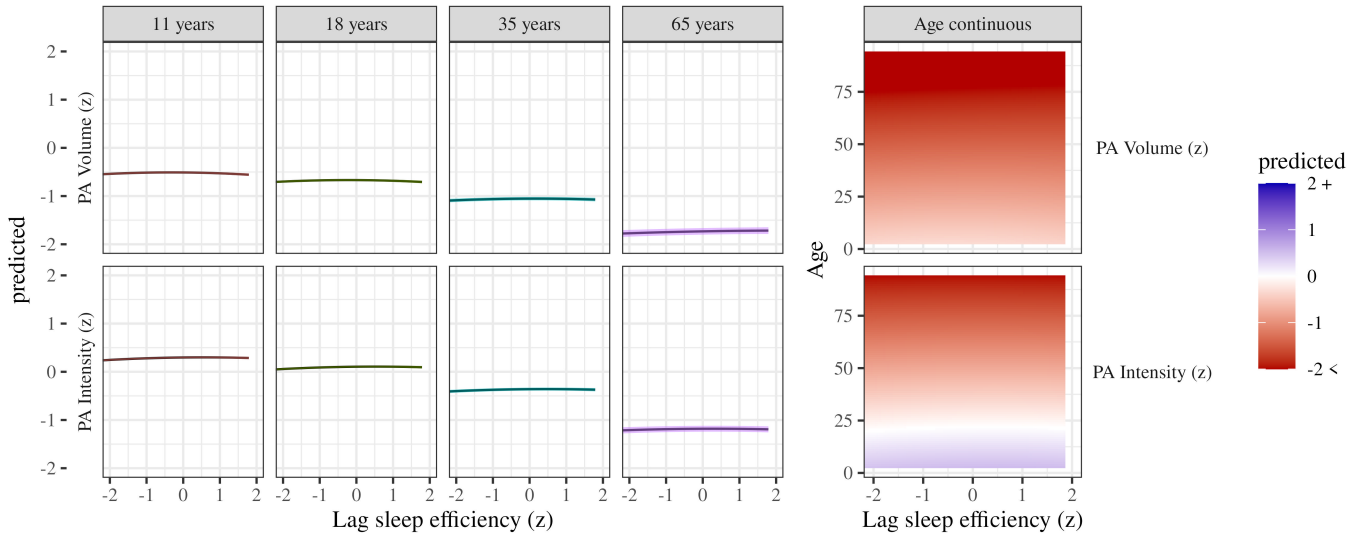

Figure 4. Physical activity by sleep efficiency

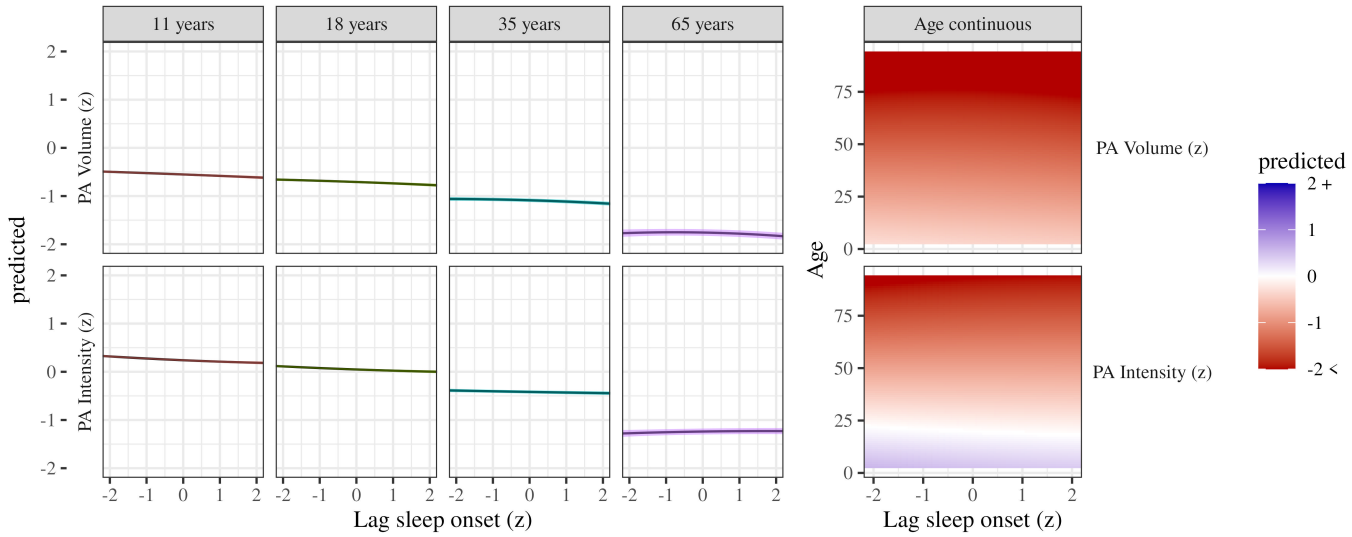

Figure 5. Physical activity by sleep onset

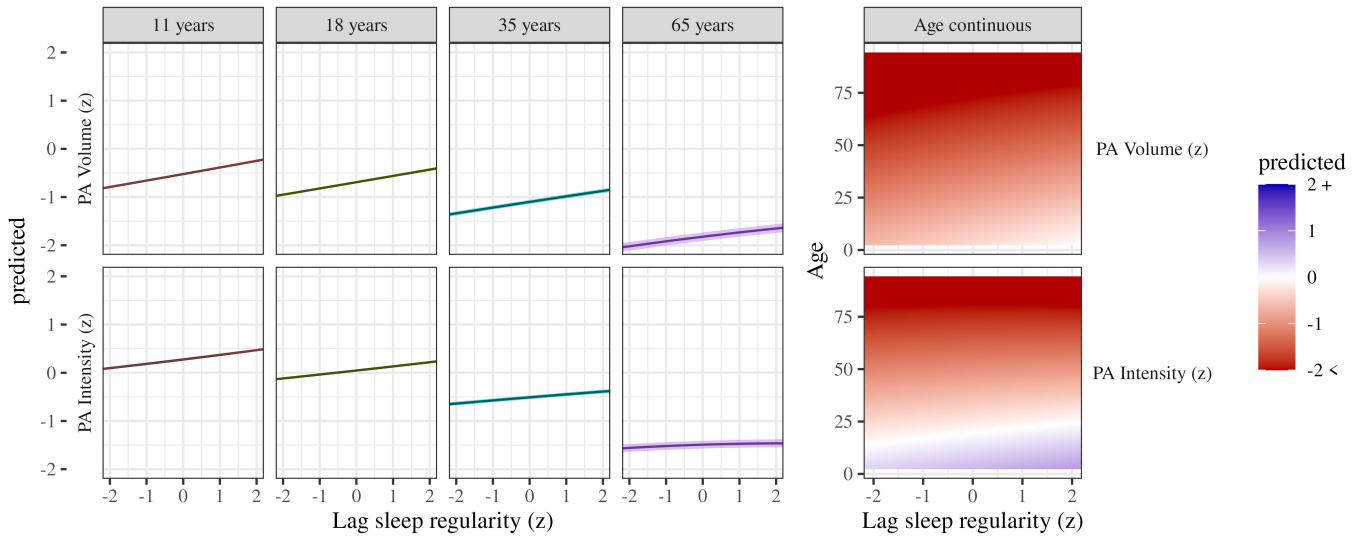

Figure 6. Physical activity by sleep regularity

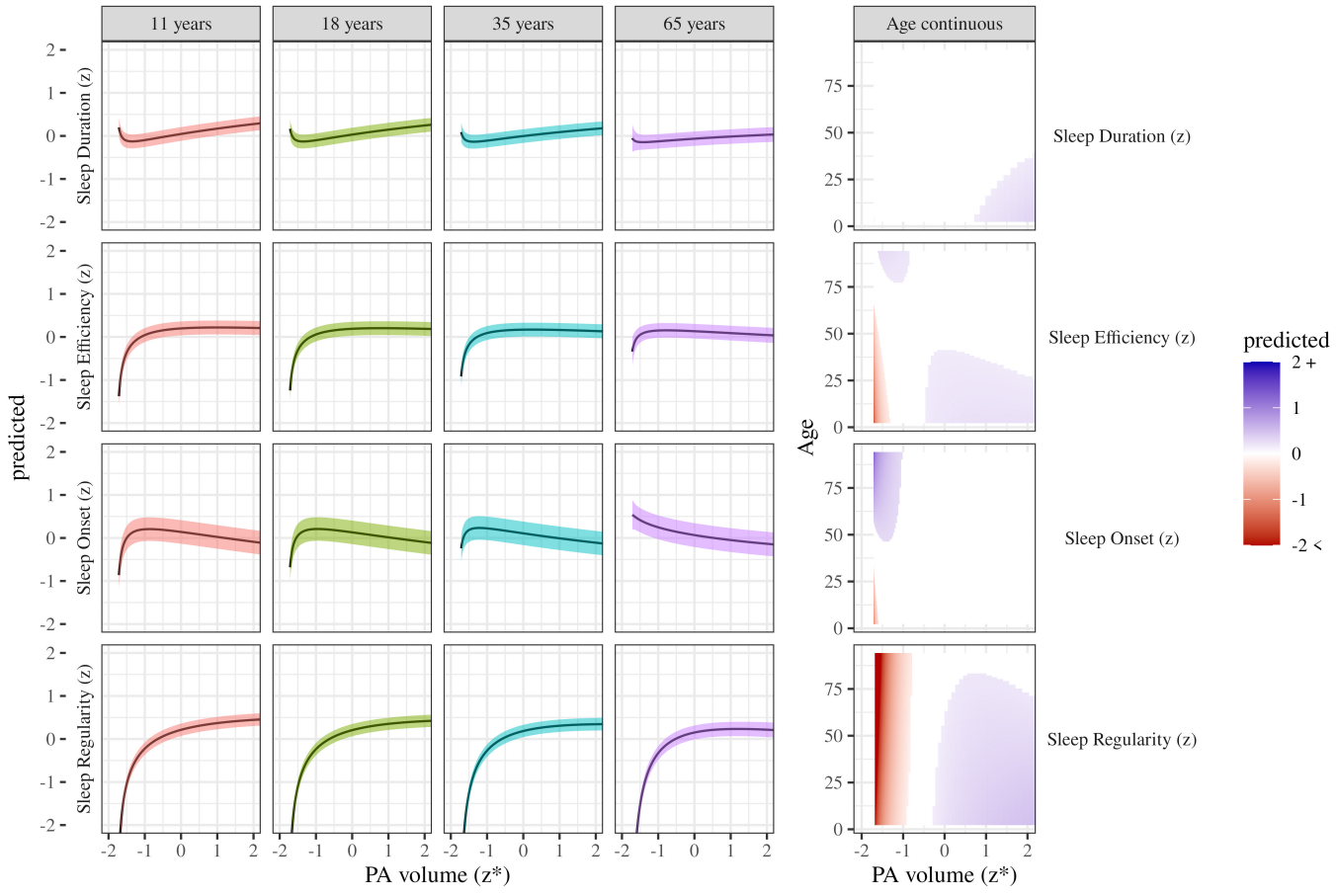

Figure 7. Sleep metrics on Physical activity volume

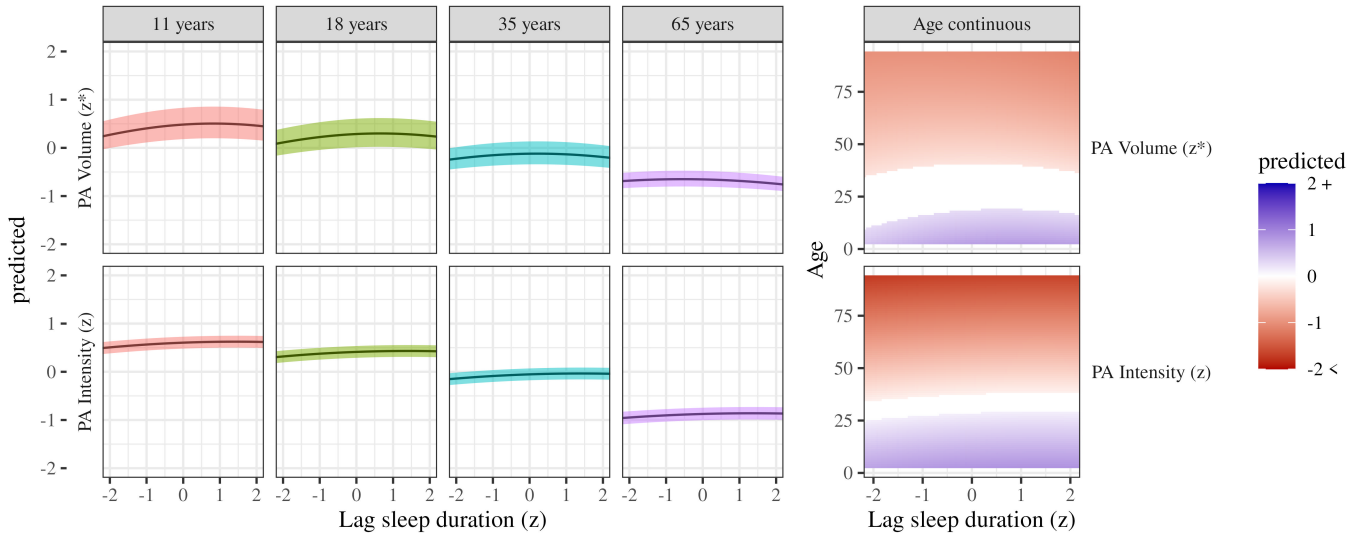

Figure 8. Physical activity by sleep duration

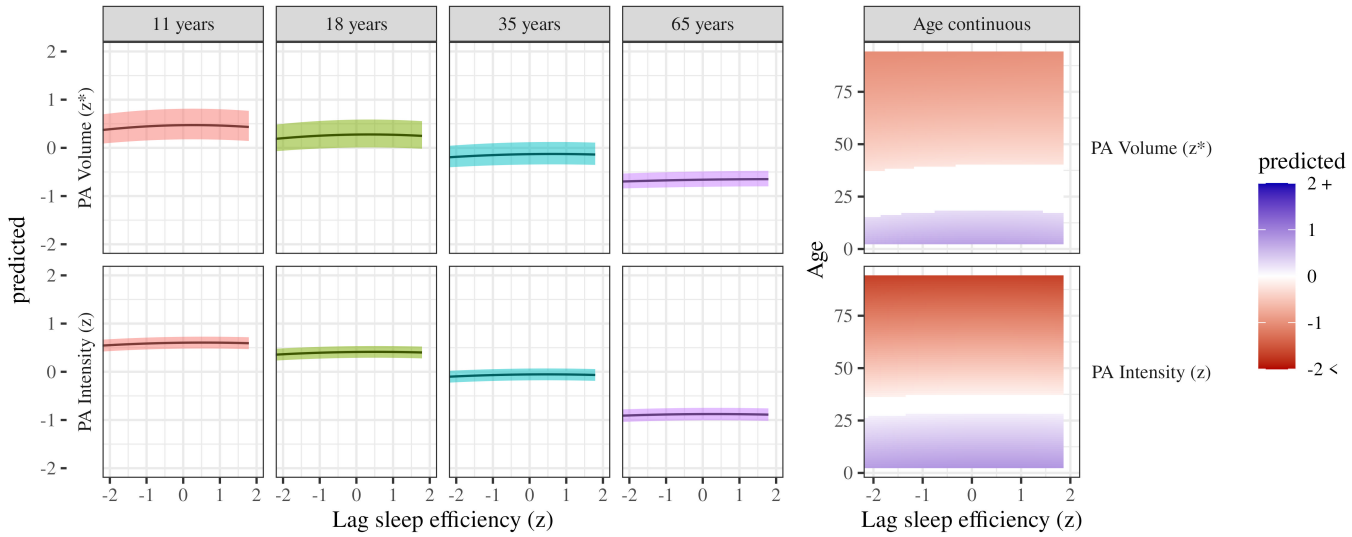

Figure 9. Physical activity by sleep efficiency

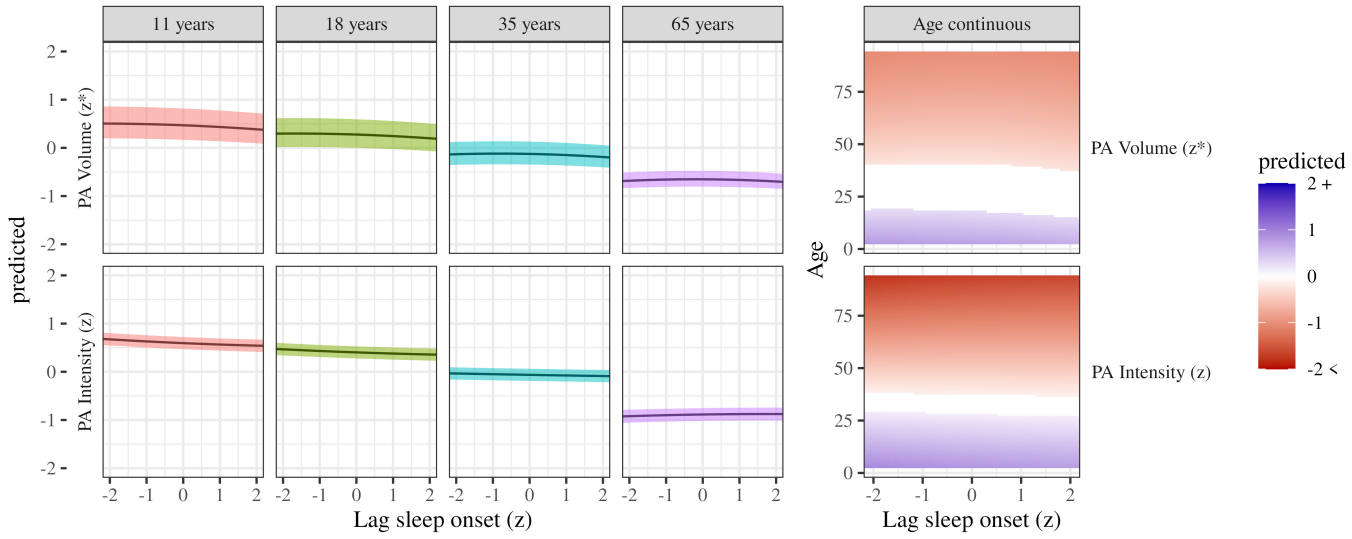

Figure 10. Physical activity by sleep onset

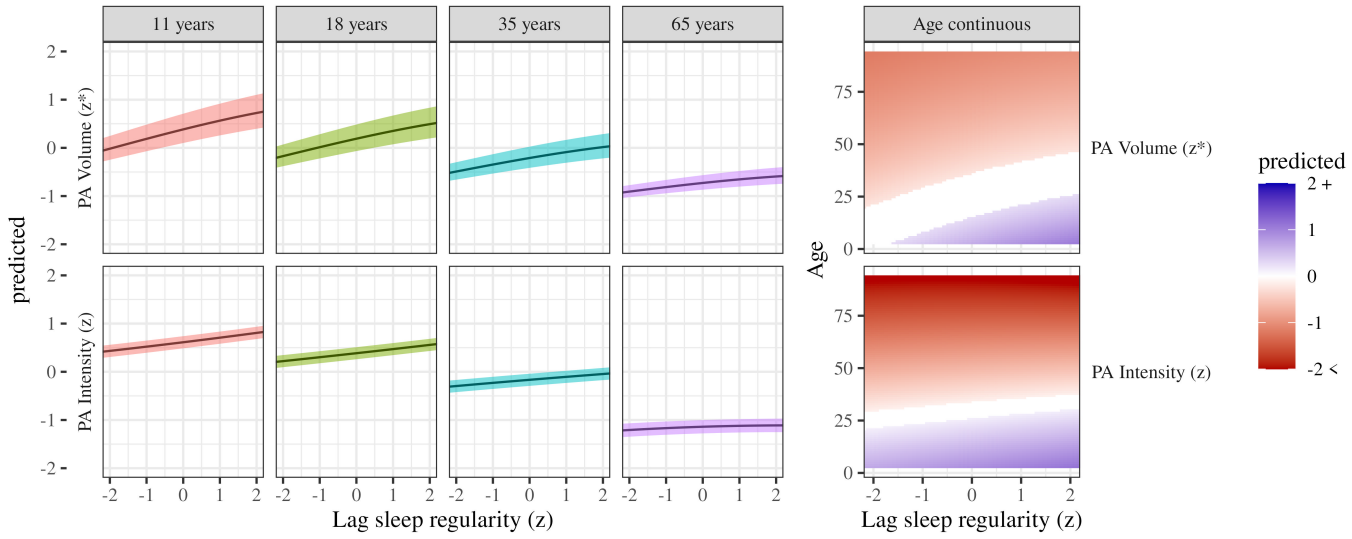

Figure 11. Physical activity by sleep regularity

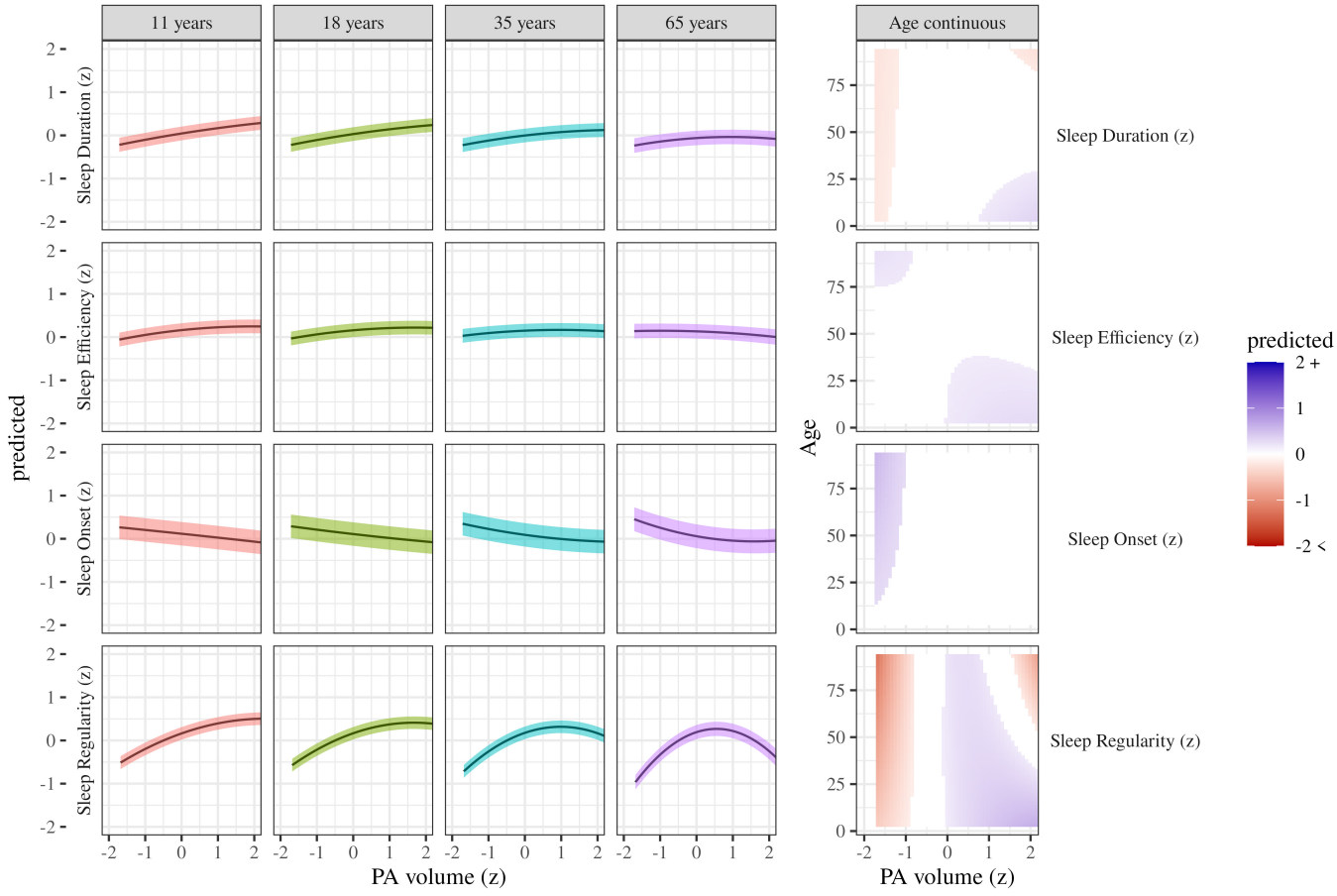

Figure 12. Sleep metrics on physical activity volume (all wear days)

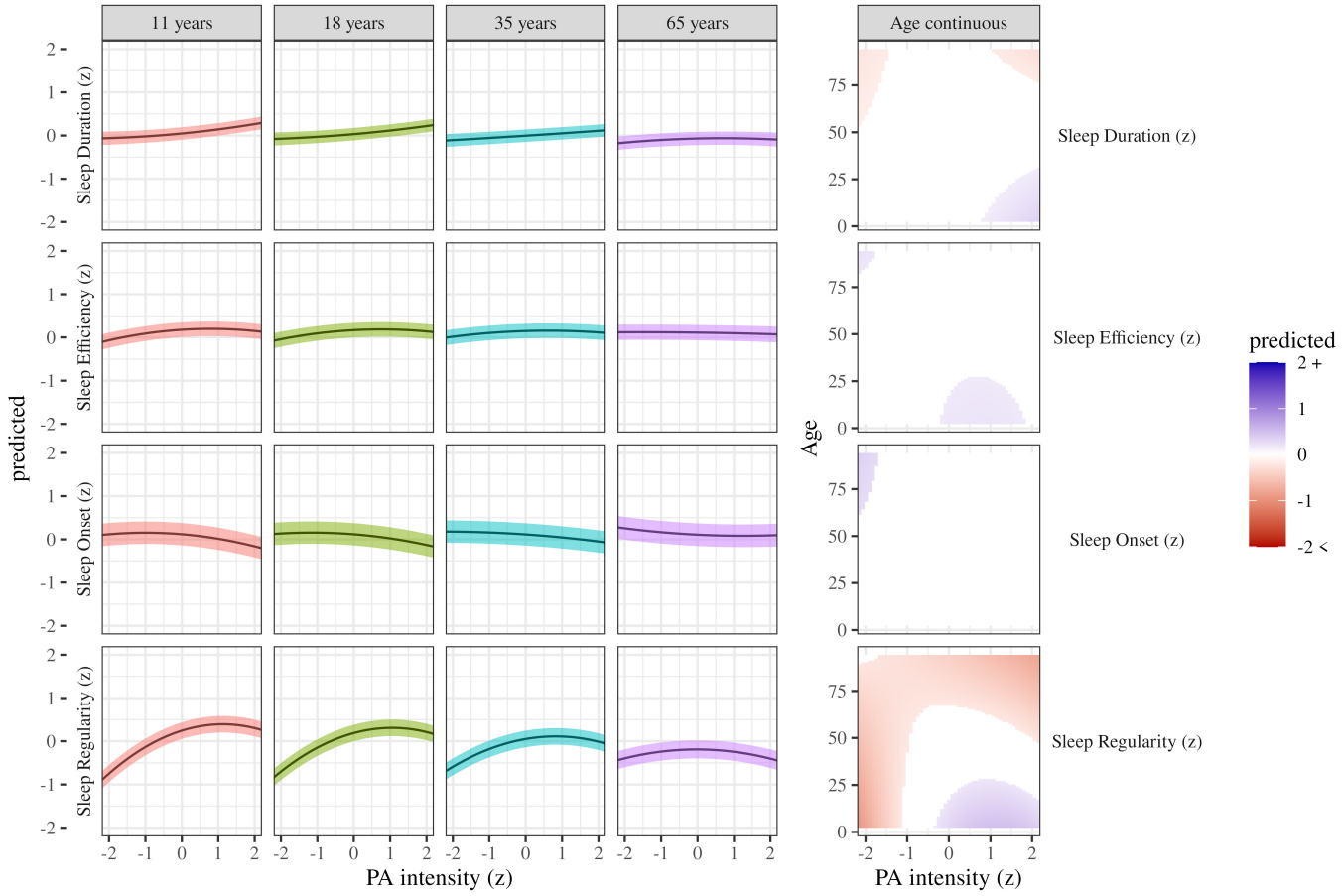

Figure 13. Sleep metrics on physical activity intensity (all wear days)

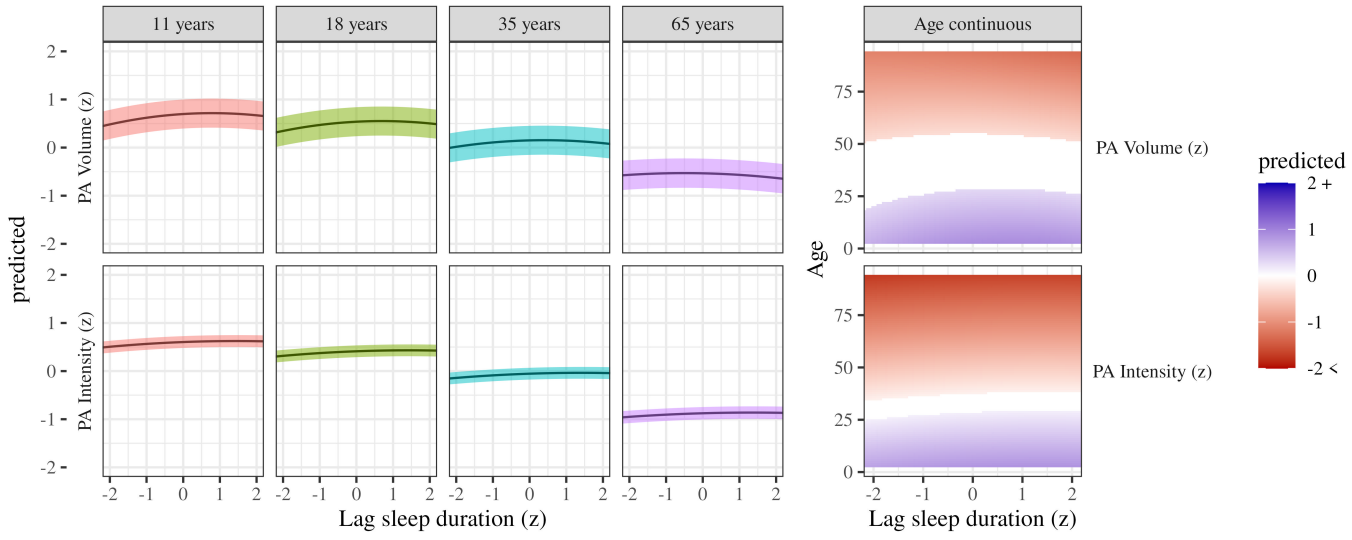

Figure 14. Physical activity by previous night sleep duration (all wear days)

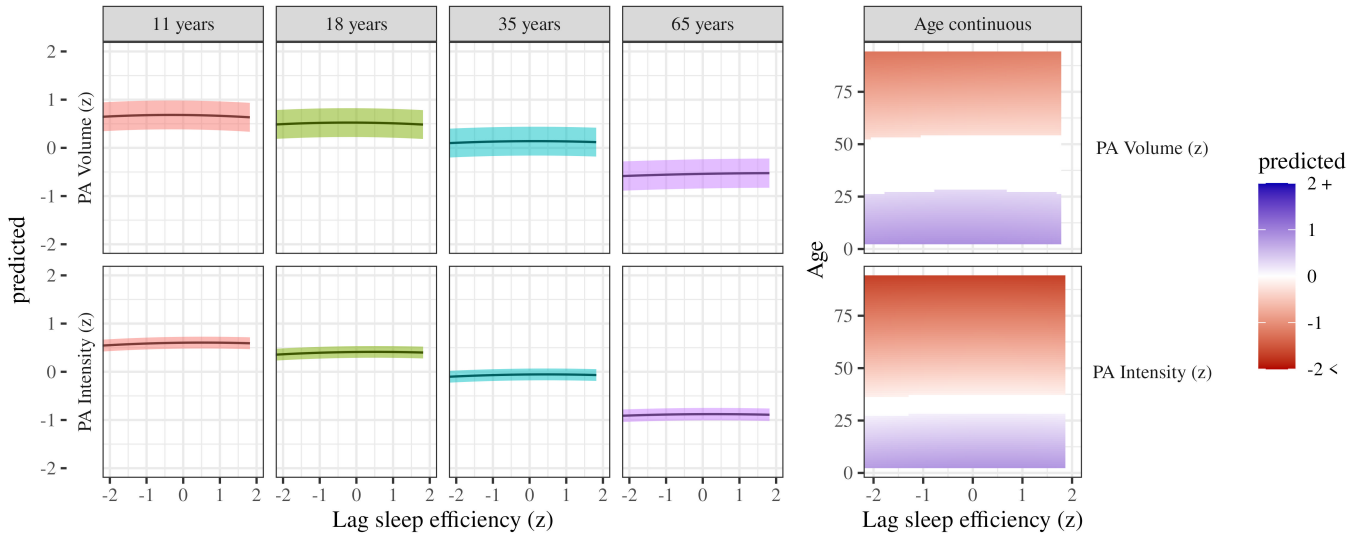

Figure 15. Physical activity by previous night sleep efficiency (all wear days)

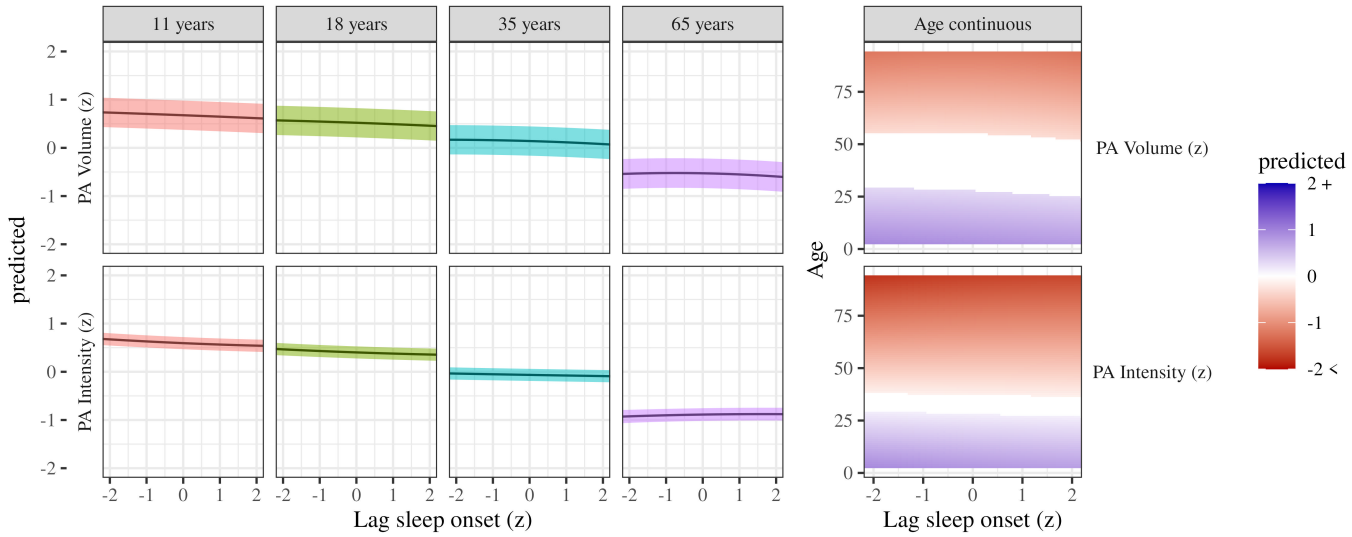

Figure 16. Physical activity by previous night sleep onset (all wear days)

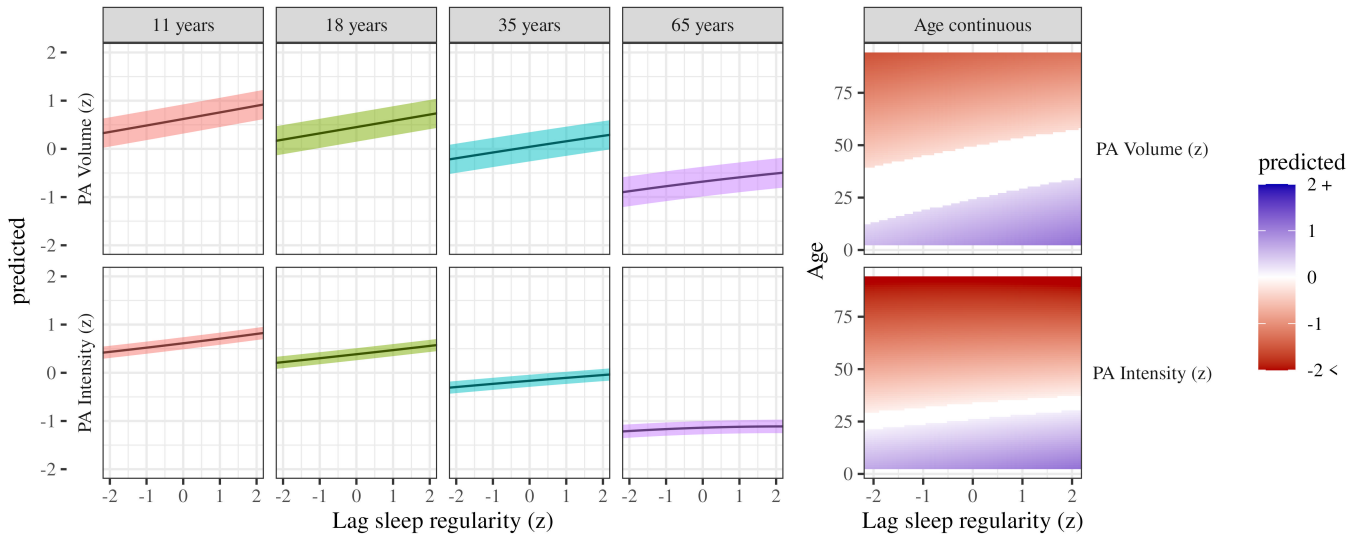

Figure 17. Physical activity by previous night sleep regularity (all wear days)

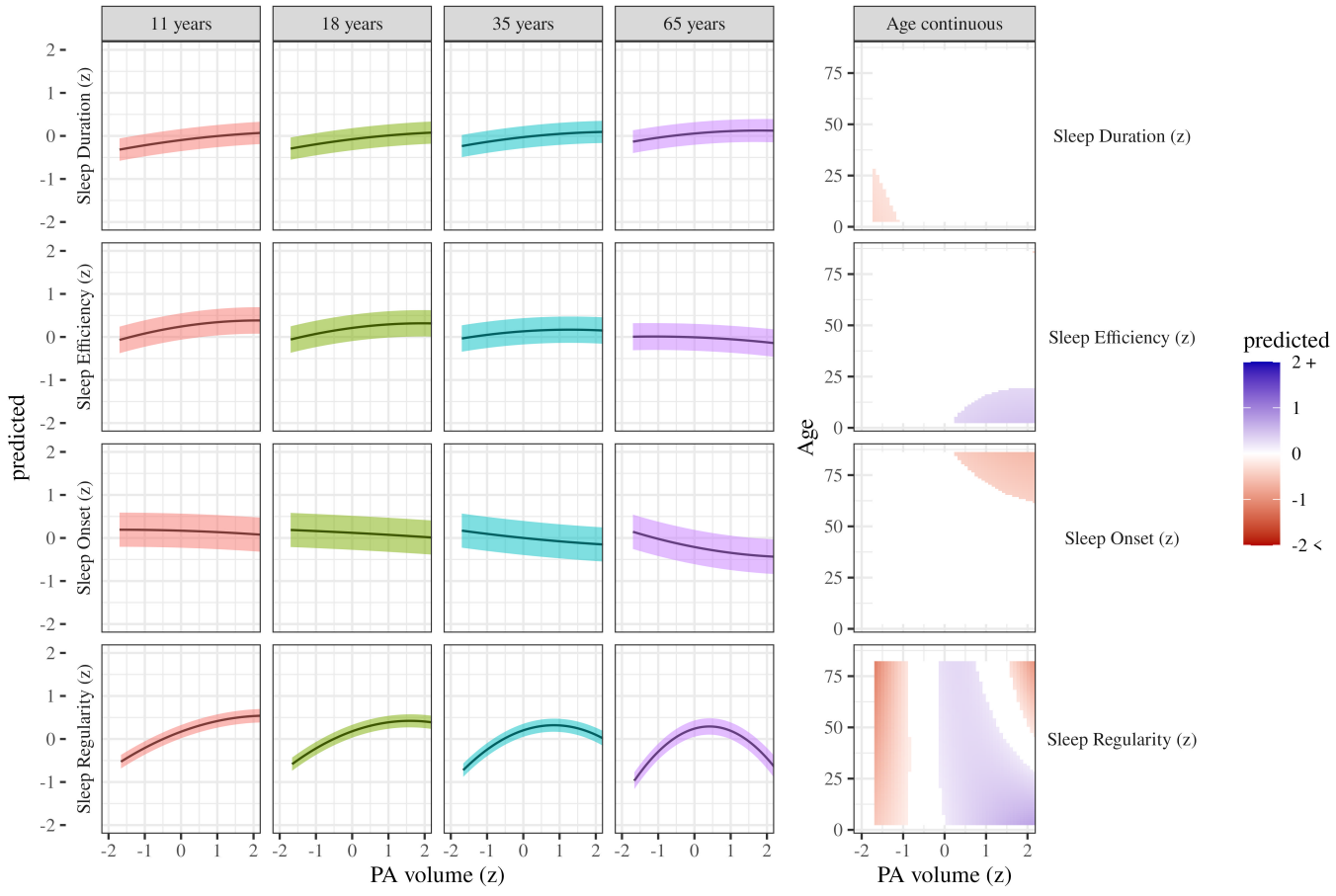

Figure 18. Sleep metrics on physical activity volume (at least seven valid days)

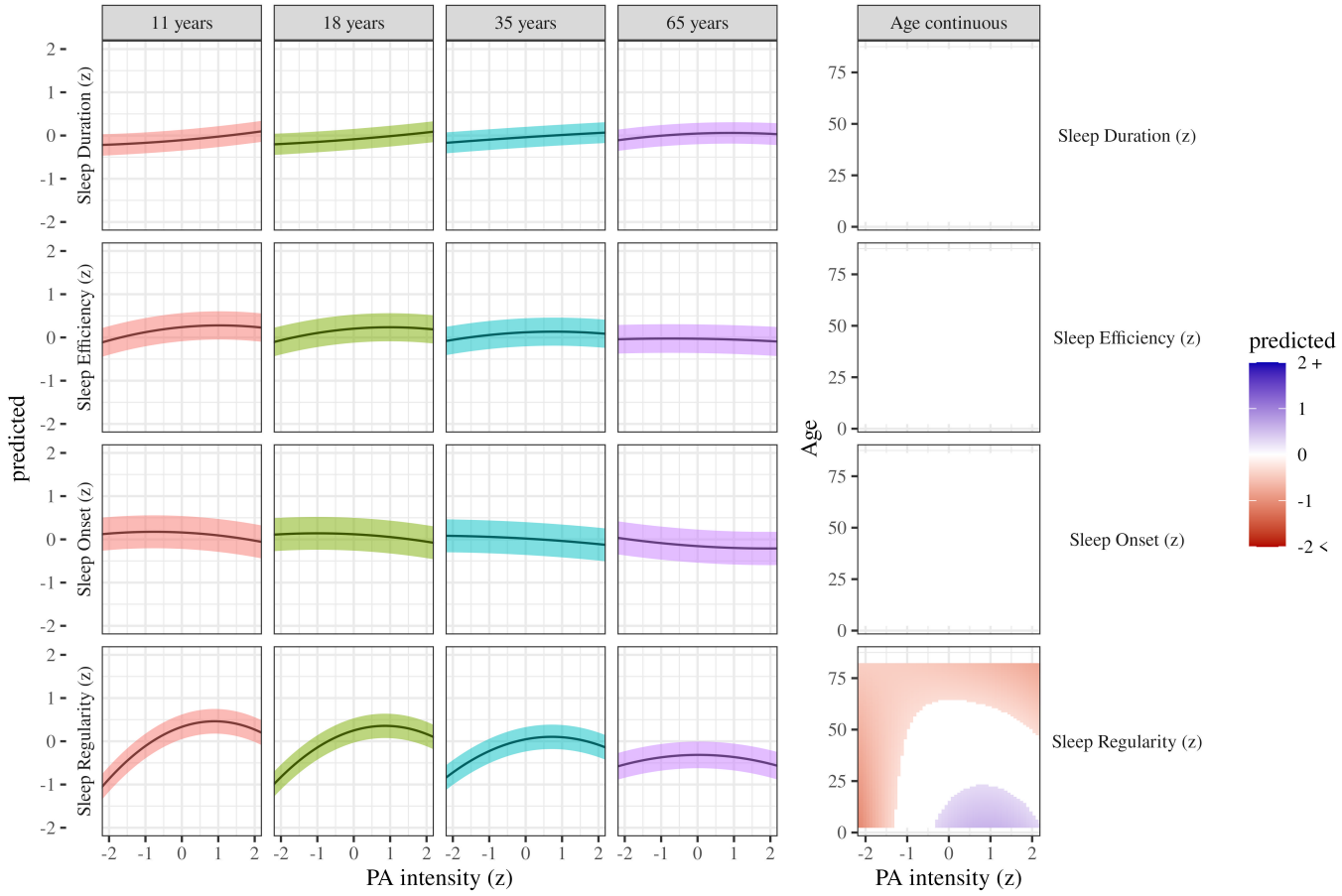

Figure 19. Sleep metrics on physical activity intensity (at least seven valid days)

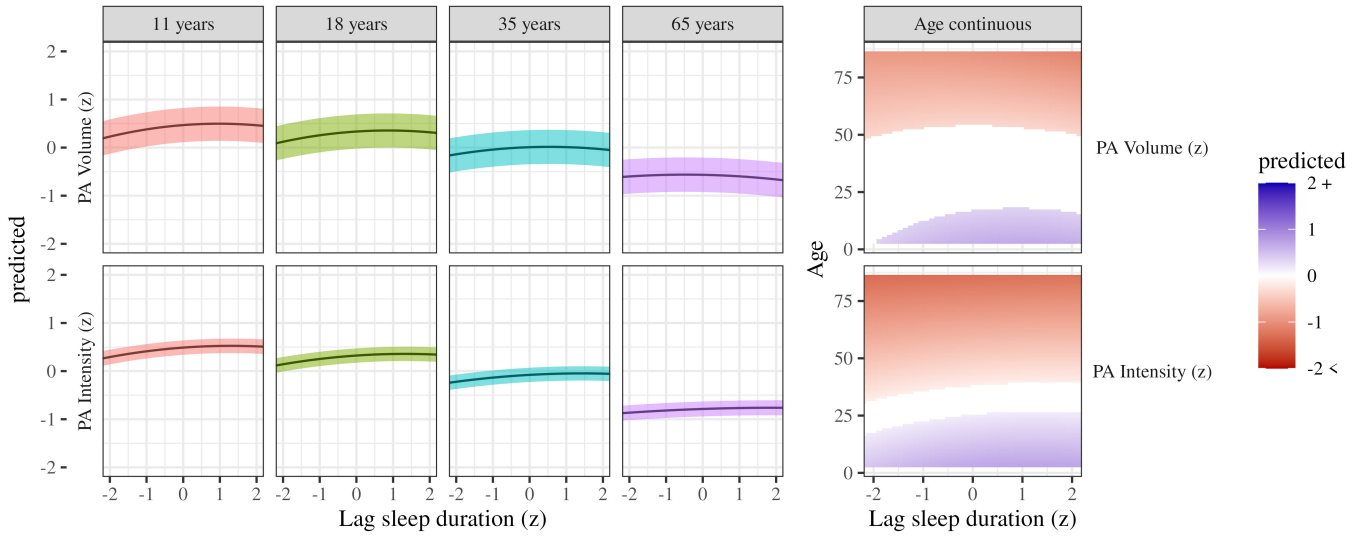

Figure 20. Physical activity by previous night sleep duration (at least seven valid days)

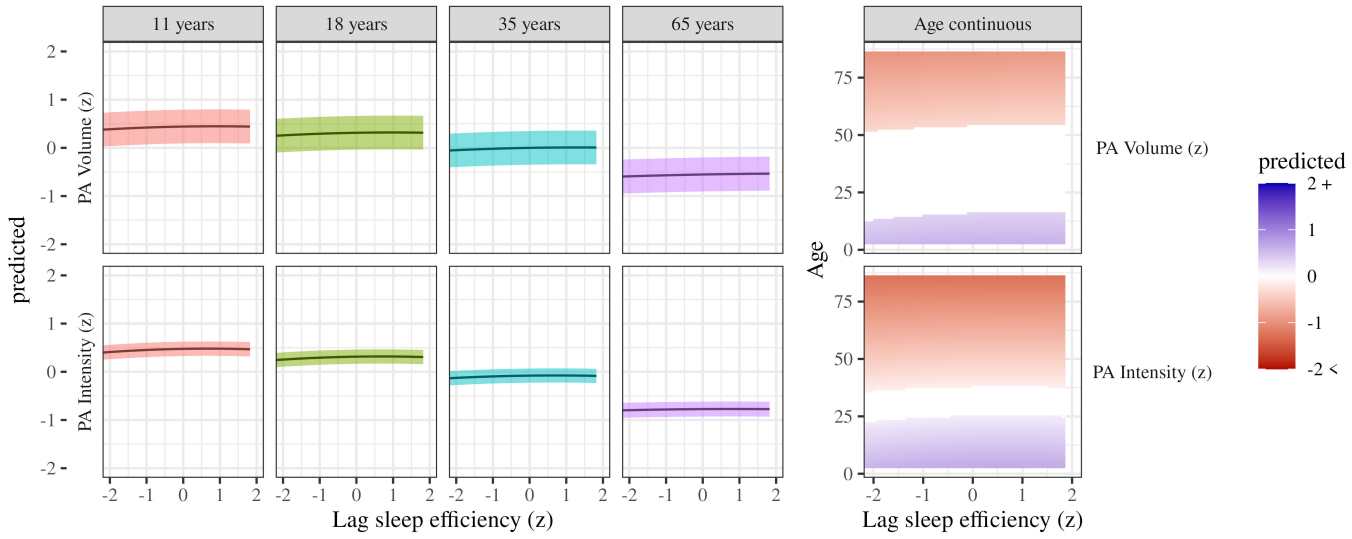

Figure 21. Physical activity by previous night sleep efficiency (at least seven valid days)

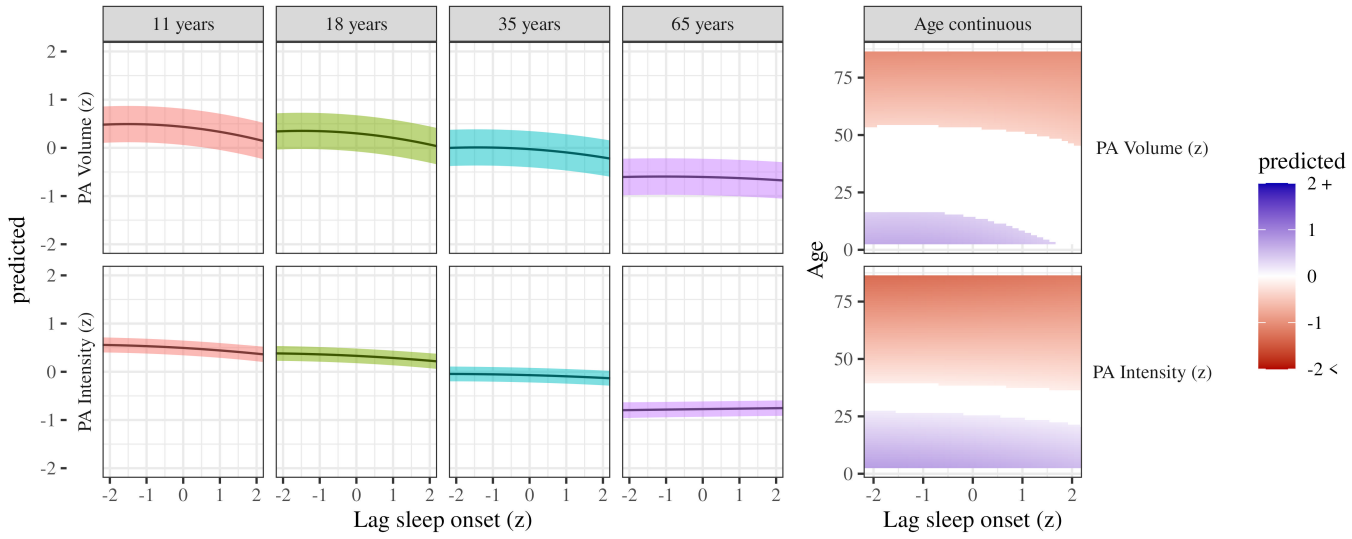

Figure 22. Physical activity by previous night sleep onset (at least seven valid days)

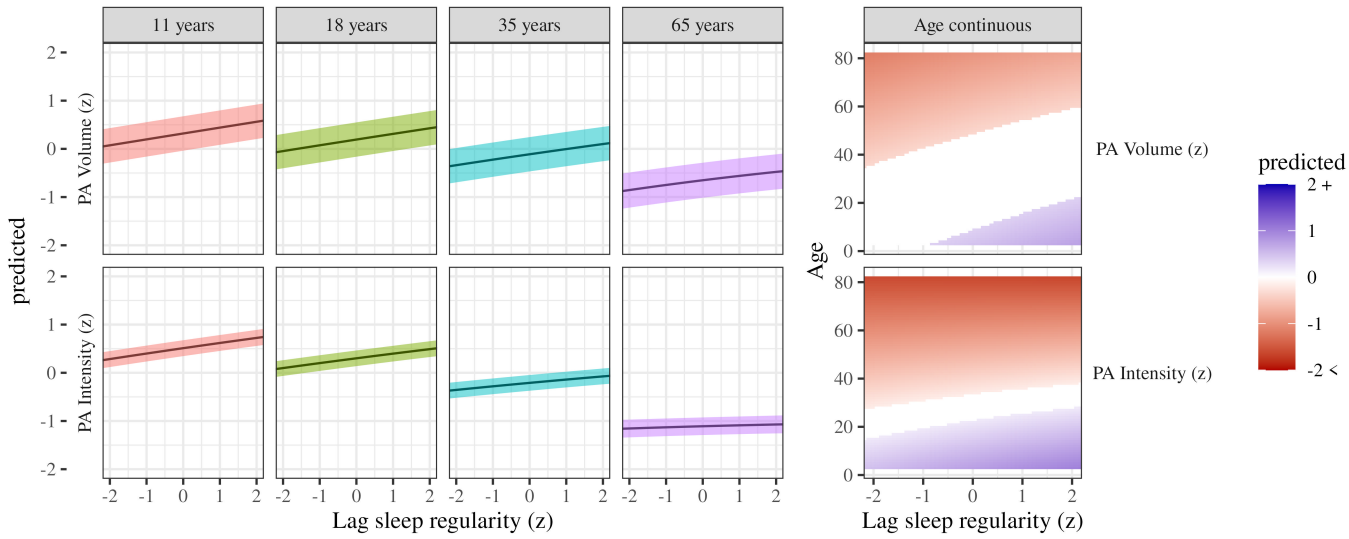

Figure 23. Physical activity by previous night sleep regularity (at least seven valid days)

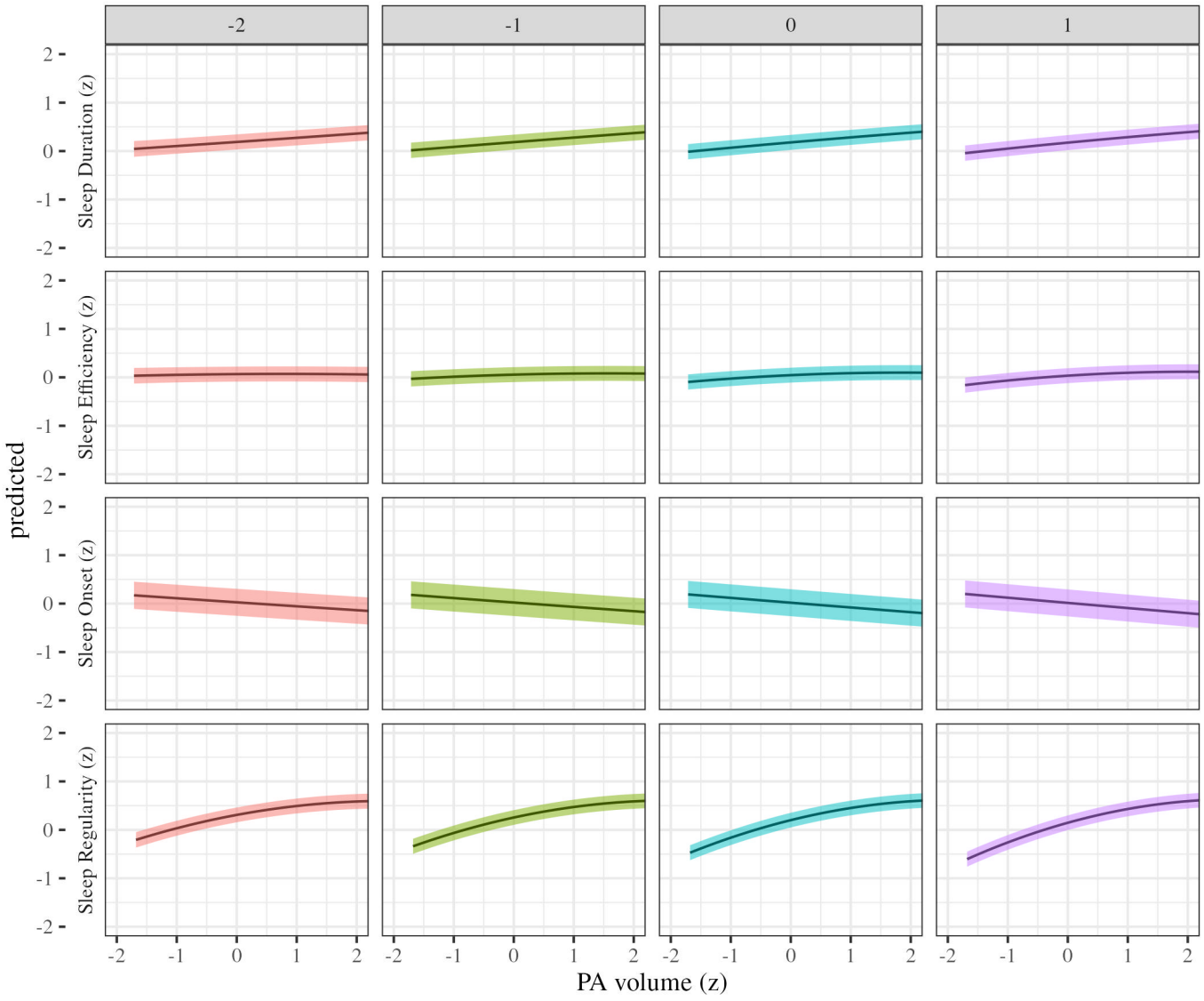

Figure 24. Sleep metrics on Physical activity volume by BMI z-score

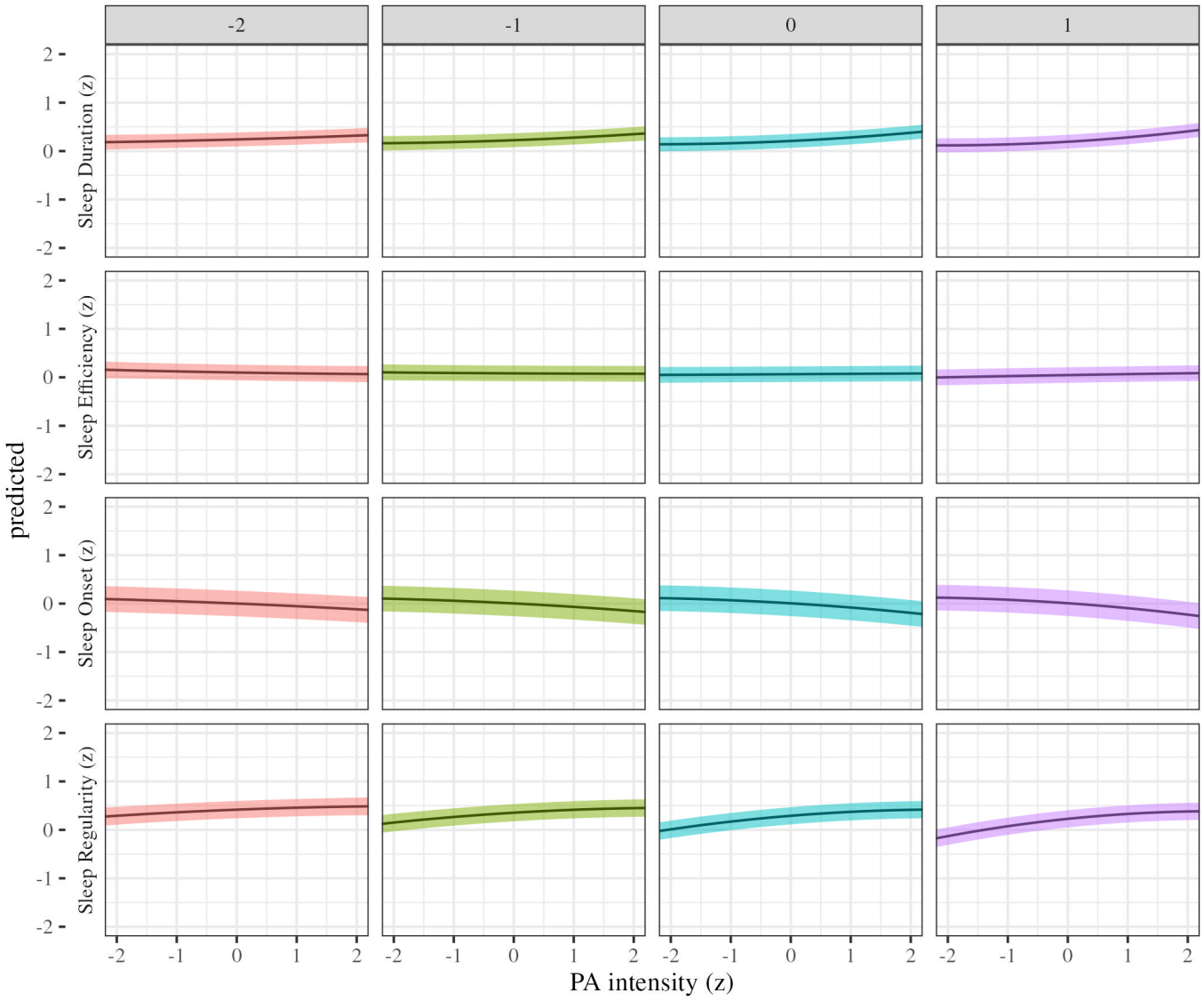

Figure 25. Sleep metrics on Physical activity intensity moderated by BMI z-score

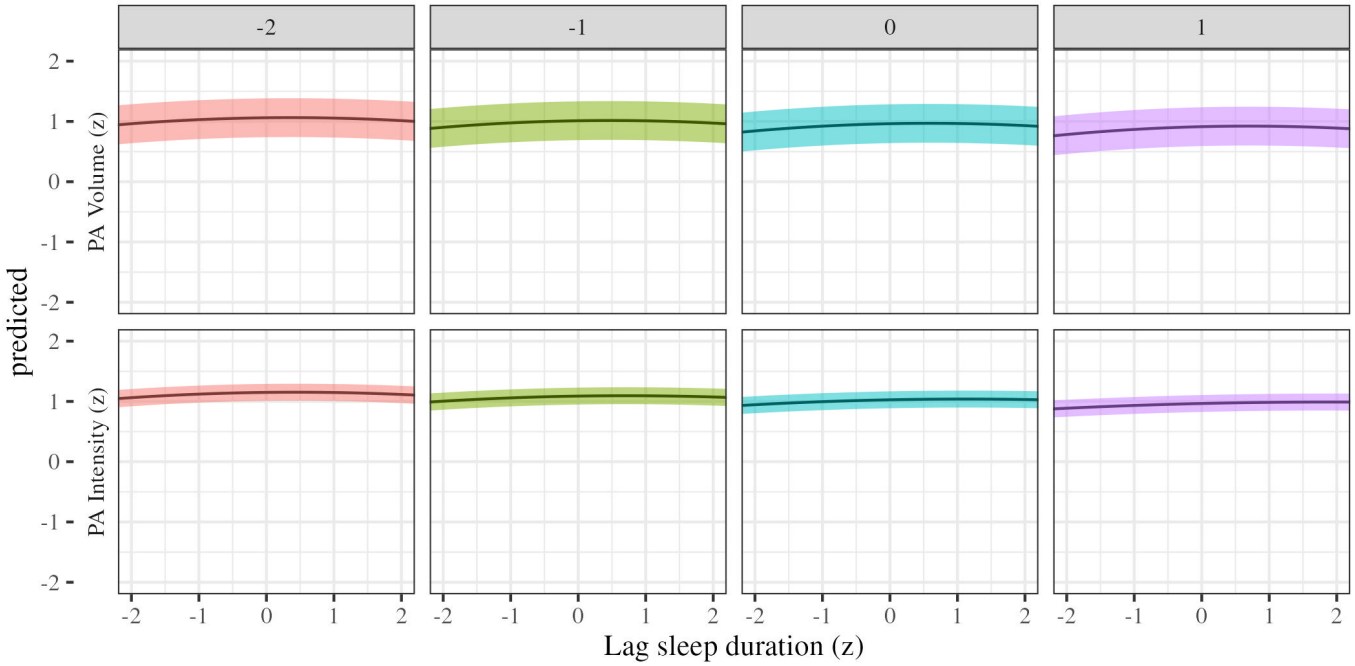

Figure 26. Physical activity by sleep duration moderated by BMI z-score

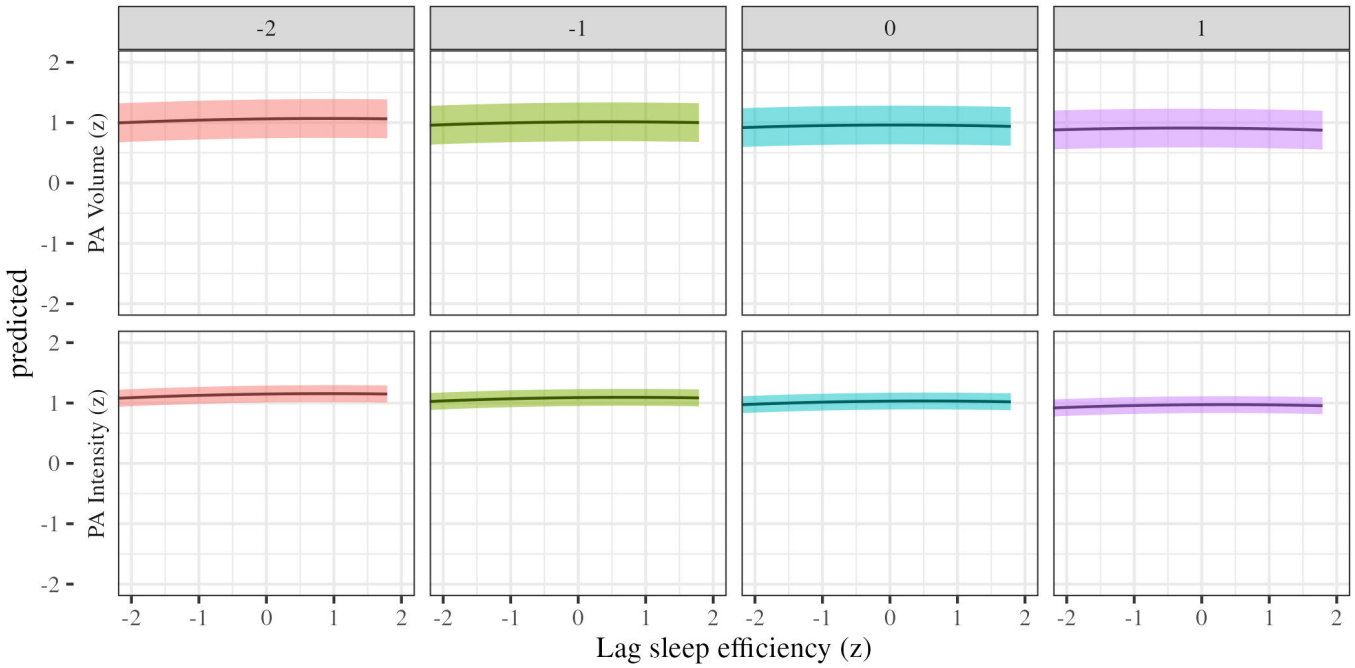

Figure 27. Physical activity by sleep efficiency moderated by BMI z-score

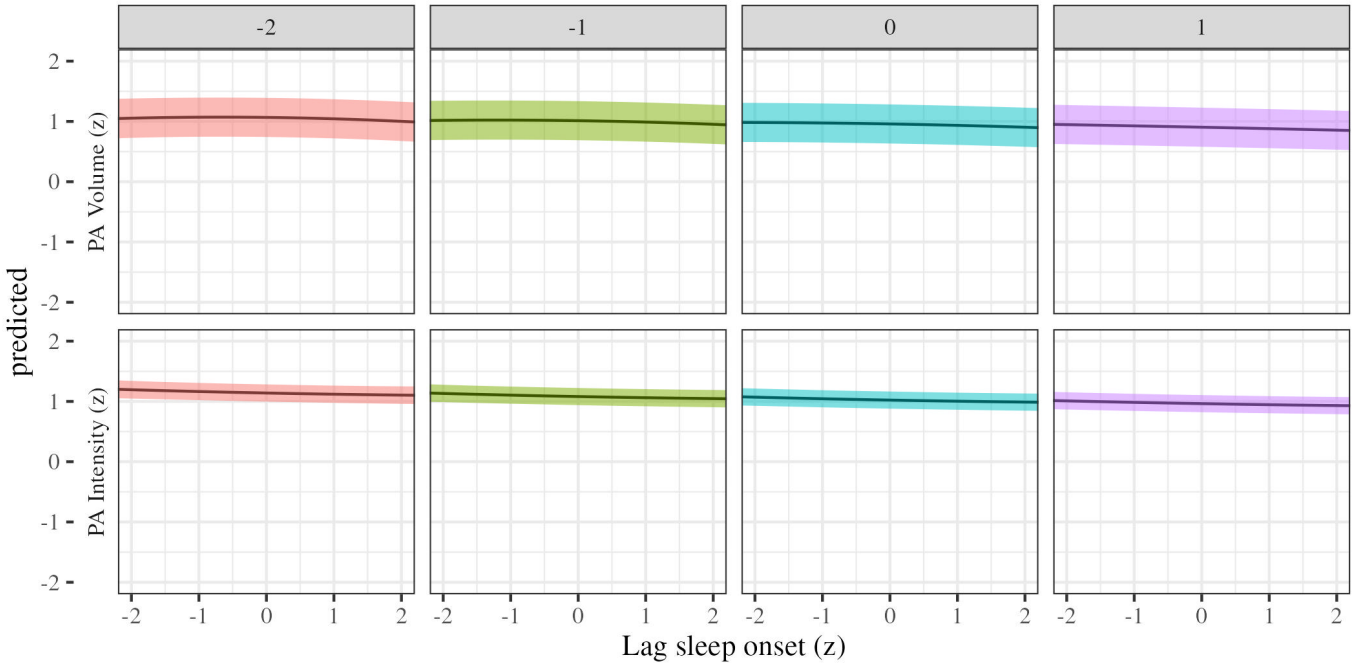

Figure 28. Physical activity by sleep onset moderated by BMI z-score

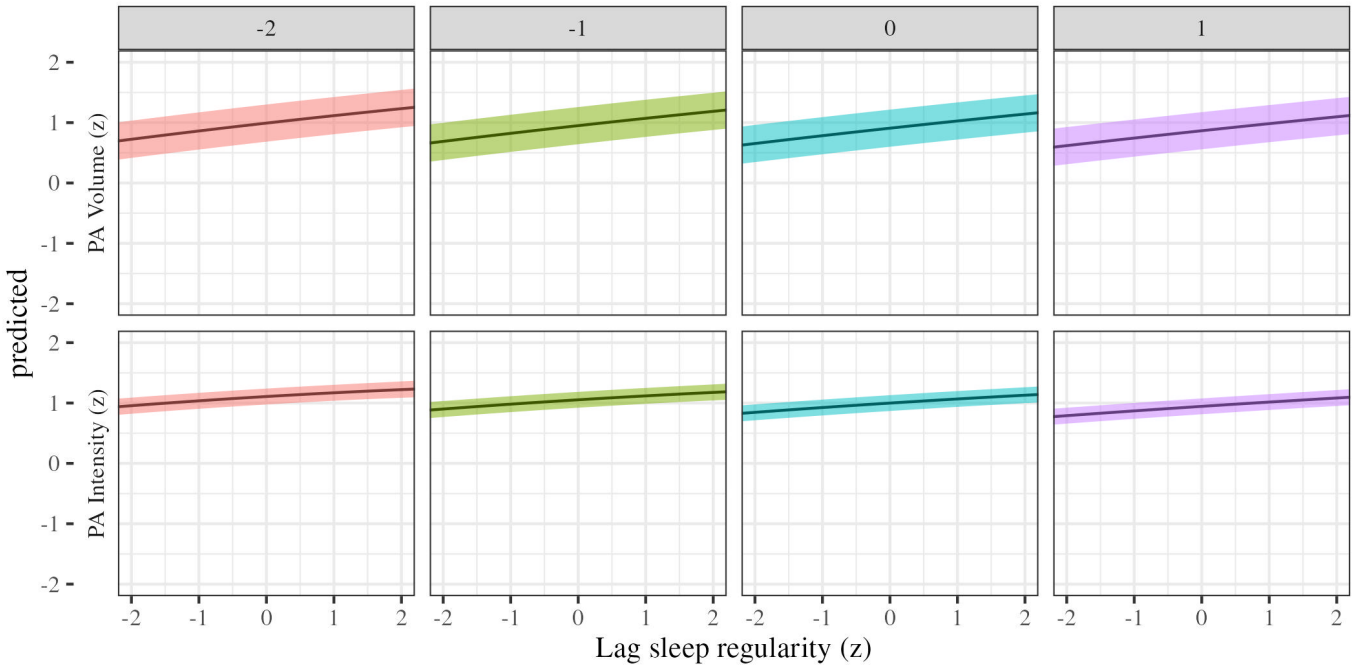

Figure 29. Physical activity by sleep regularity moderated by BMI z-score

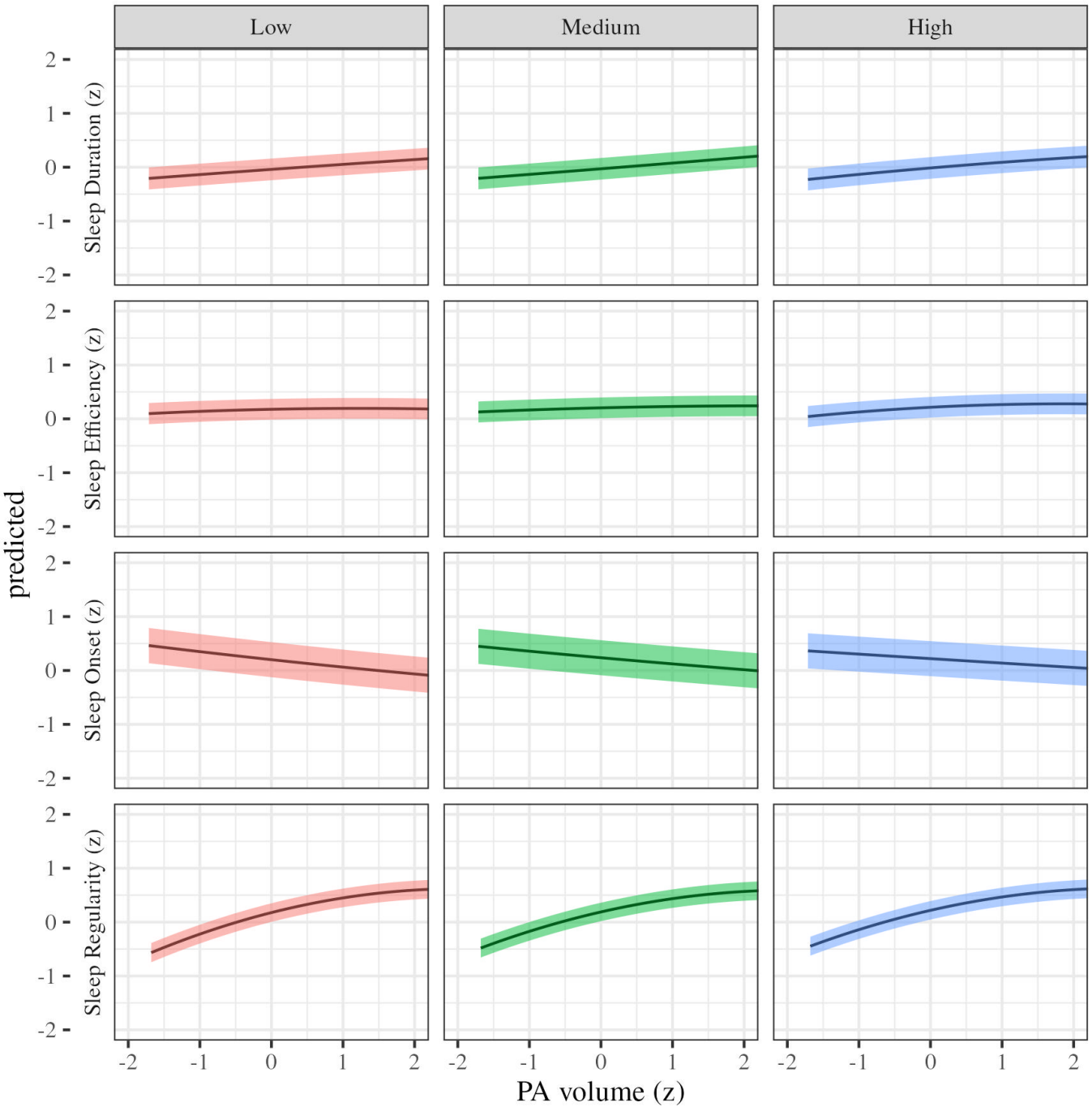

Figure 30. Sleep metrics on Physical activity volume by SES

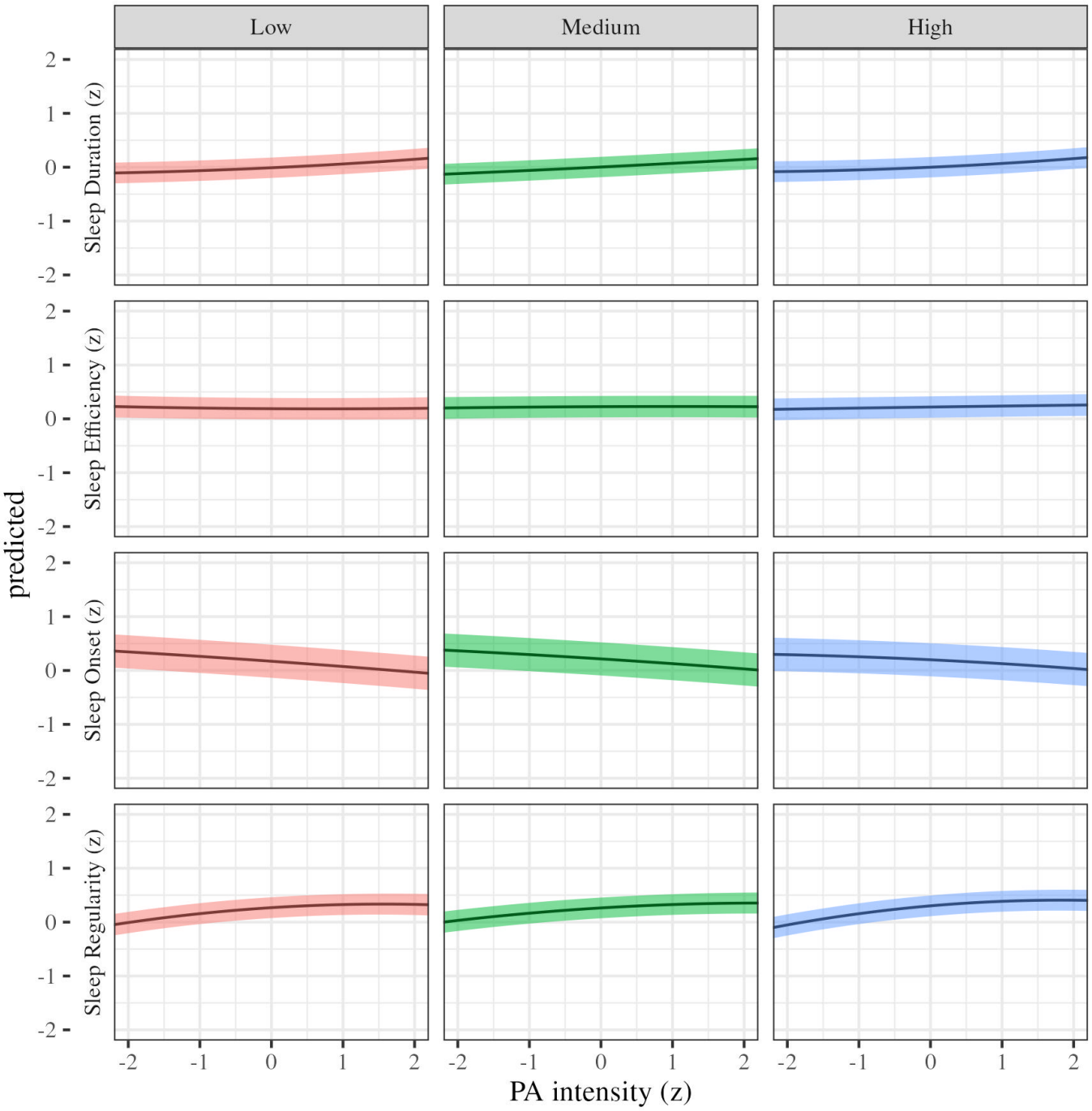

Figure 31. Sleep metrics on Physical activity intensity moderated by SES

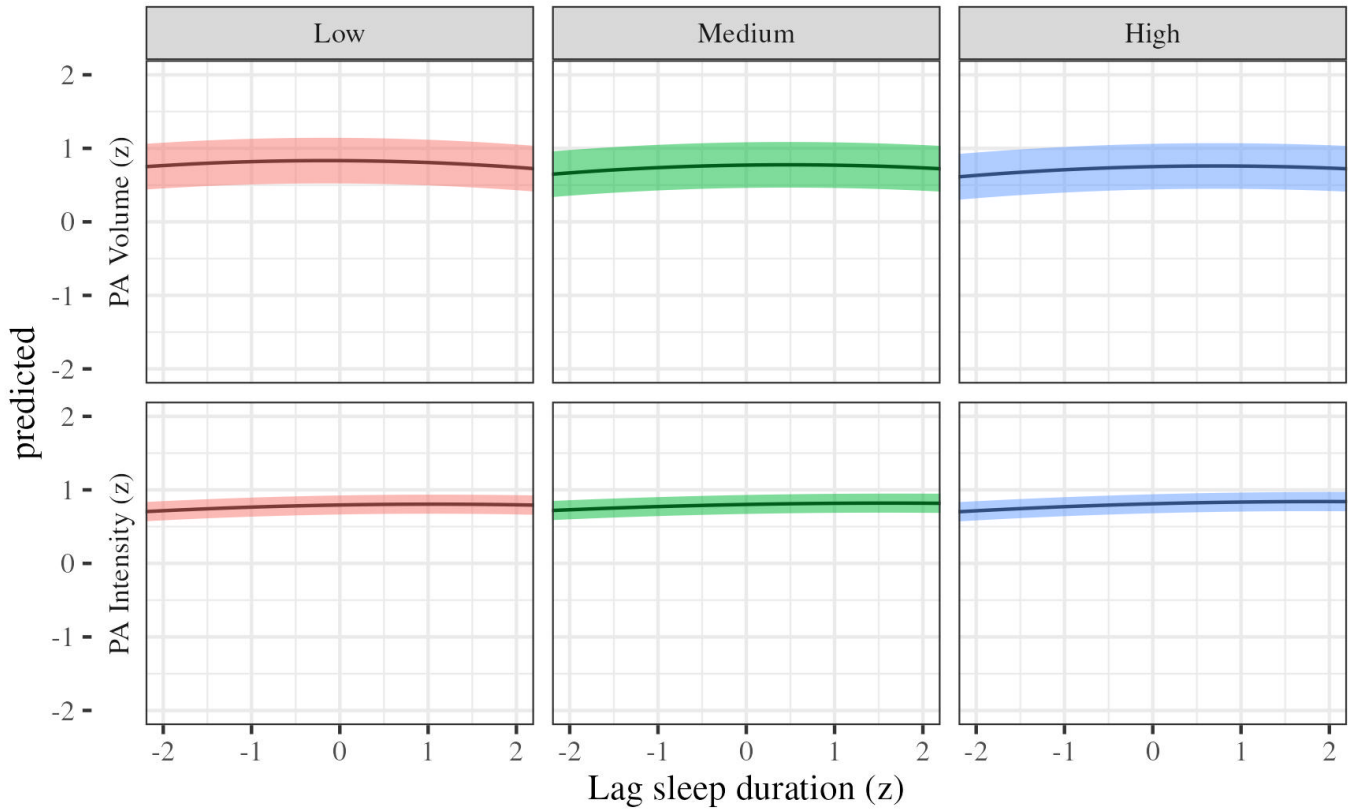

Figure 32. Physical activity by sleep duration moderated by SES

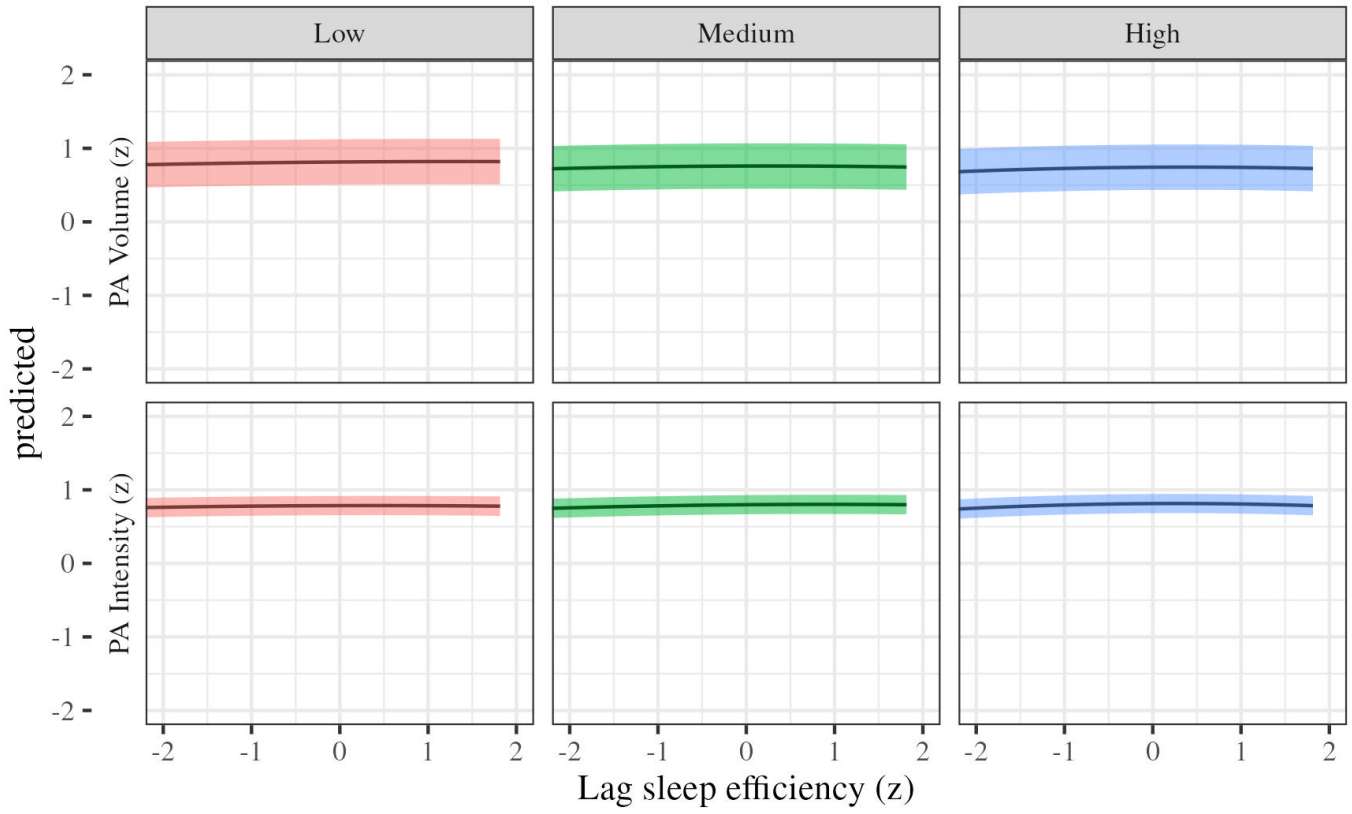

Figure 33. Physical activity by sleep efficiency moderated by SES

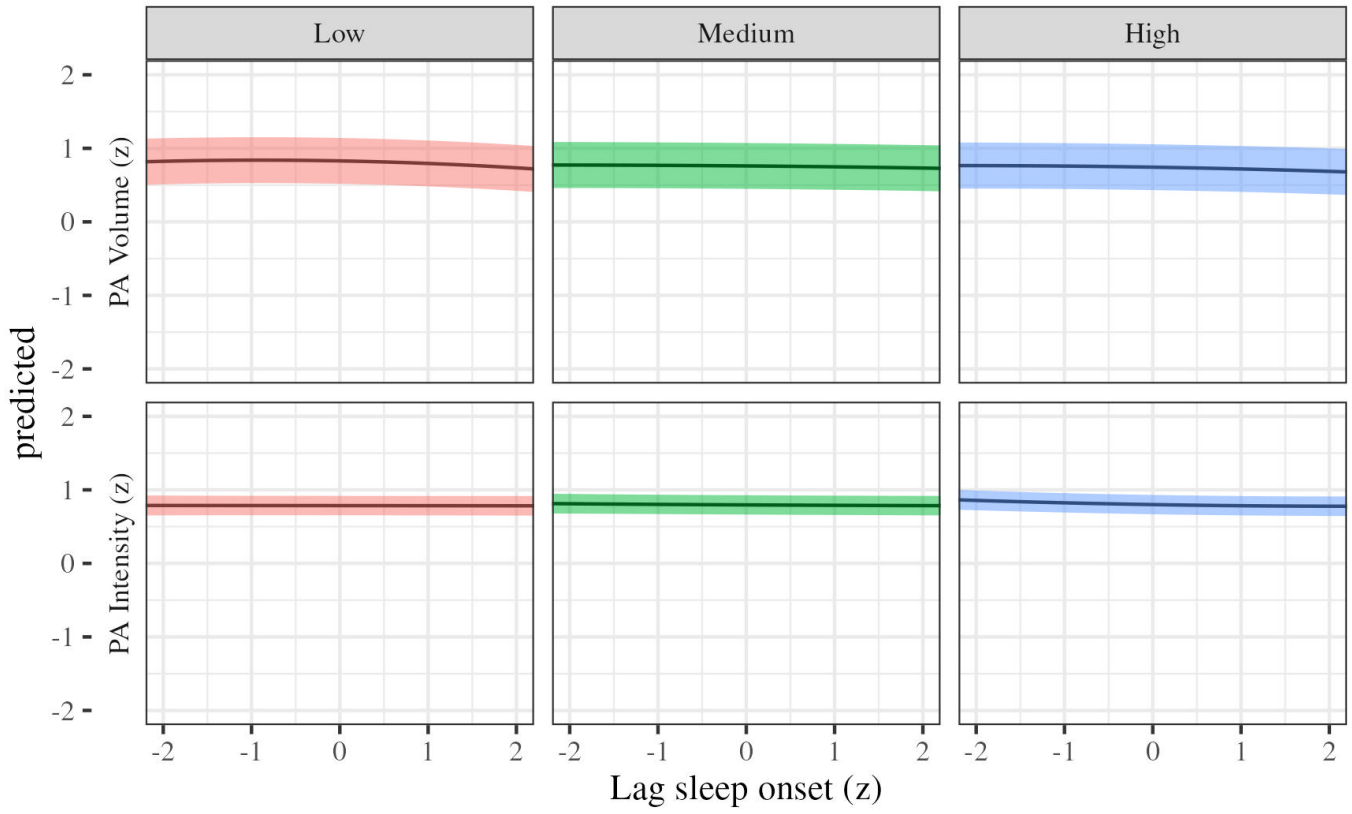

Figure 34. Physical activity by sleep onset moderated by SES

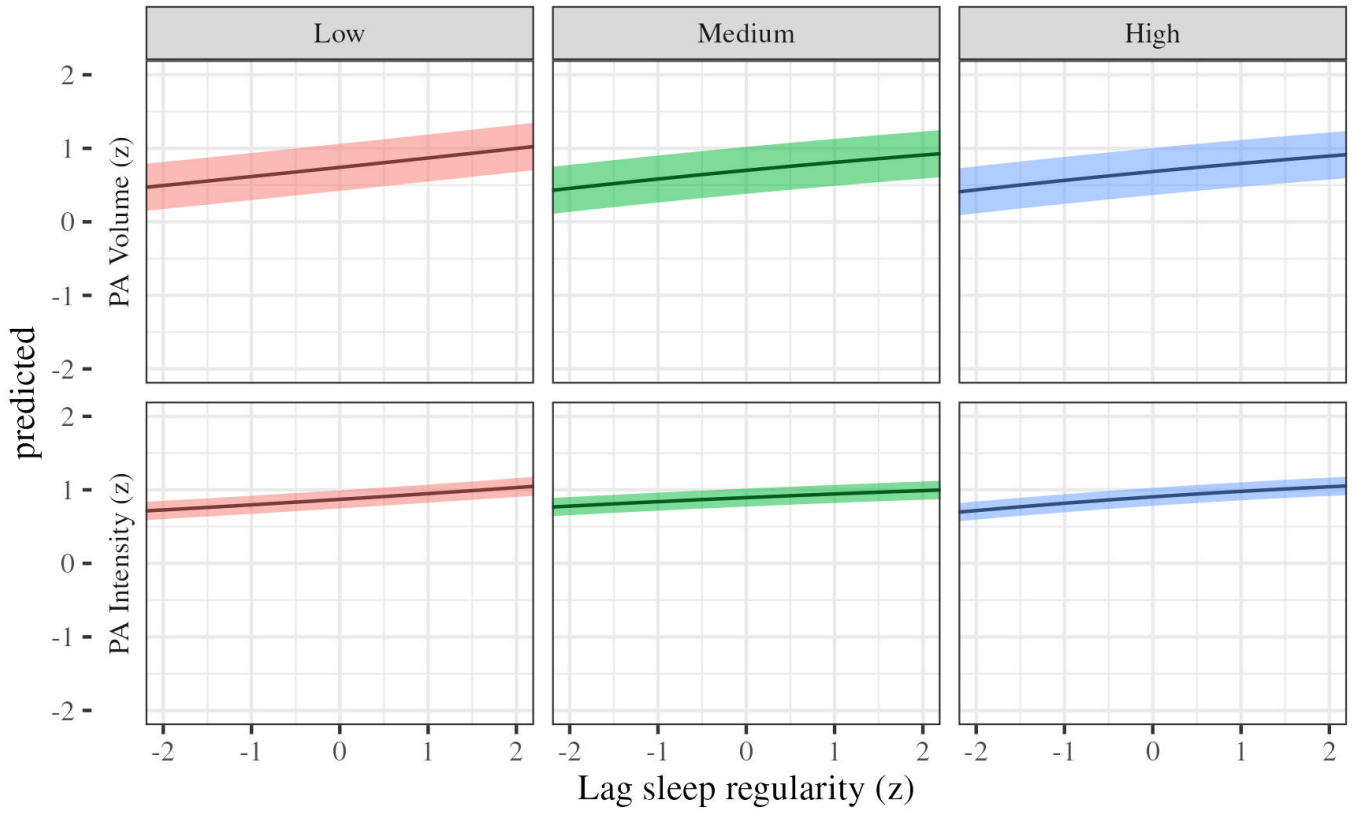

Figure 35. Physical activity by sleep regularity moderated by SES

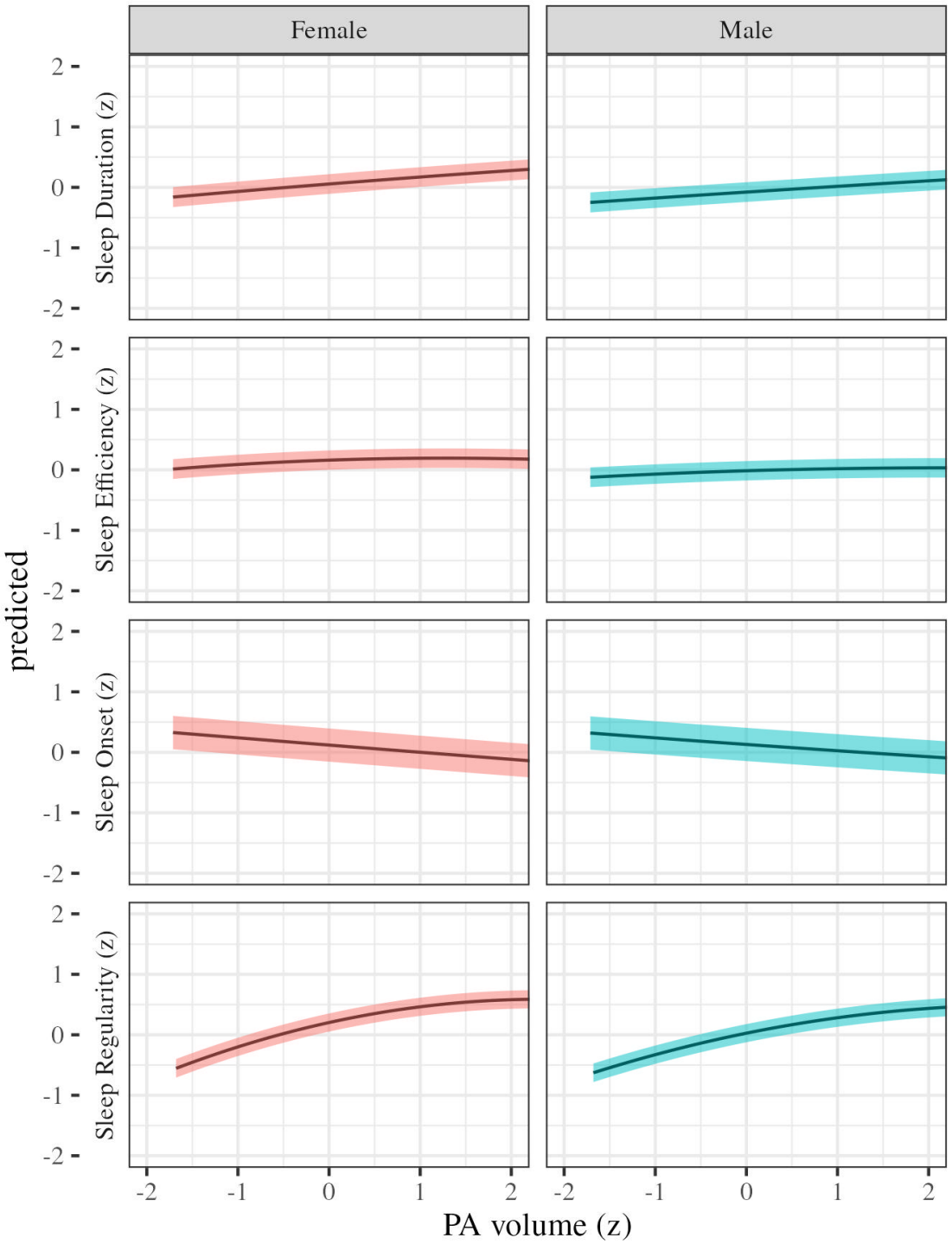

Figure 36. Sleep metrics on Physical activity volume by sex

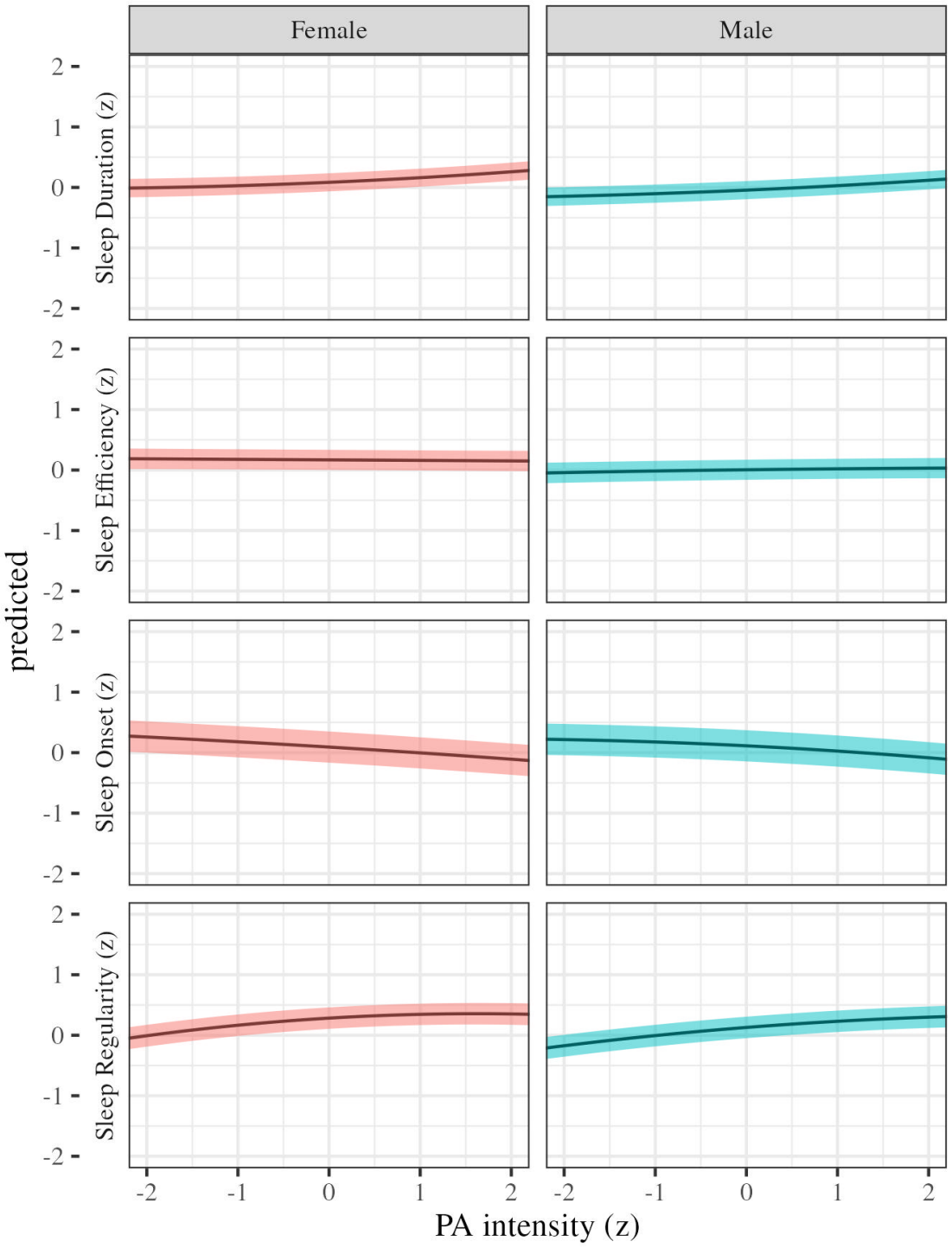

Figure 37. Sleep metrics on Physical activity intensity moderated by sex

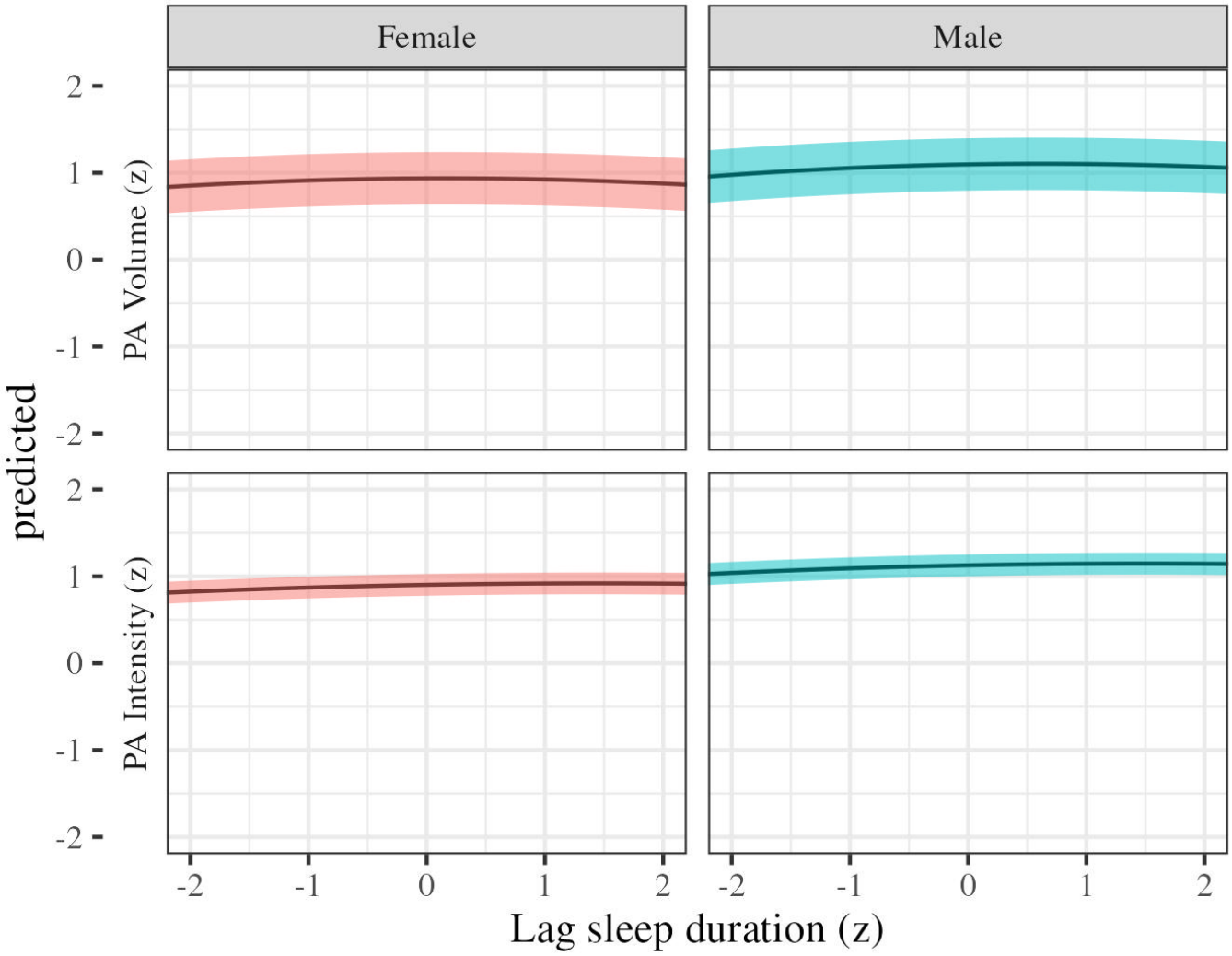

Figure 38. Physical activity by sleep duration moderated by sex

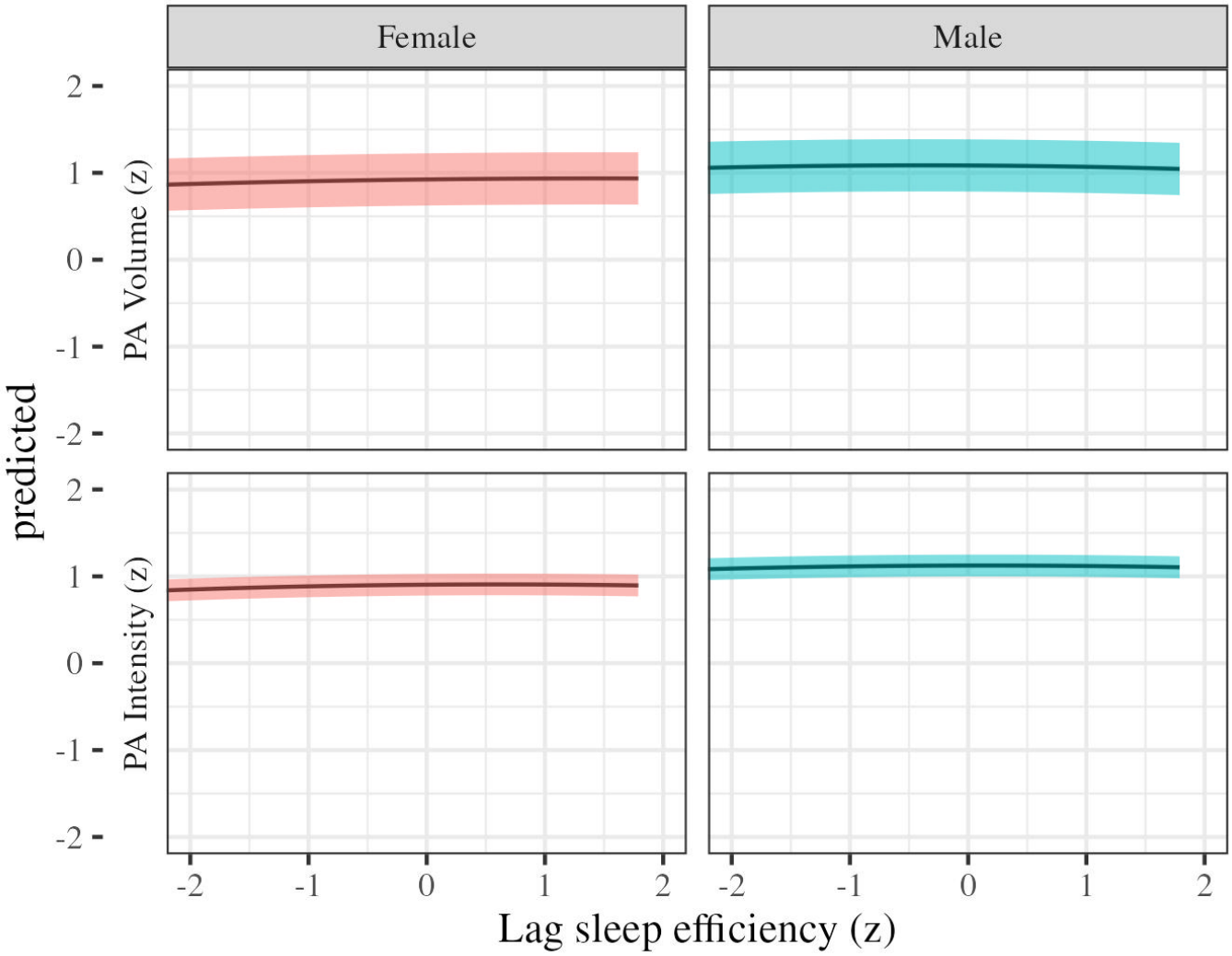

Figure 39. Physical activity by sleep efficiency moderated by sex

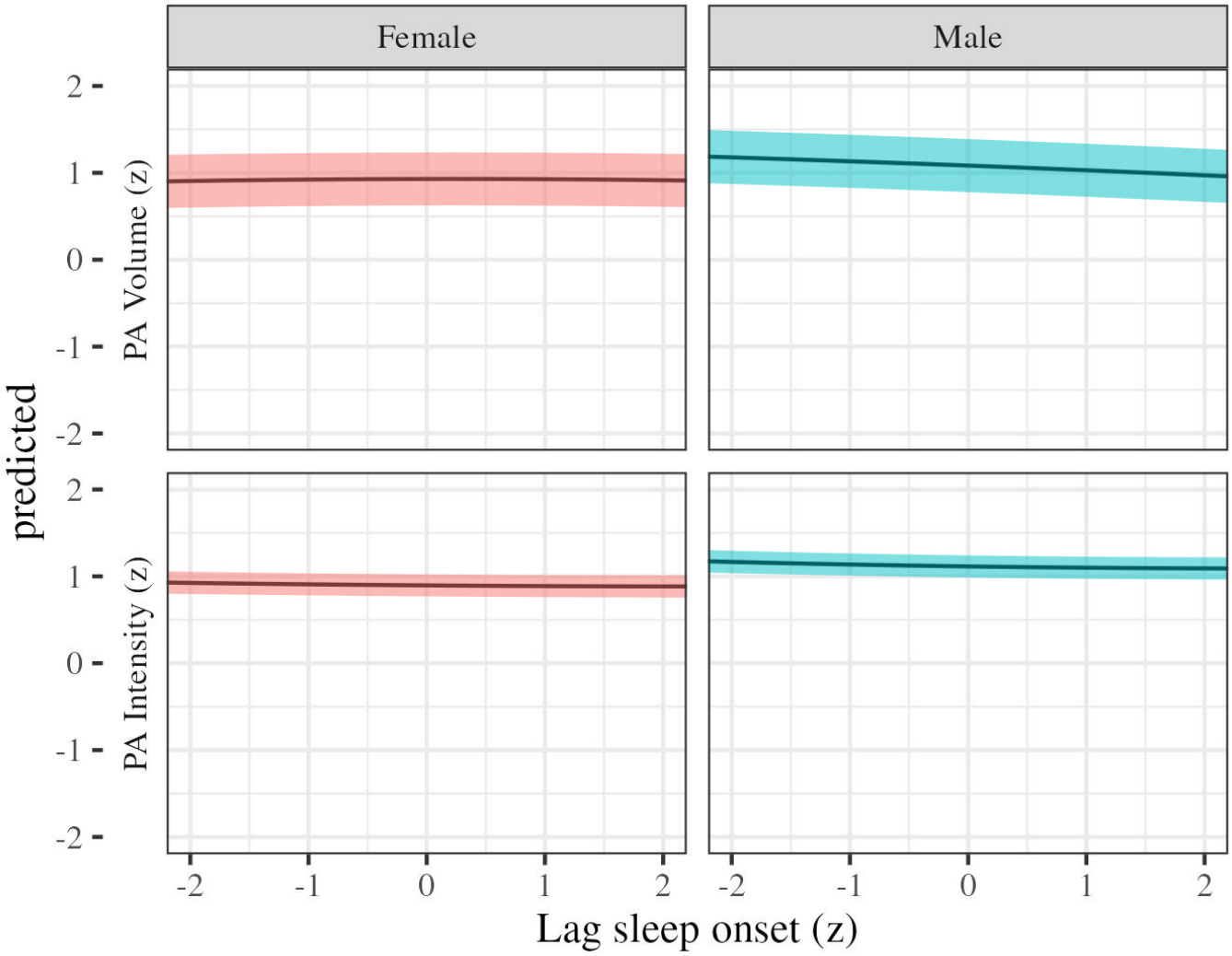

Figure 40. Physical activity by sleep onset moderated by sex

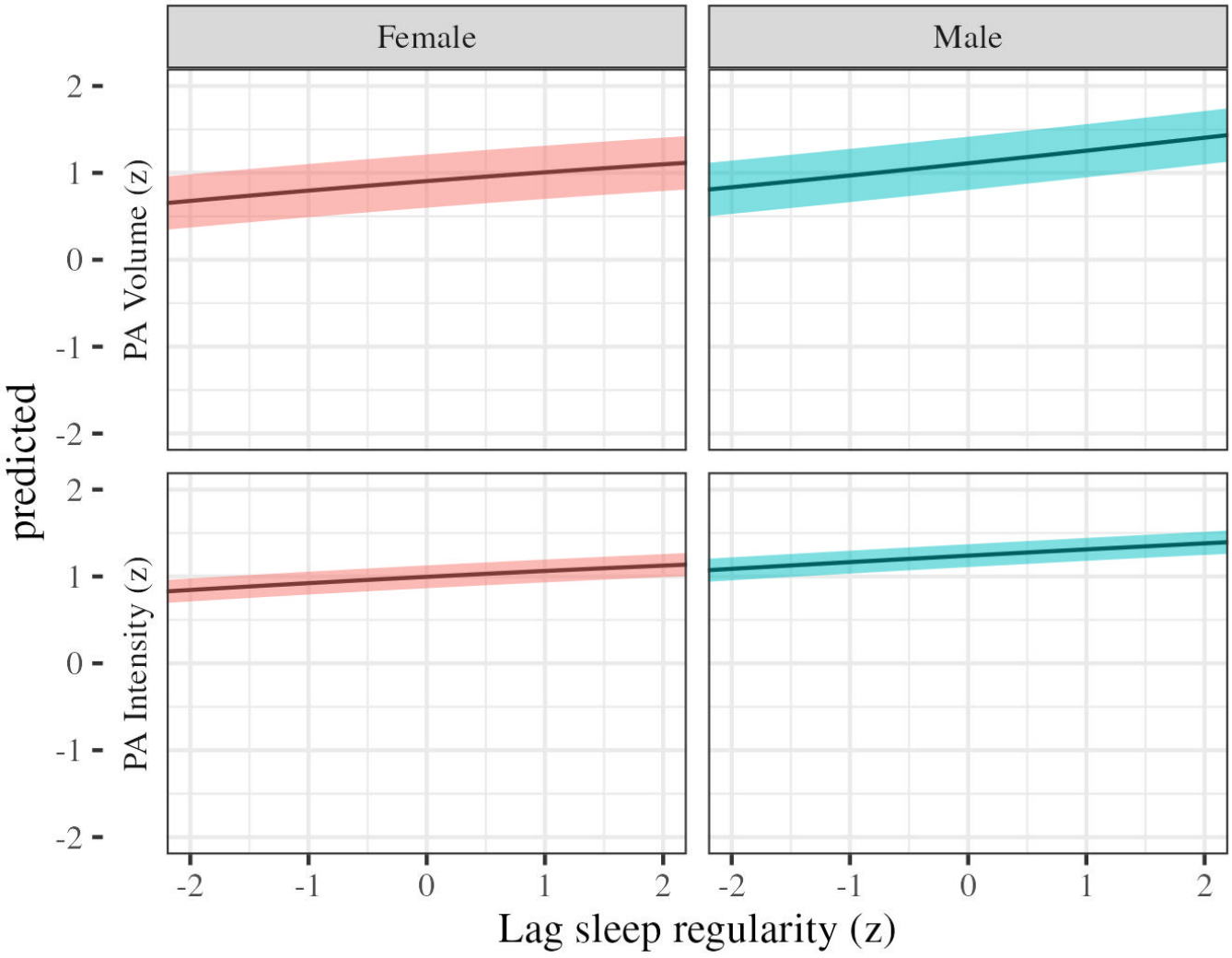

Figure 41. Physical activity by sleep regularity moderated by sex

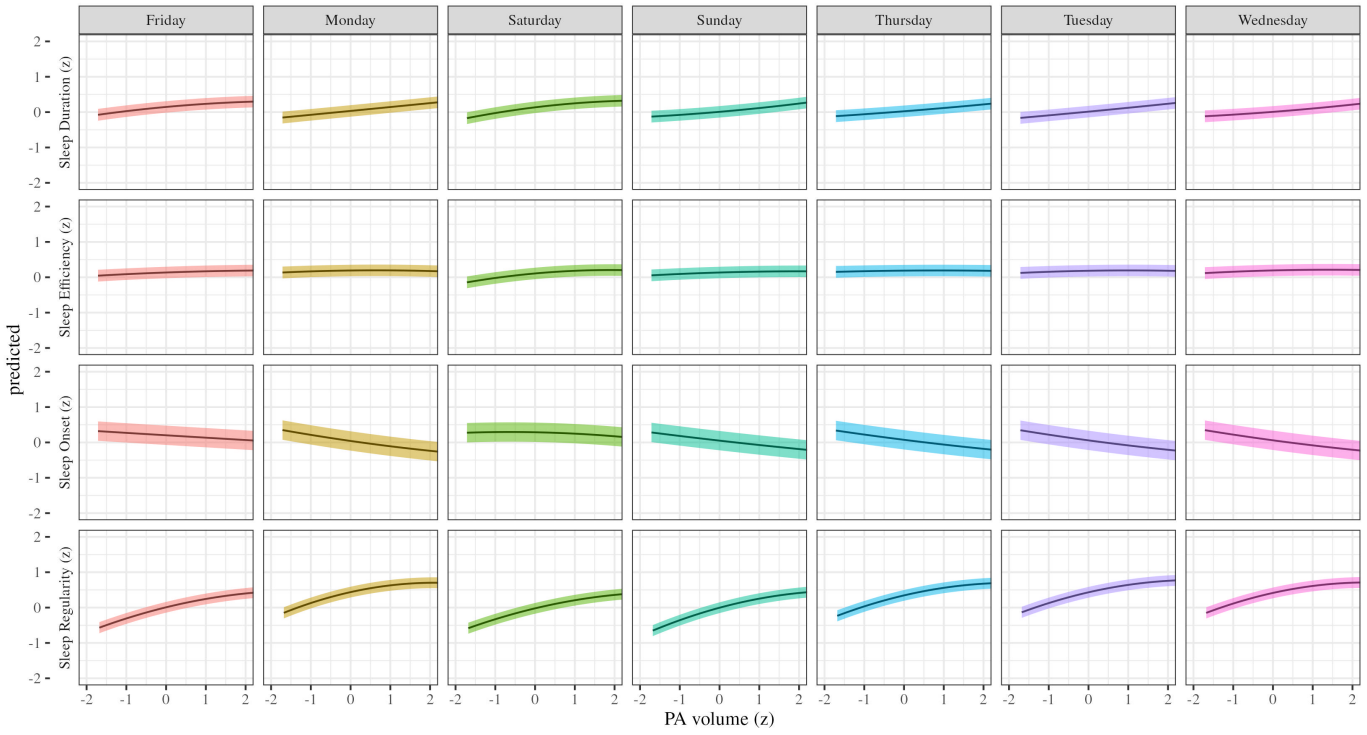

Figure 42. Sleep metrics on Physical activity volume by weekday

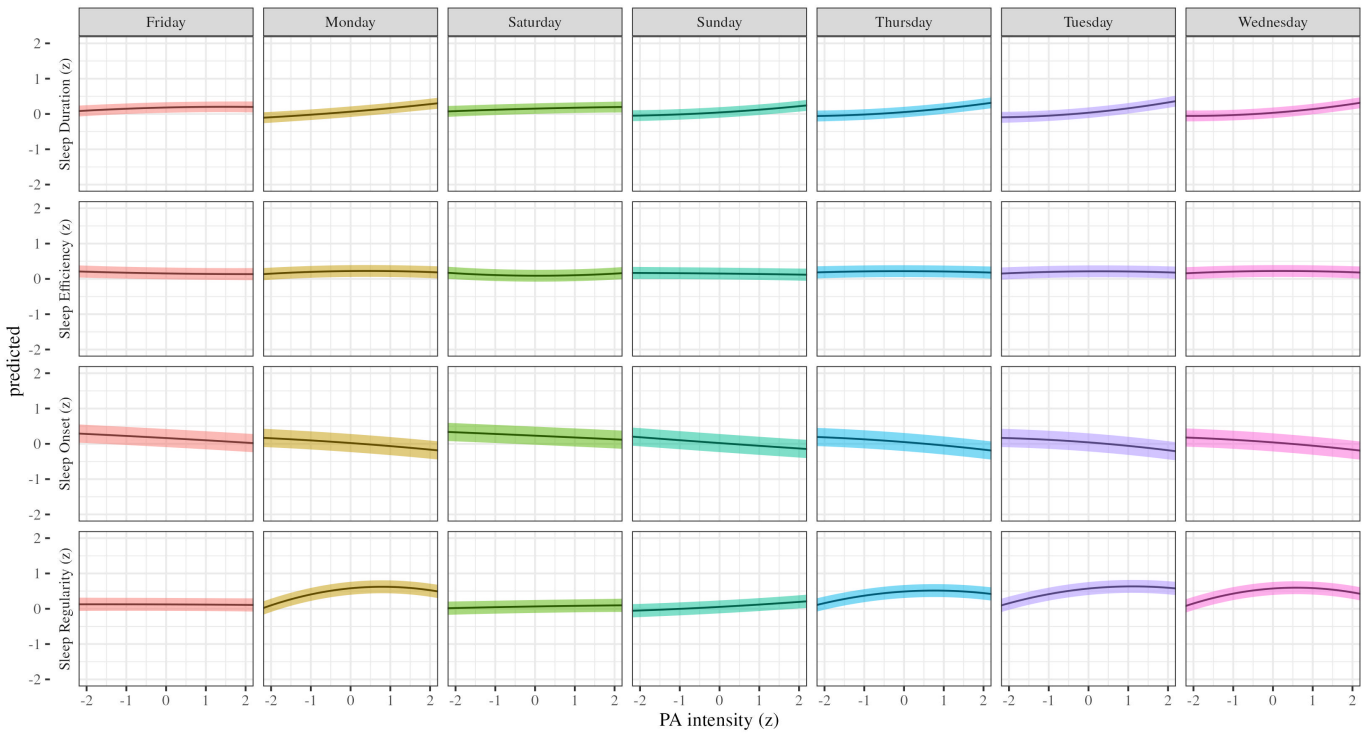

Figure 43. Sleep metrics on Physical activity intensity moderated by weekday

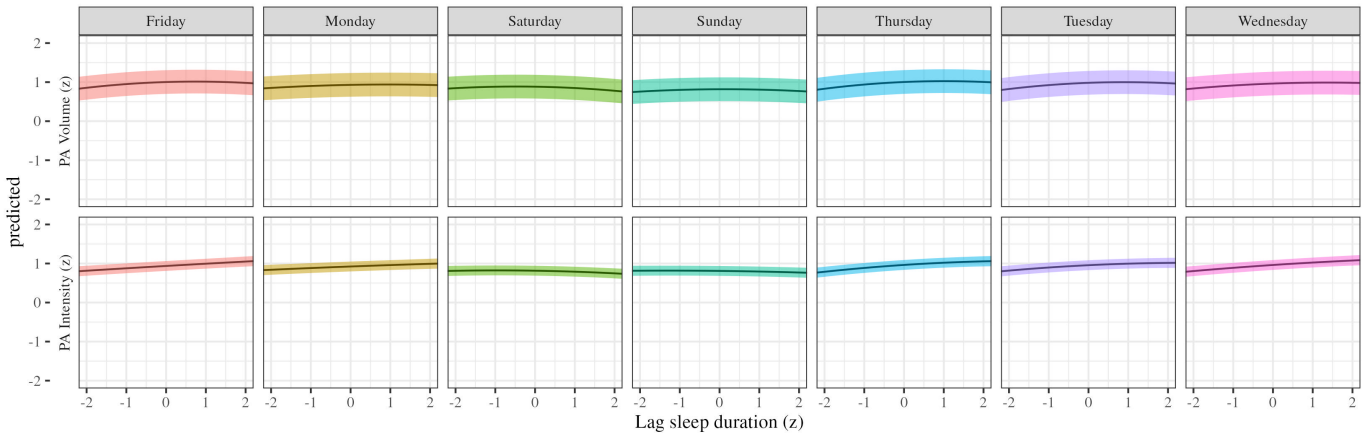

Figure 44. Physical activity by sleep duration moderated by weekday

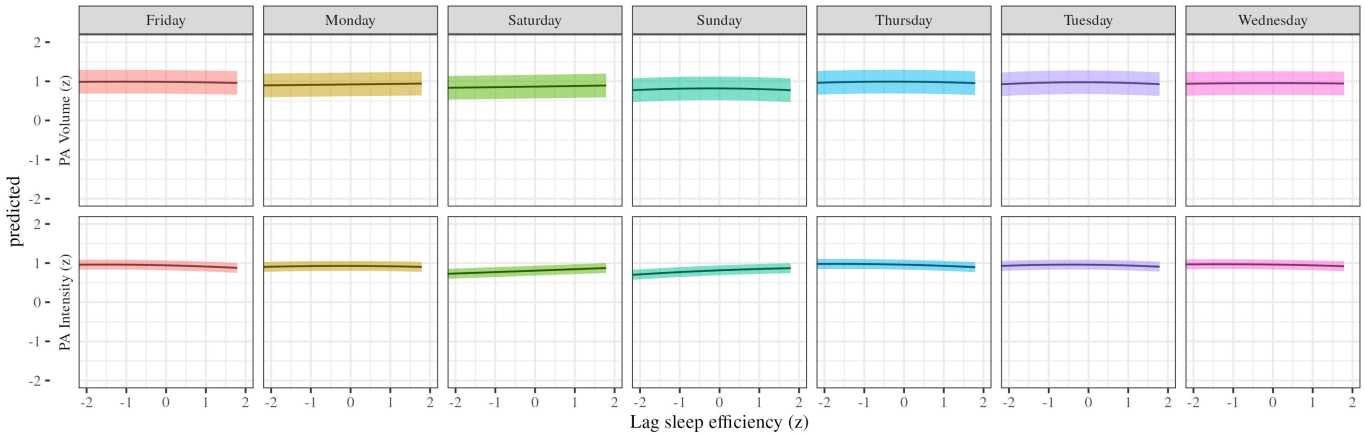

Figure 45. Physical activity by sleep efficiency moderated by weekday

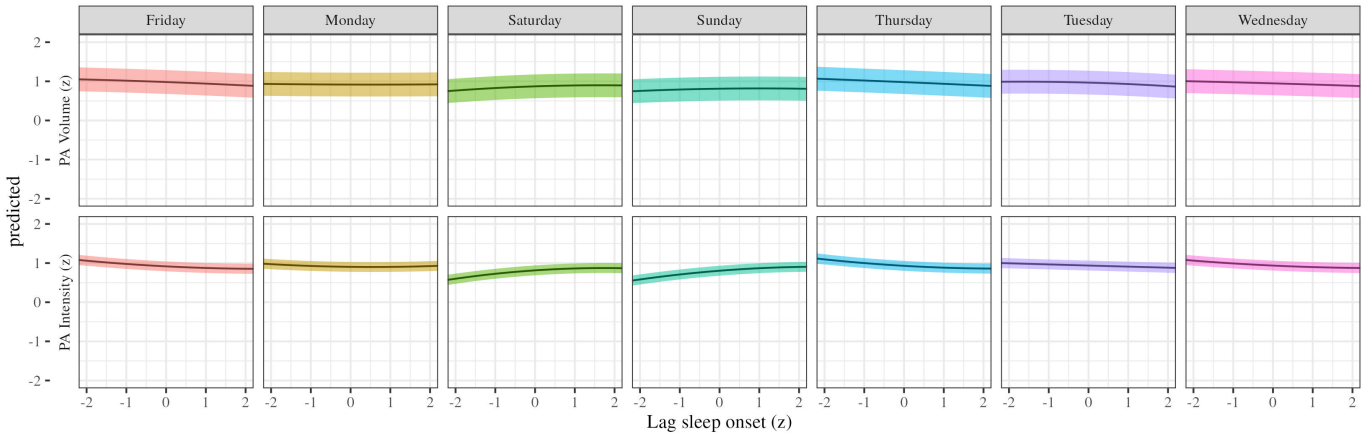

Figure 46. Physical activity by sleep onset moderated by weekday

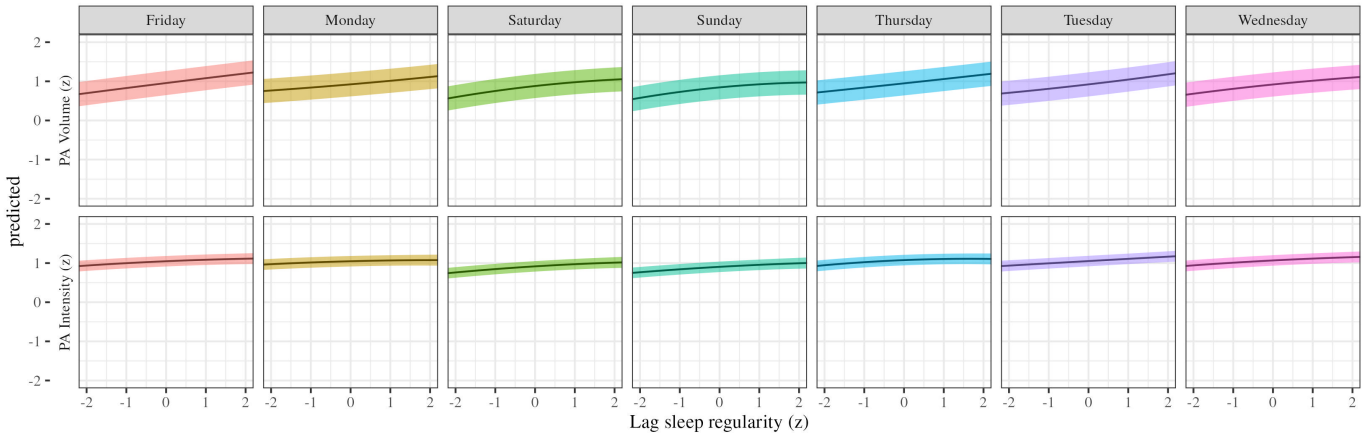

Figure 47. Physical activity by sleep regularity moderated by weekday

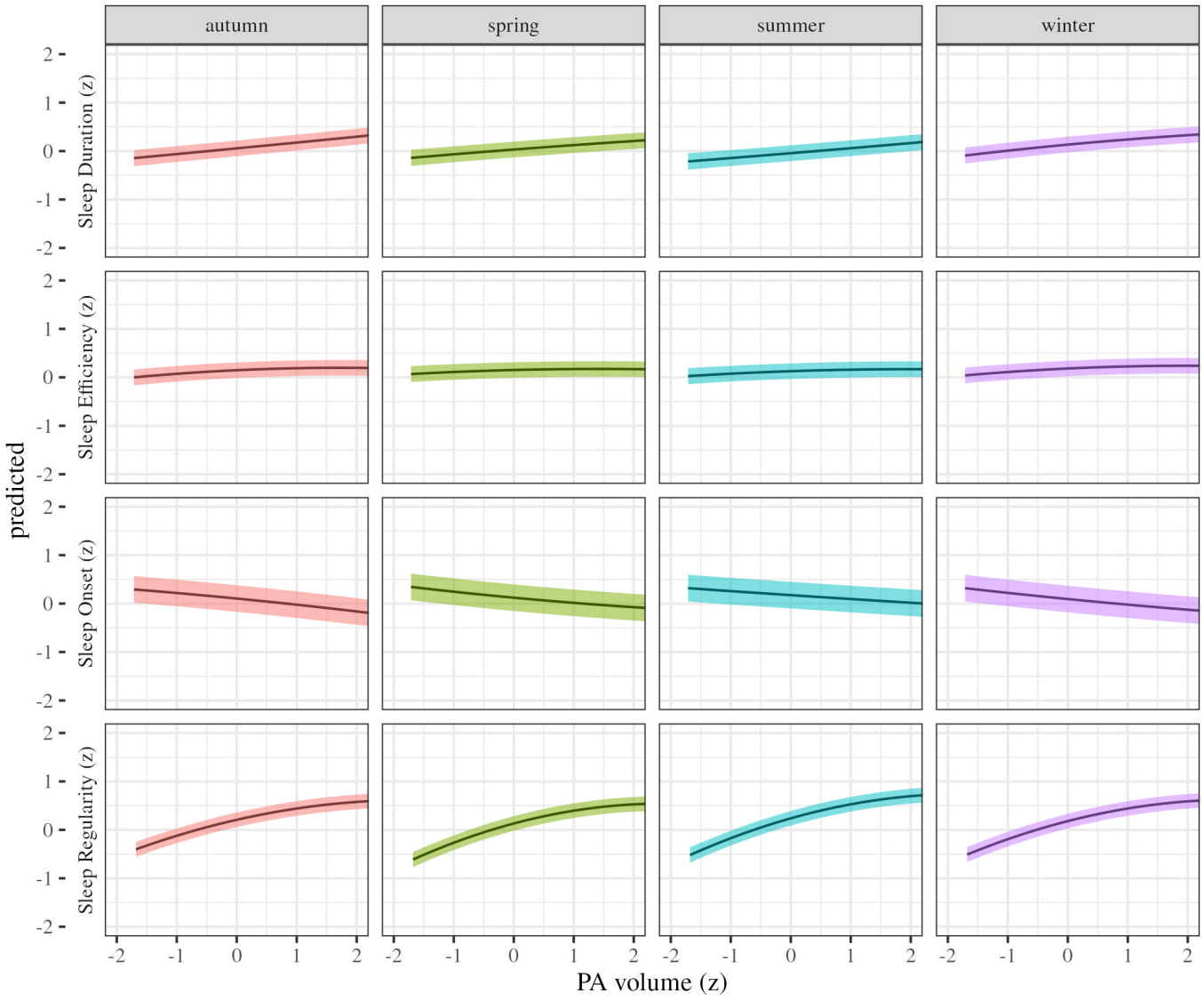

Figure 48. Sleep metrics on Physical activity volume by season

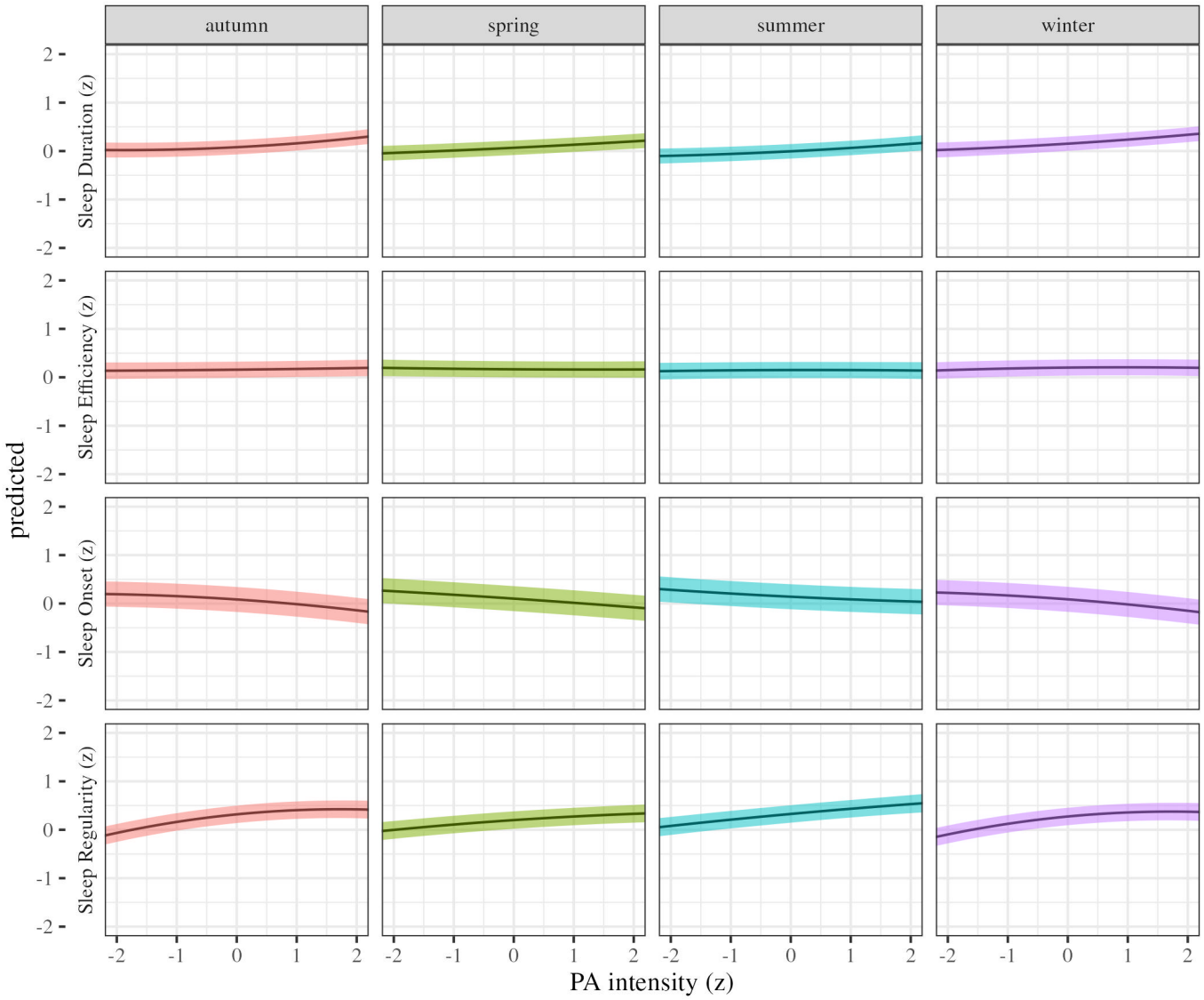

Figure 49. Sleep metrics on Physical activity intensity moderated by season

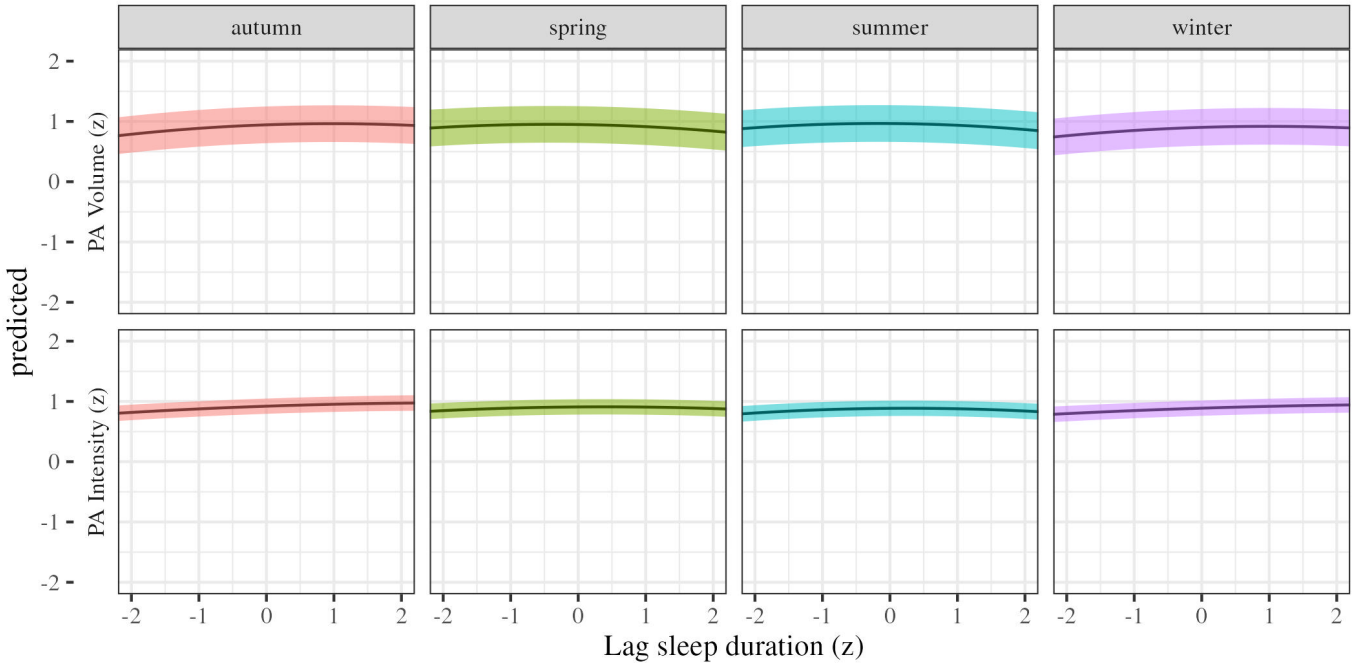

Figure 50. Physical activity by sleep duration moderated by season

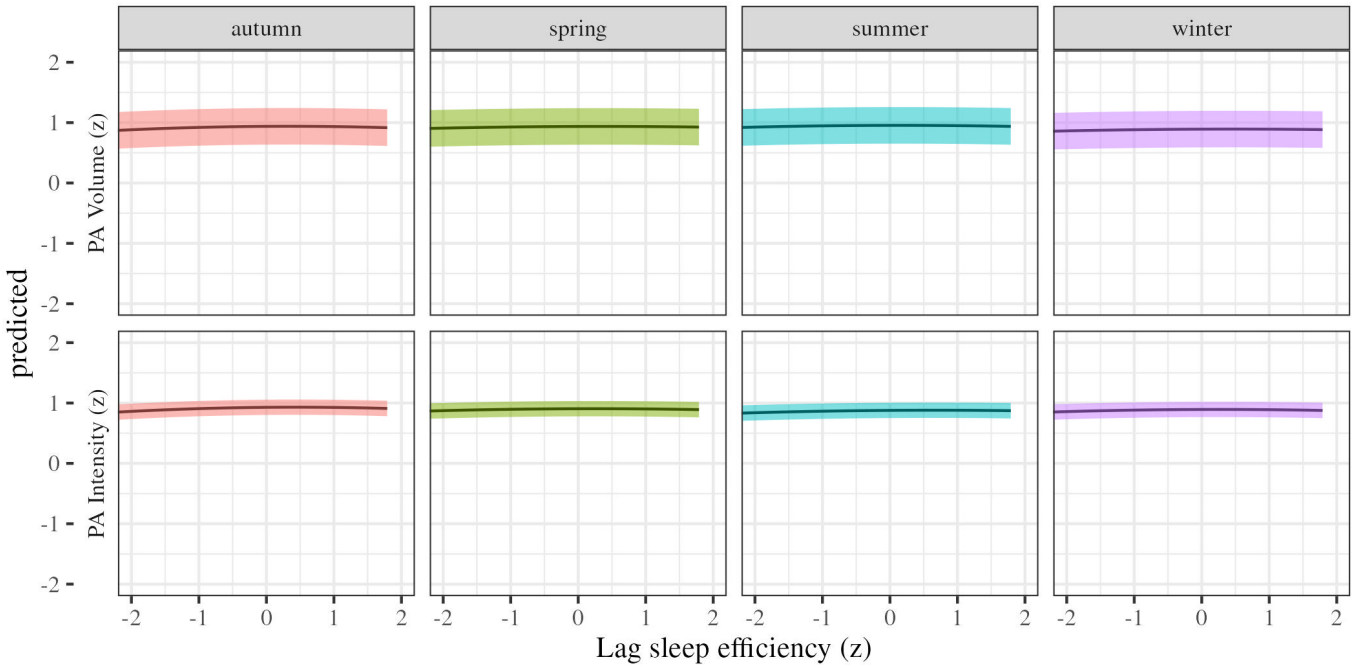

Figure 51. Physical activity by sleep efficiency moderated by season

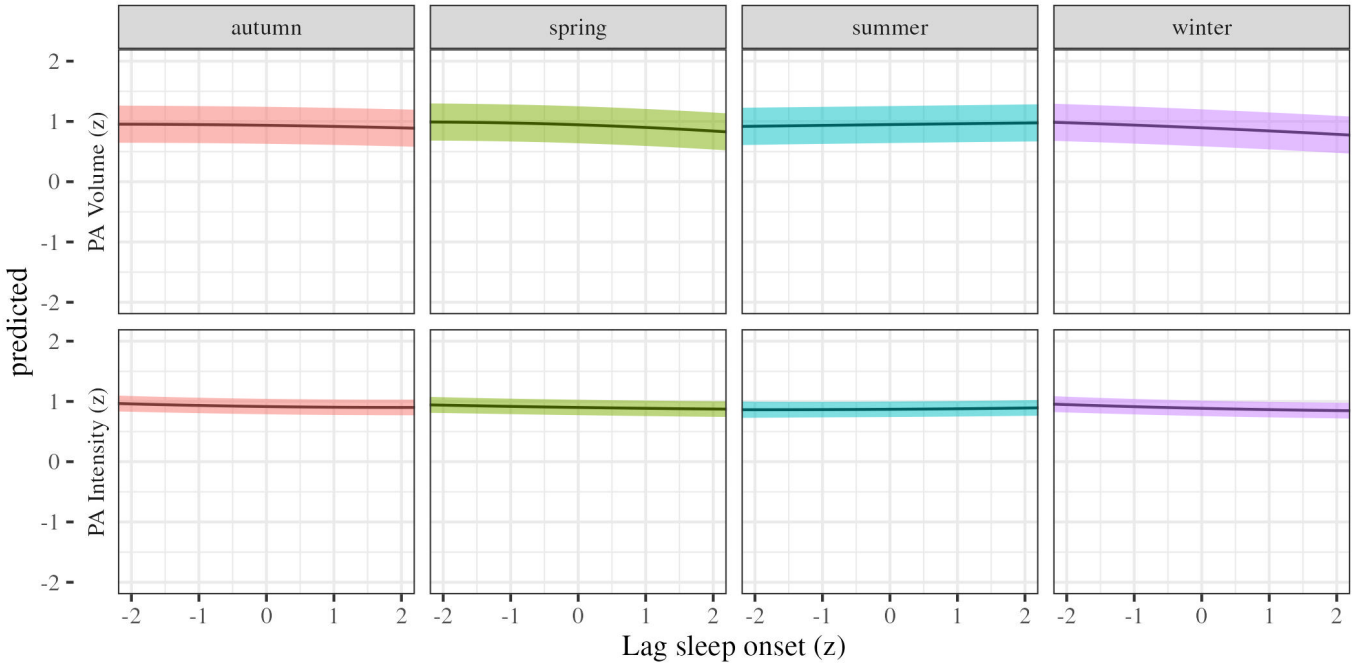

Figure 52. Physical activity by sleep onset moderated by season

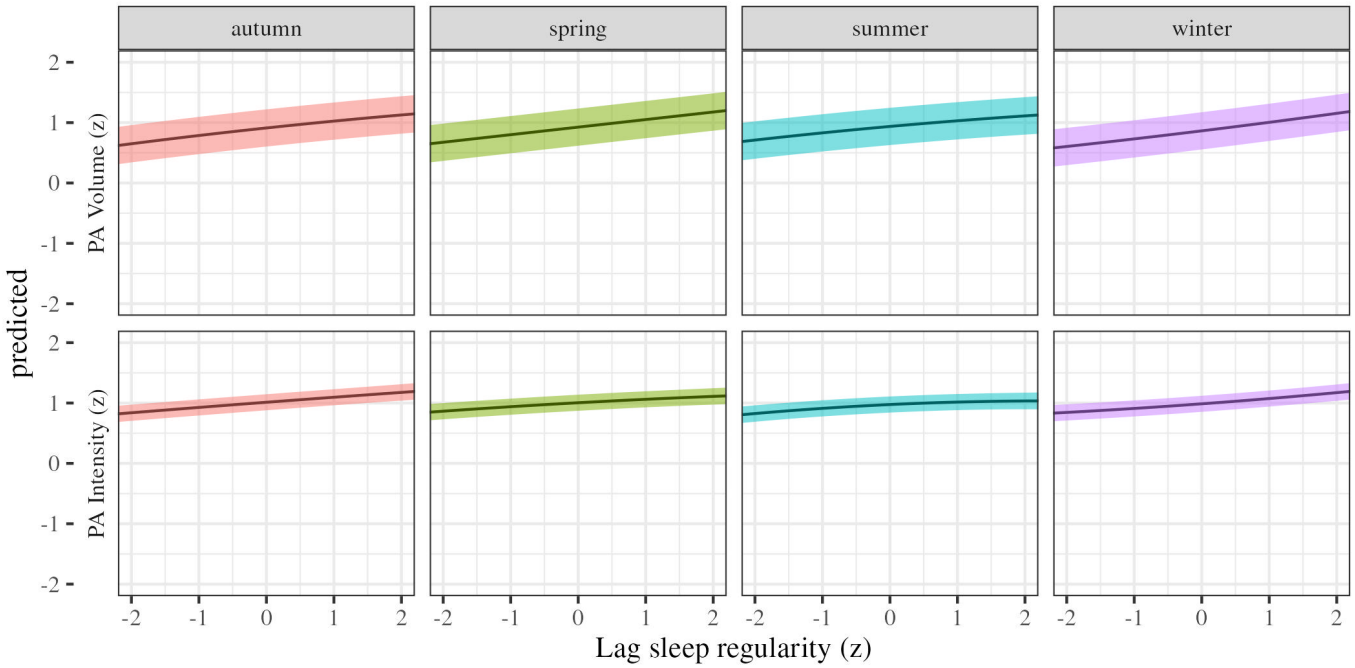

Figure 53. Physical activity by sleep regularity moderated by season

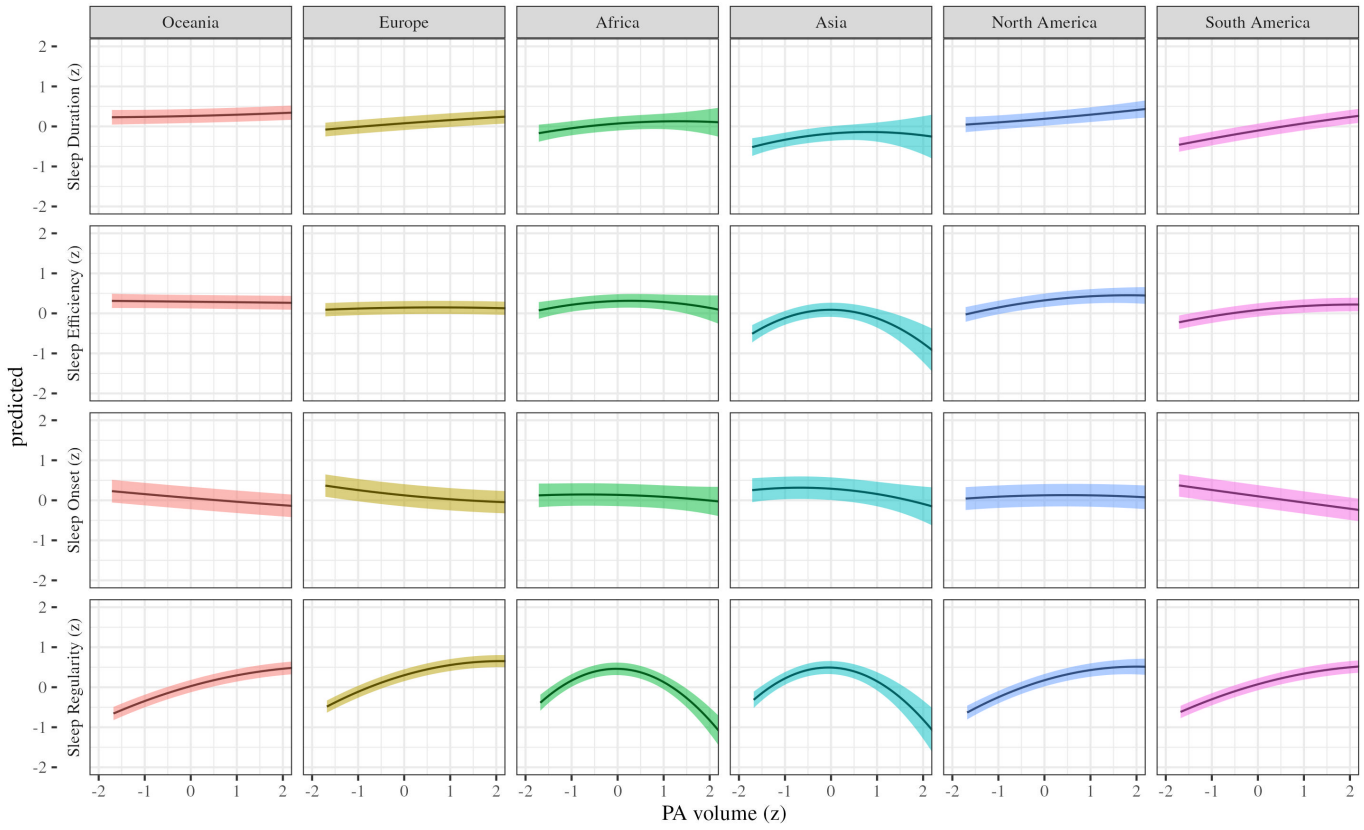

Figure 54. Sleep metrics on Physical activity volume by region

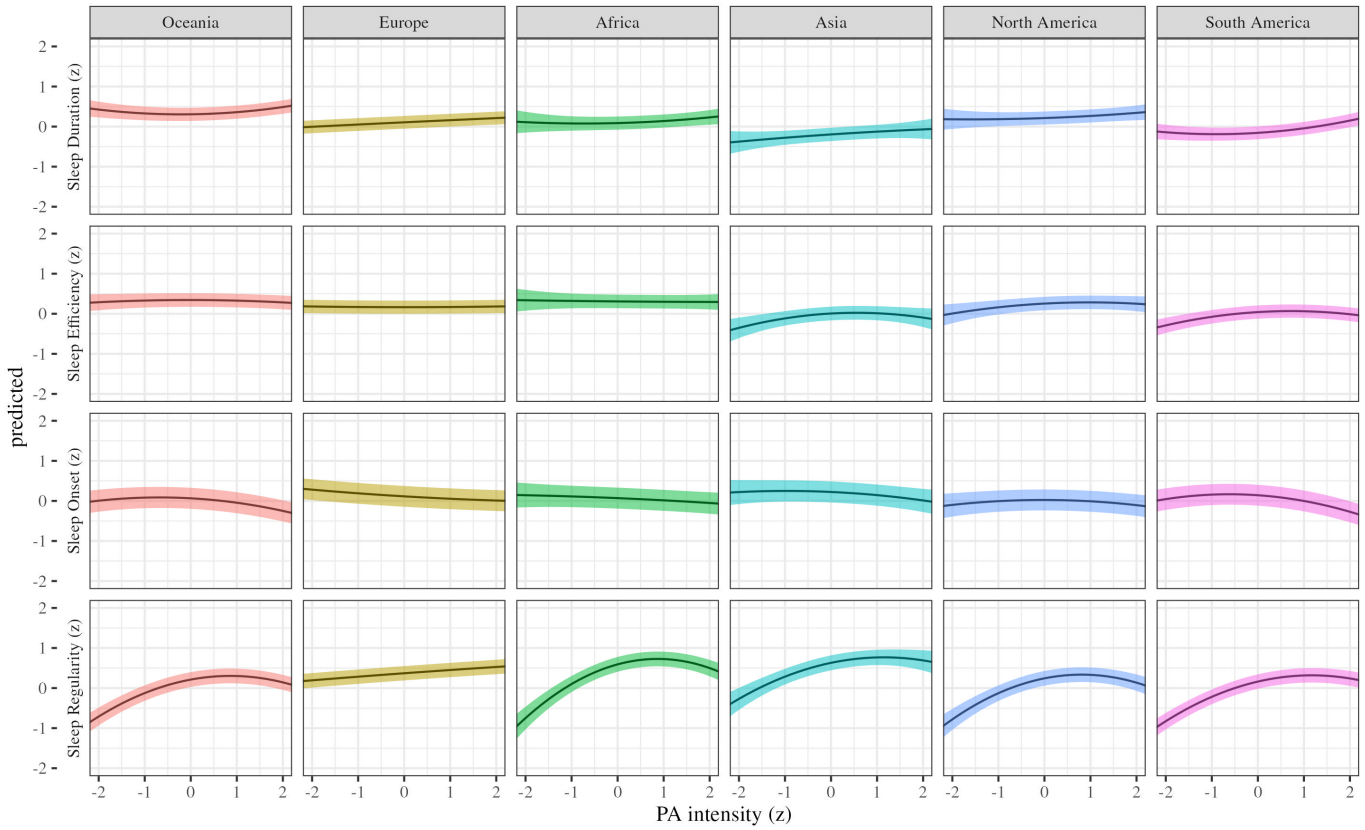

Figure 55. Sleep metrics on Physical activity intensity moderated by region

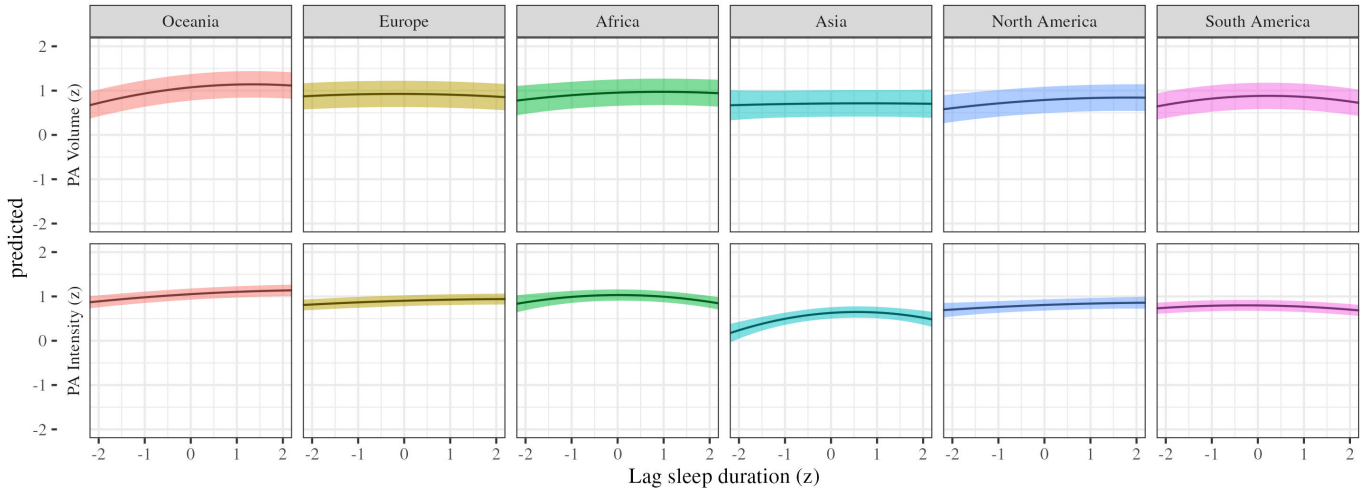

Figure 56. Physical activity by sleep duration moderated by region

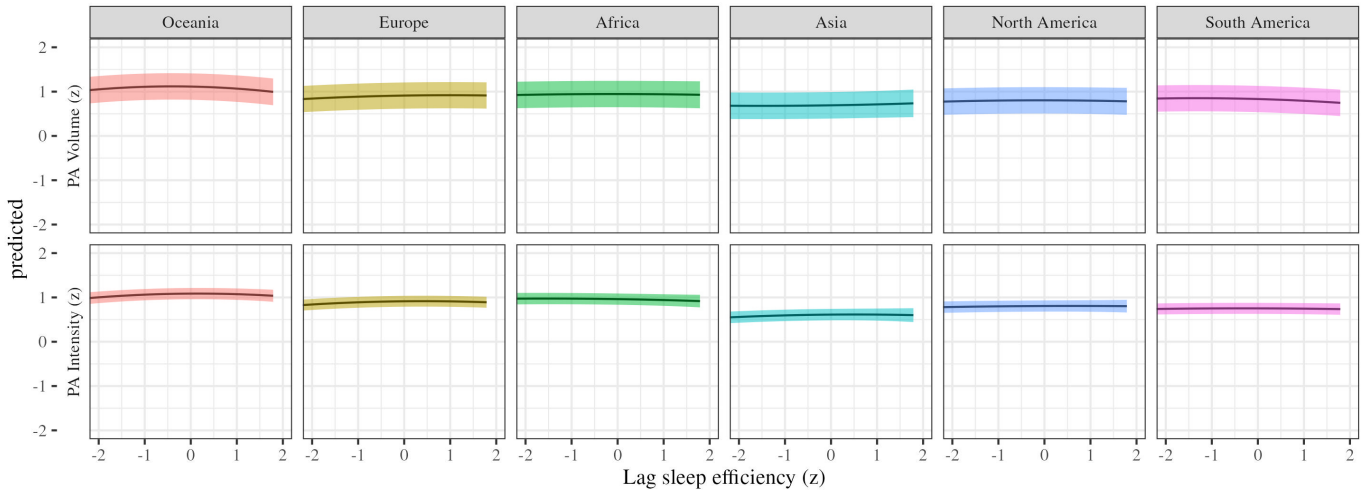

Figure 57. Physical activity by sleep efficiency moderated by region

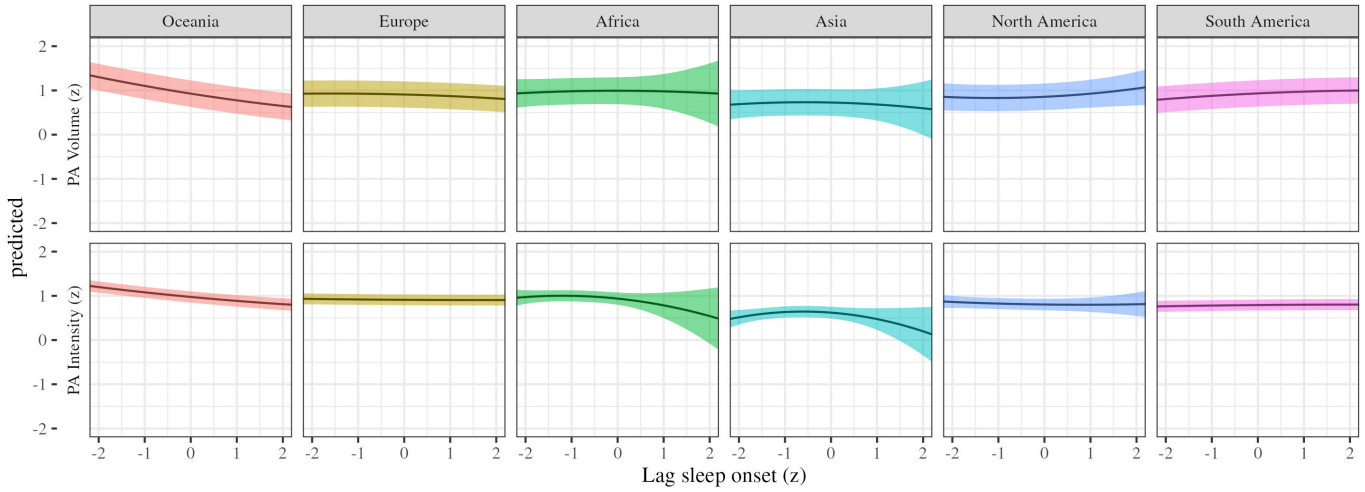

Figure 58. Physical activity by sleep onset moderated by region

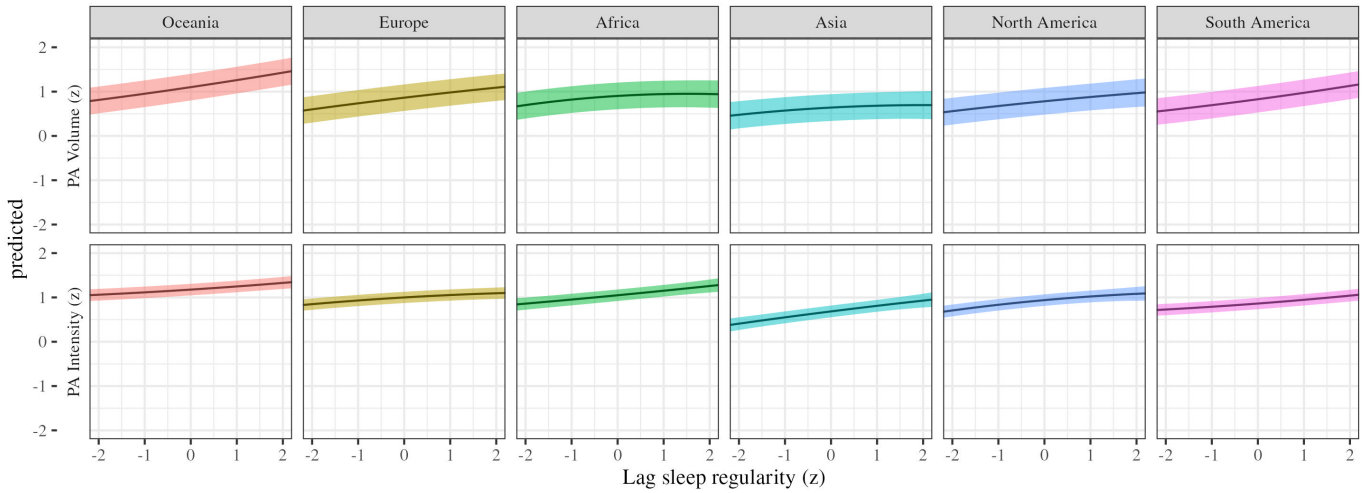

Figure 59. Physical activity by sleep regularity moderated by region

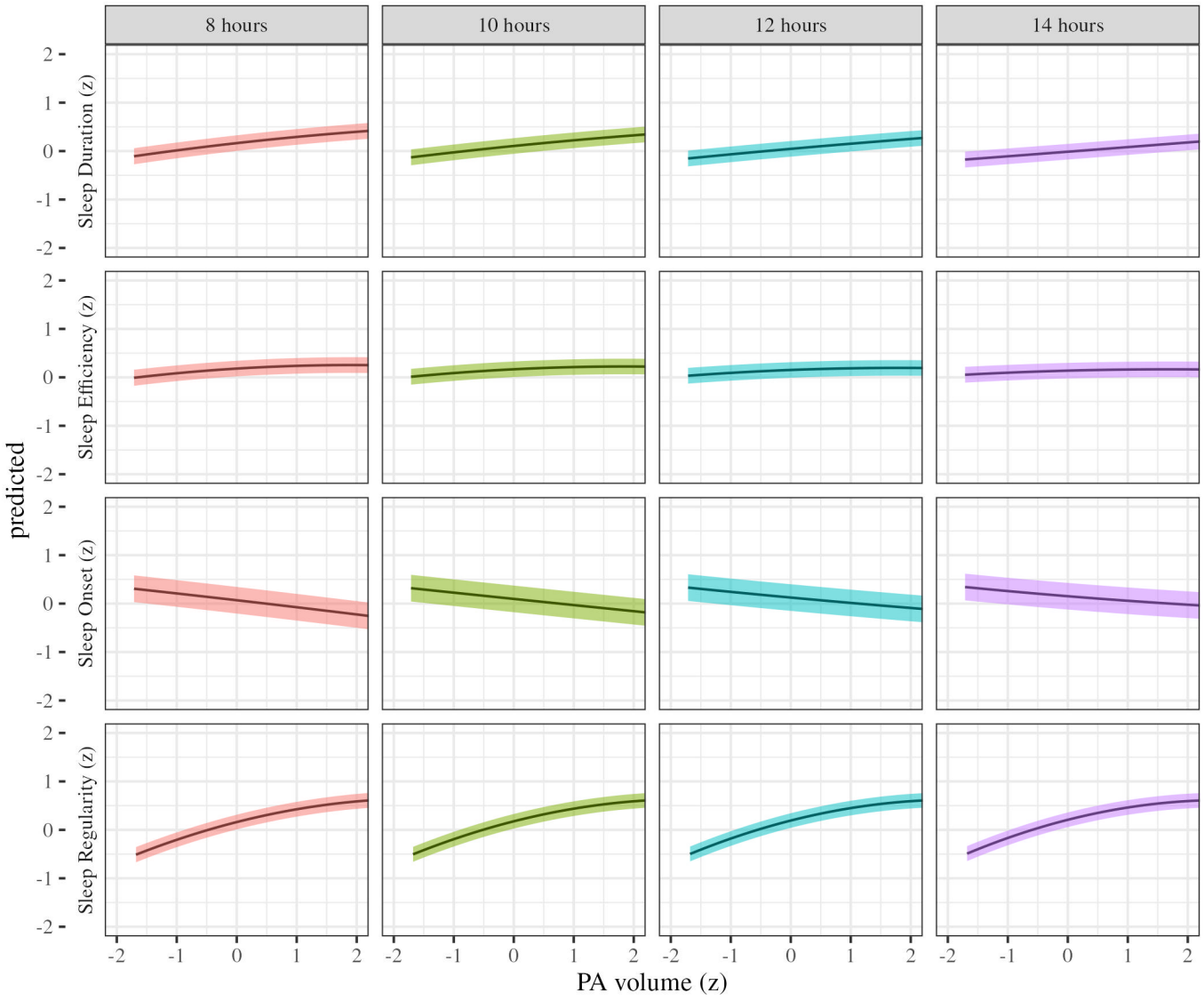

Figure 60. Sleep metrics on Physical activity volume by daylight hours

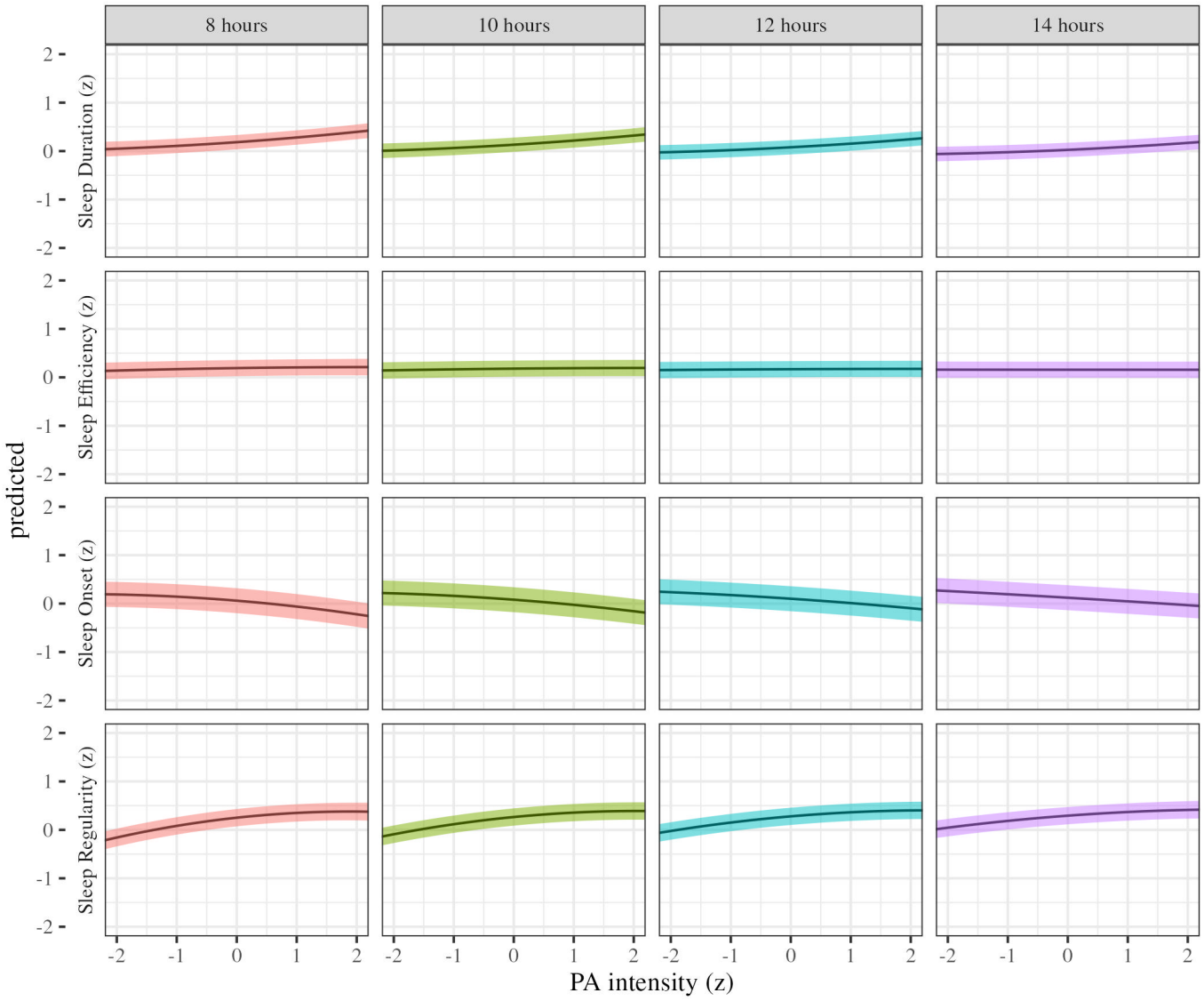

Figure 61. Sleep metrics on Physical activity intensity moderated by daylight hours

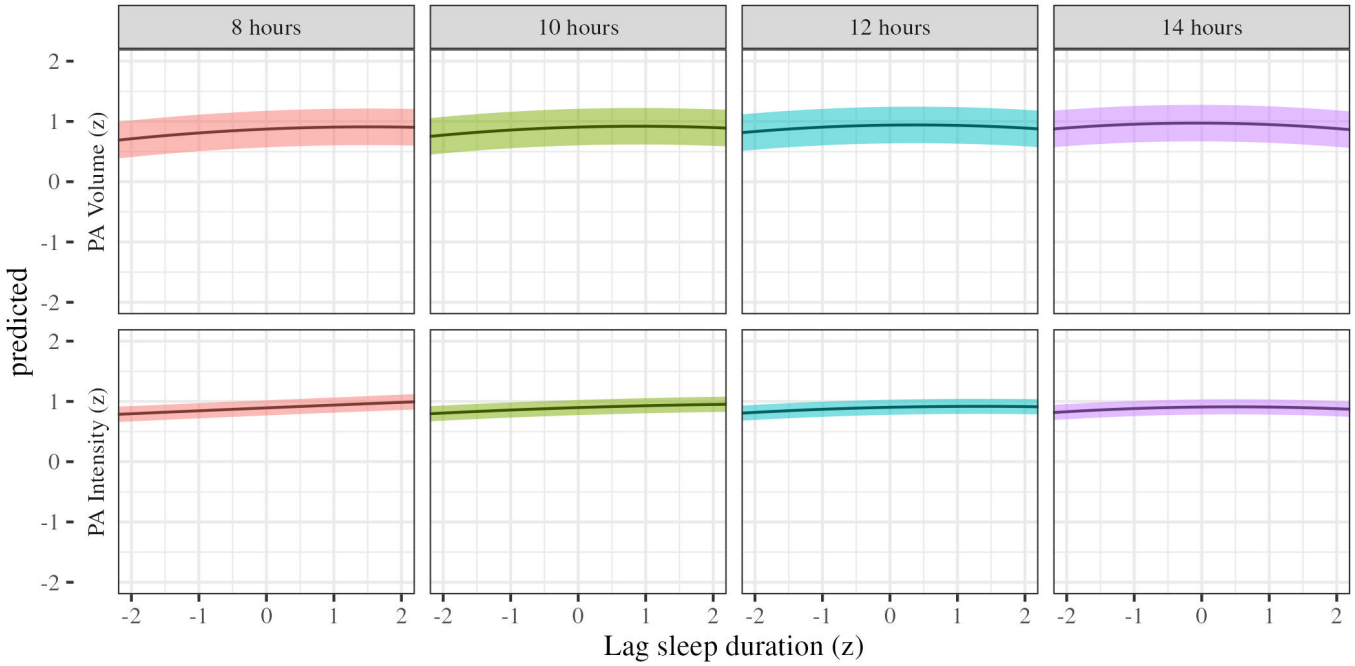

Figure 62. Physical activity by sleep duration moderated by daylight hours

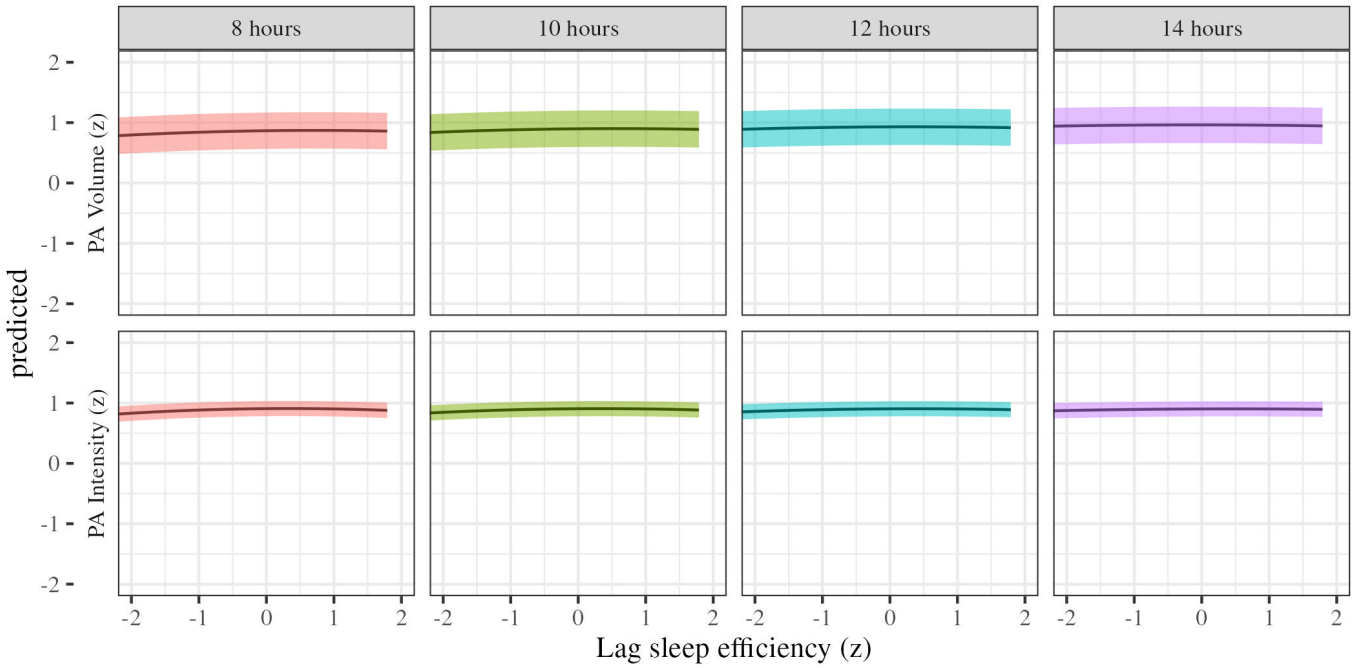

Figure 63. Physical activity by sleep efficiency moderated by daylight hours

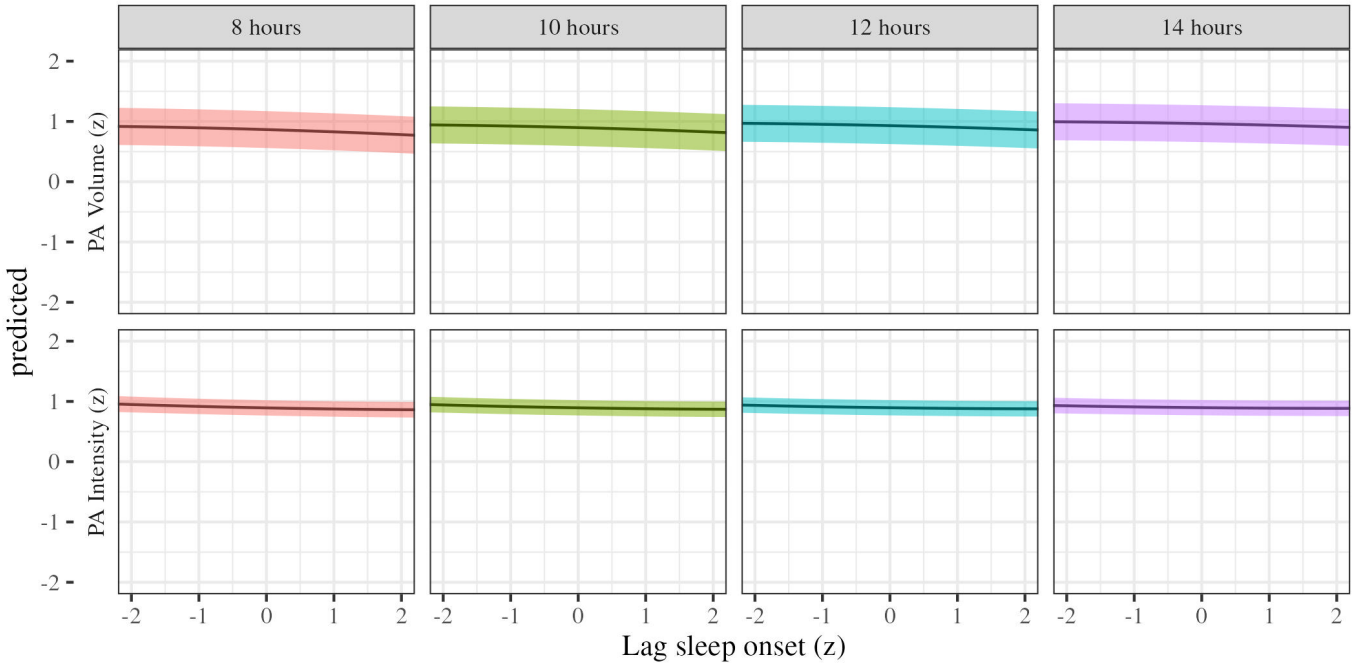

Figure 64. Physical activity by sleep onset moderated by daylight hours

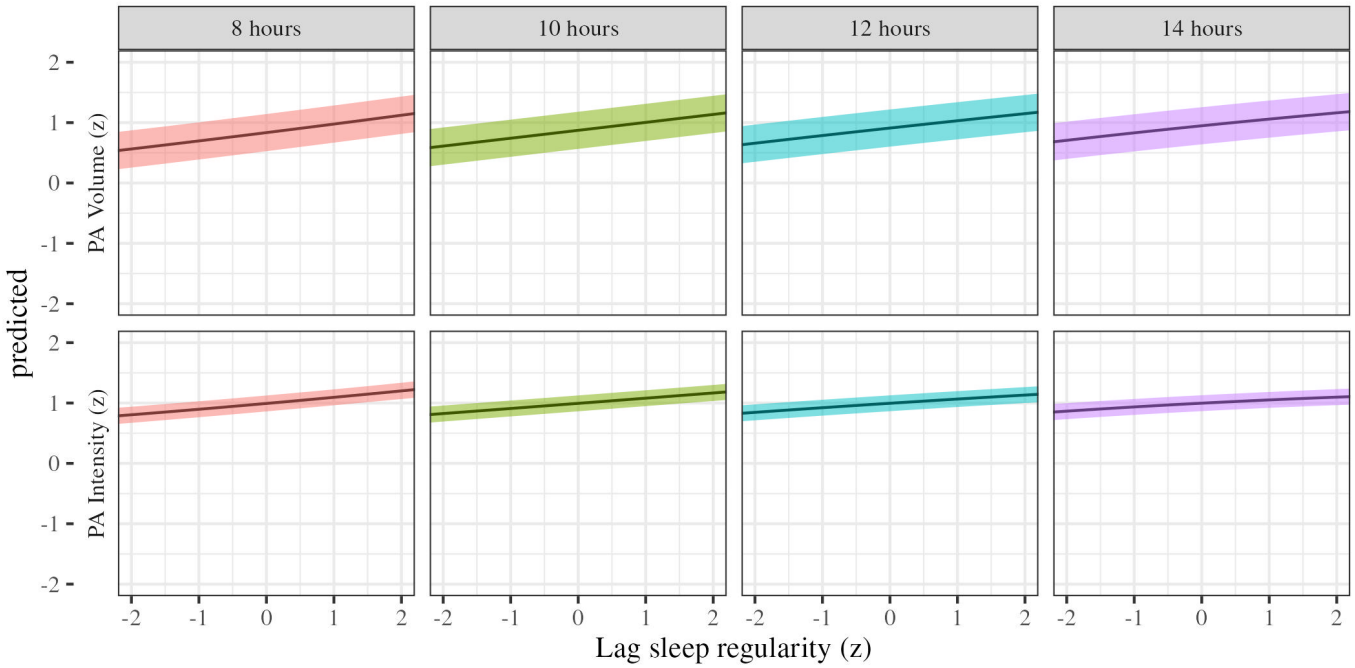

Figure 65. Physical activity by sleep regularity moderated by daylight hours

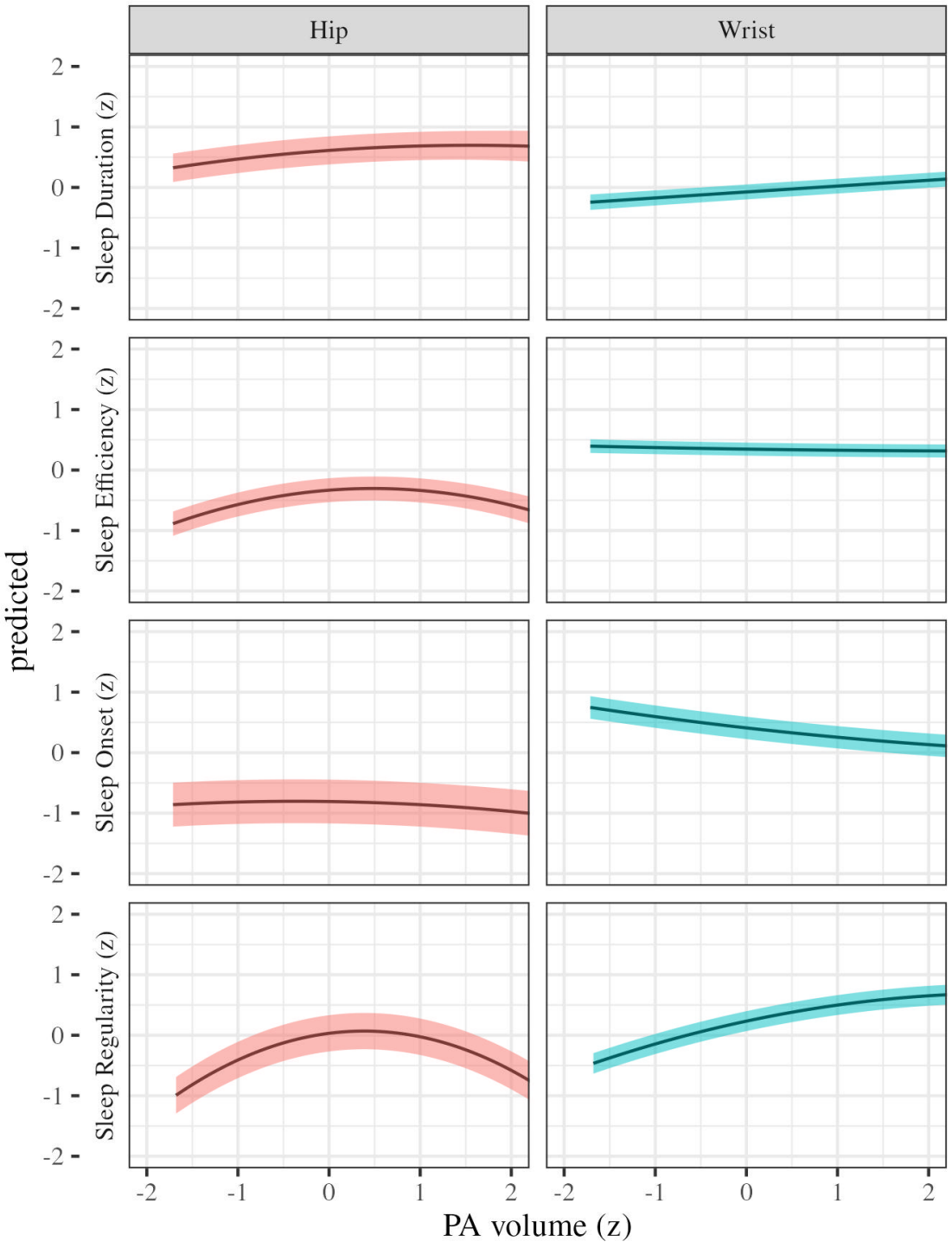

Figure 66. Sleep metrics on Physical activity volume by wear location

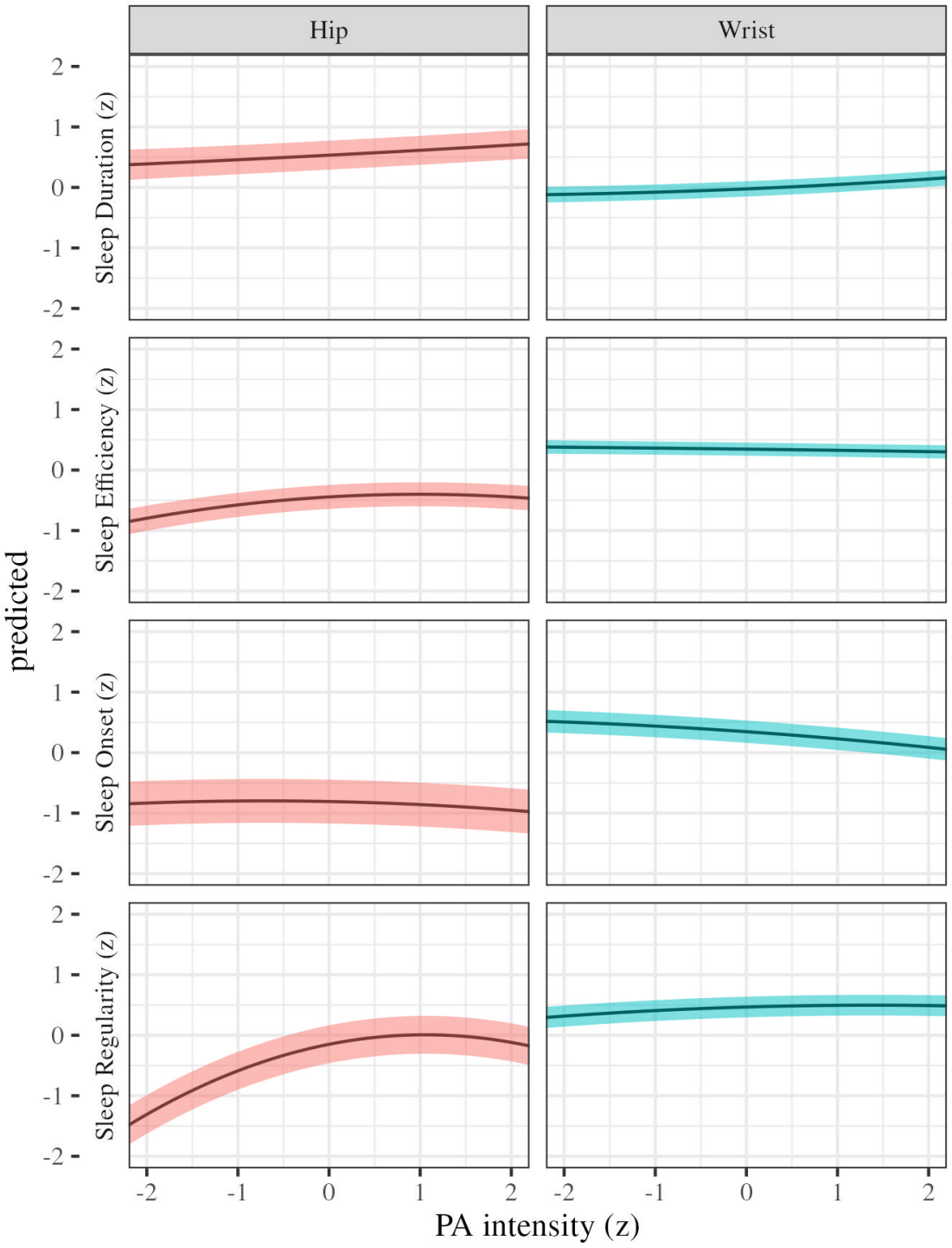

Figure 67. Sleep metrics on Physical activity intensity moderated by wear location

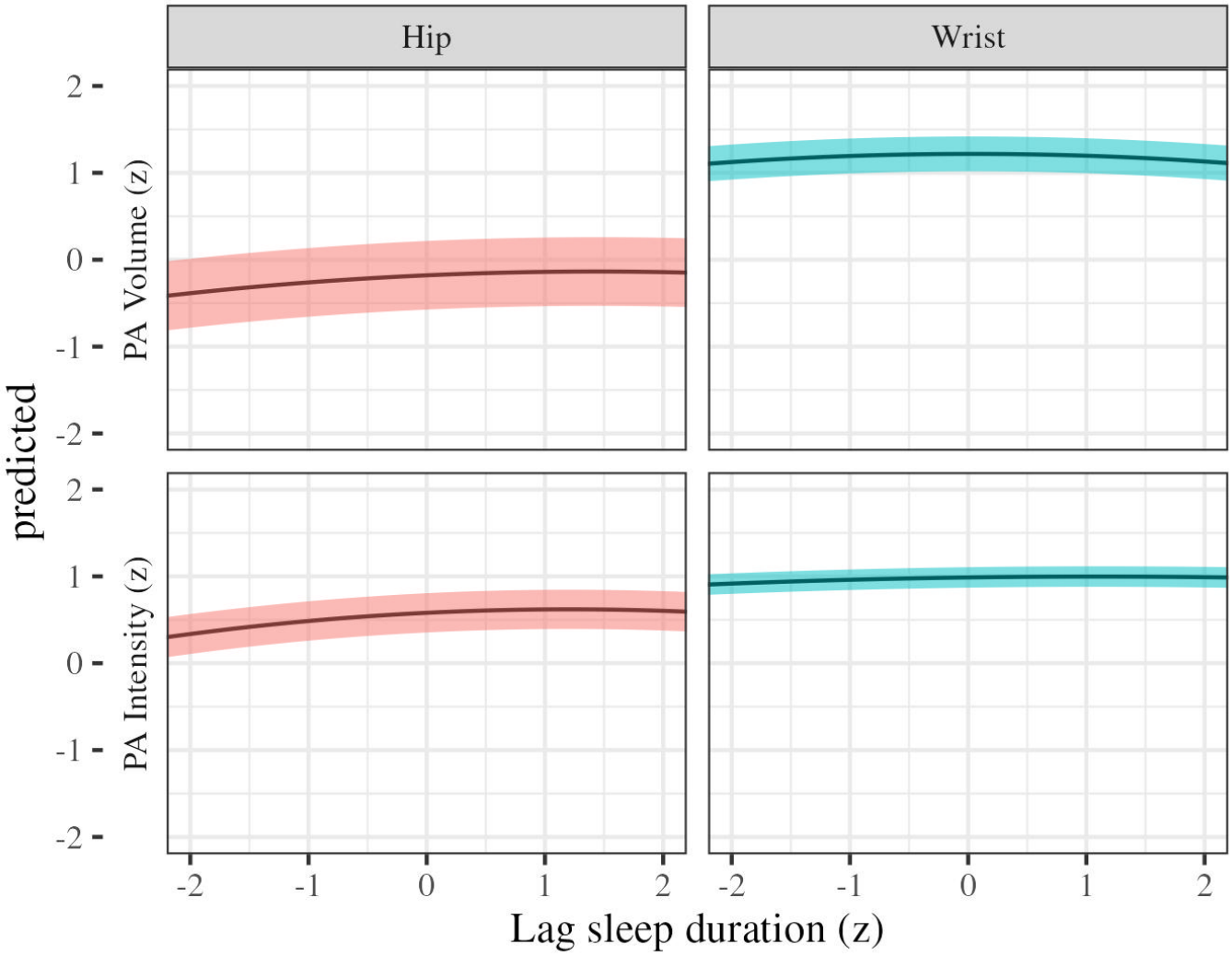

Figure 68. Physical activity by sleep duration moderated by wear location

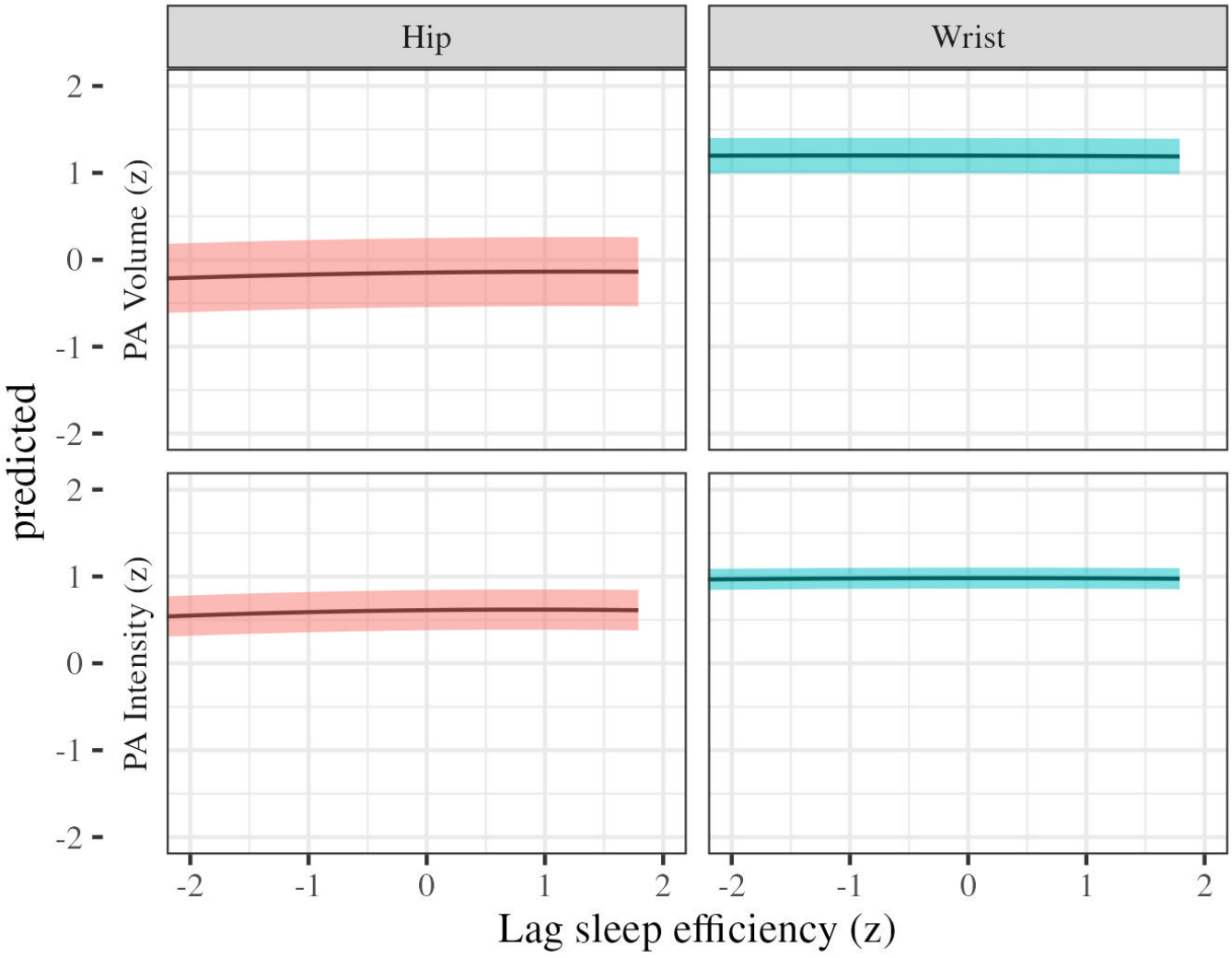

Figure 69. Physical activity by sleep efficiency moderated by wear location

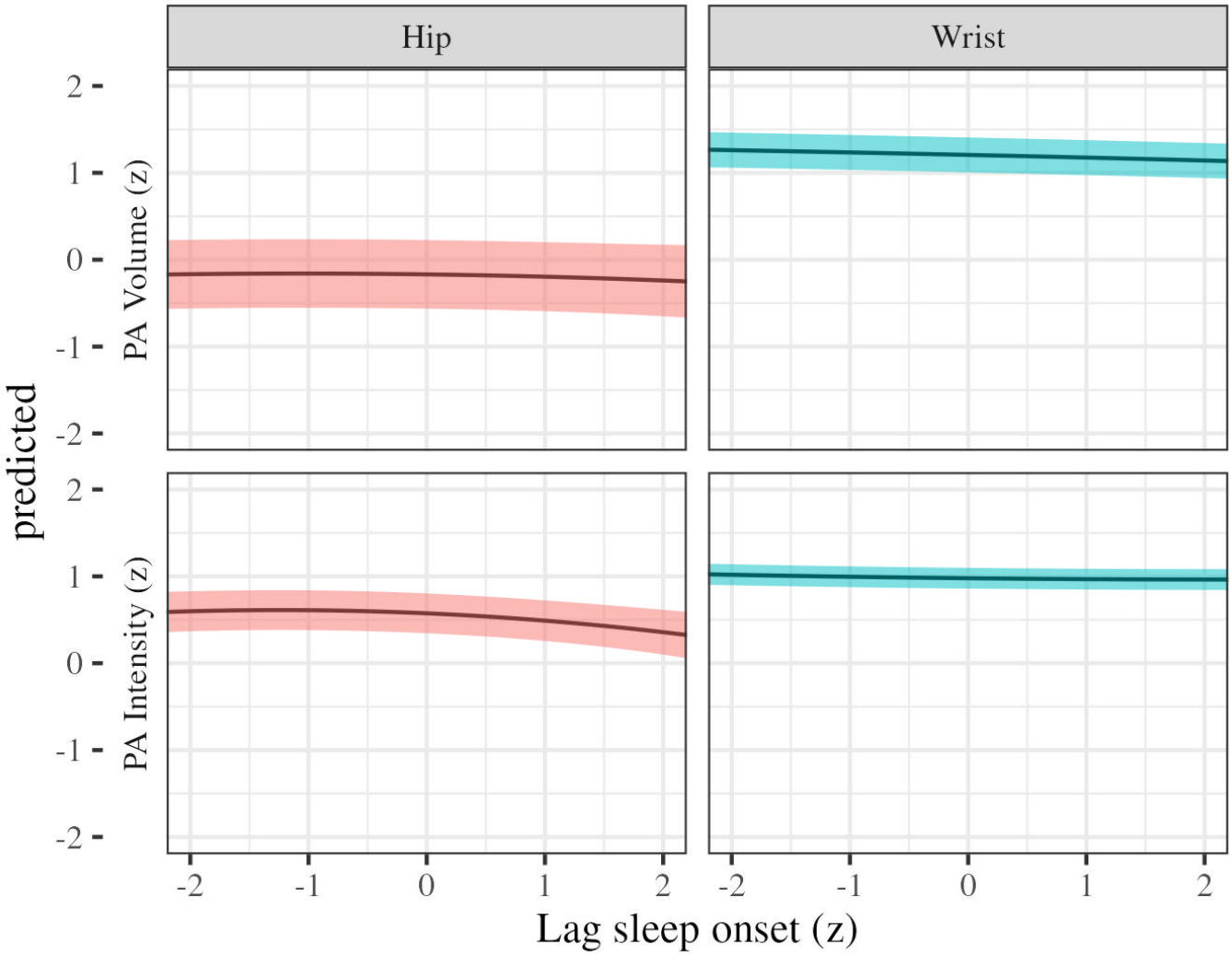

Figure 70. Physical activity by sleep onset moderated by wear location

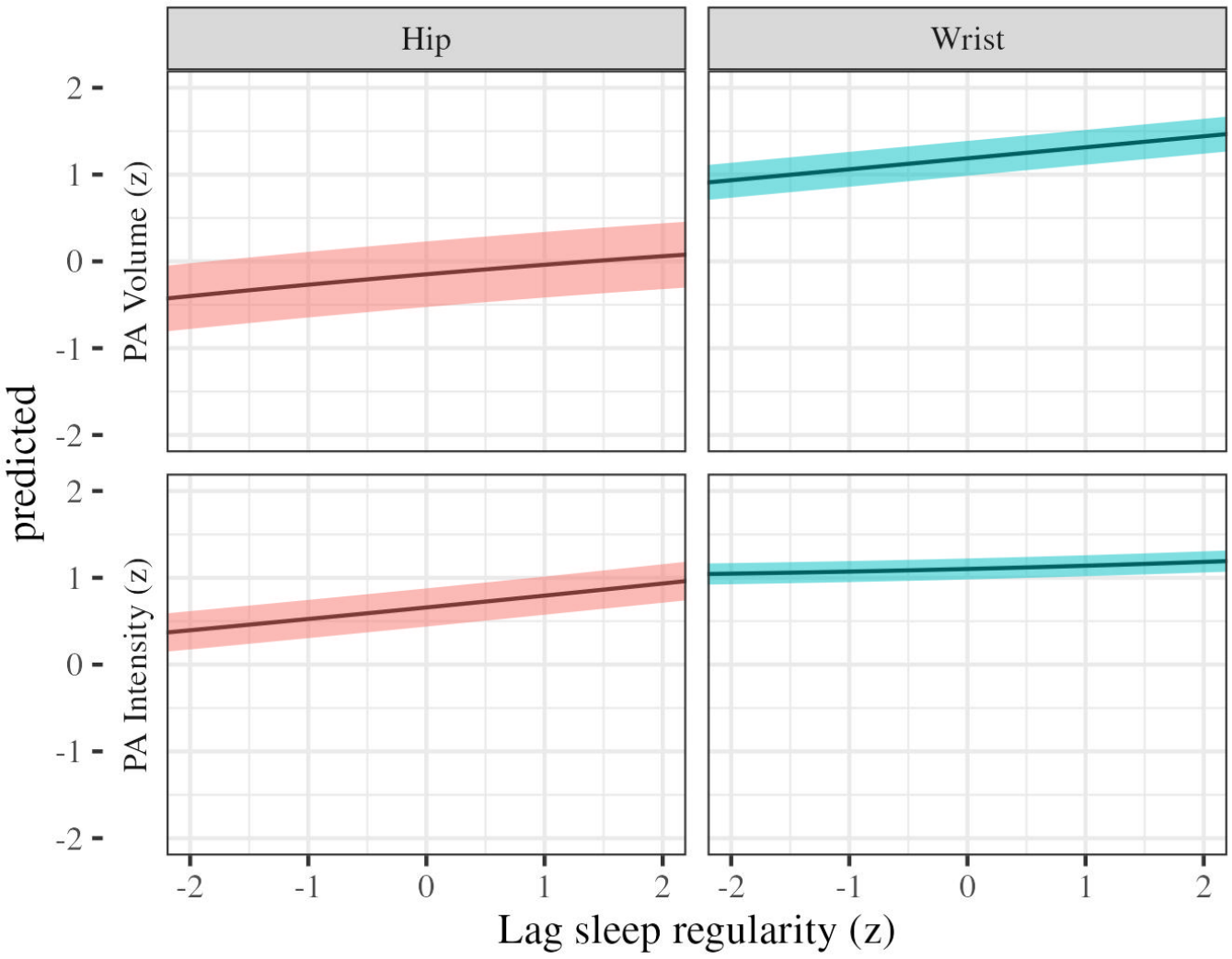

Figure 71. Physical activity by sleep regularity moderated by wear location

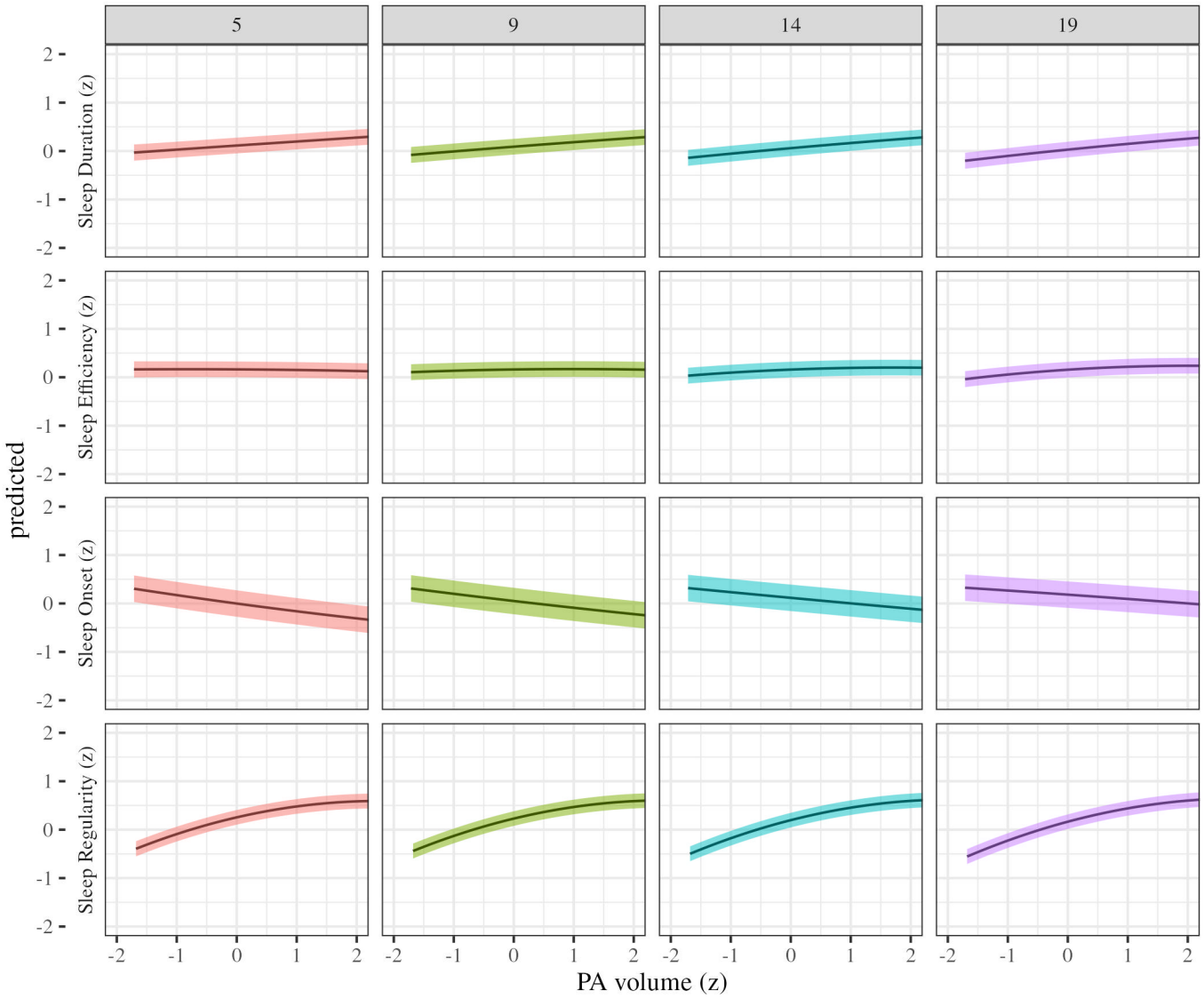

Figure 72. Sleep metrics on Physical activity volume by most active hour

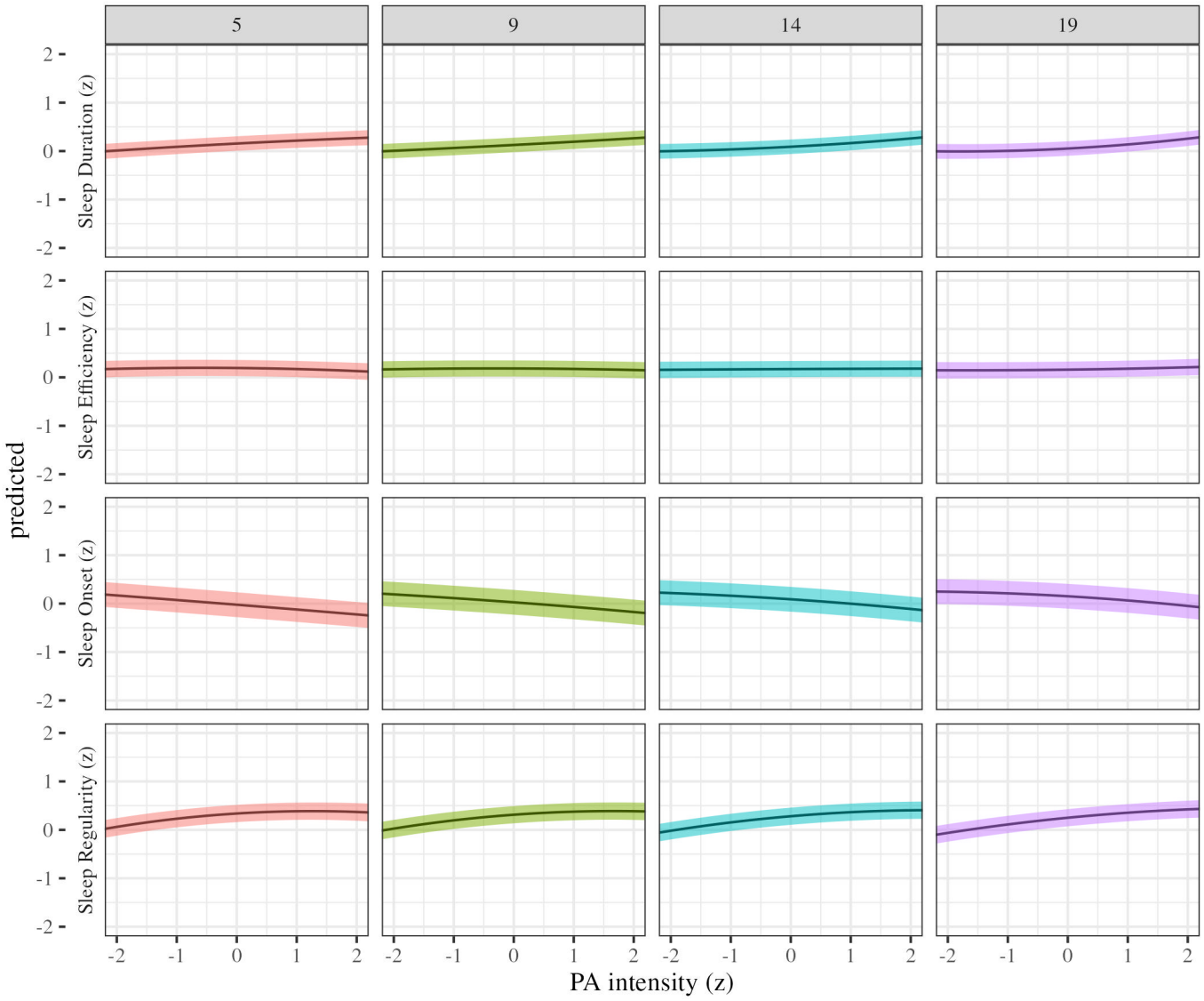

Figure 73. Sleep metrics on Physical activity intensity moderated by most active hour

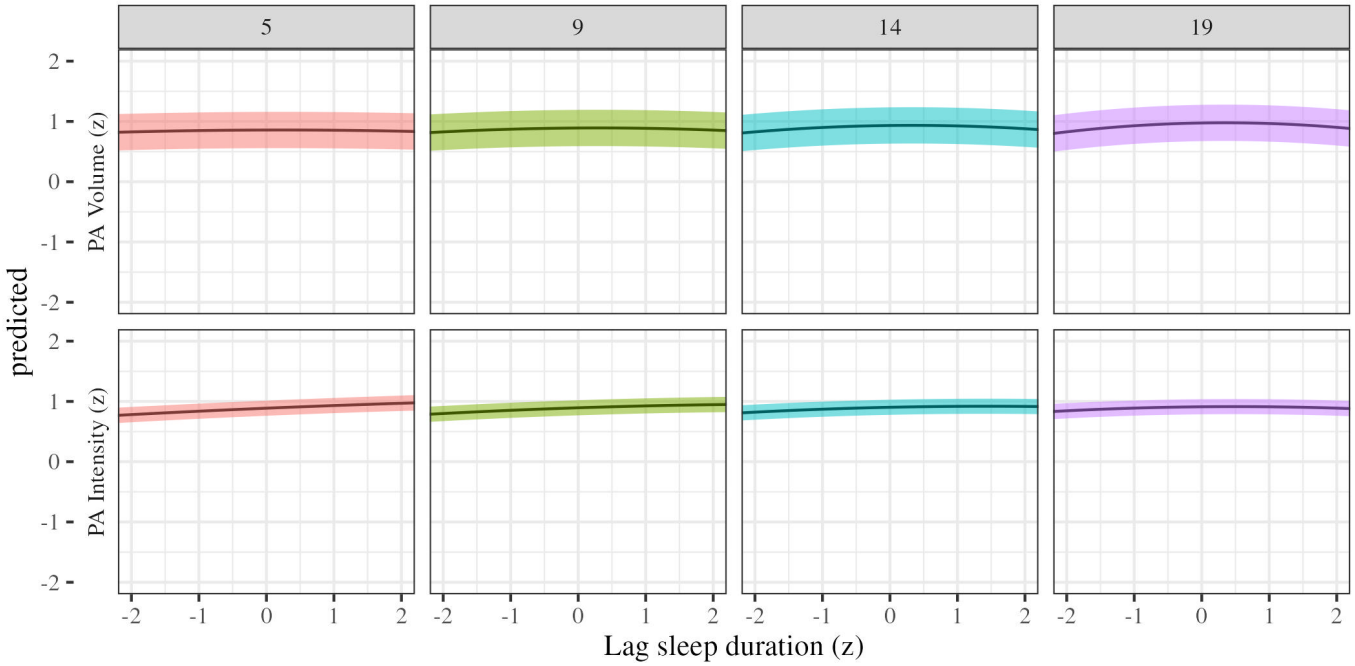

Figure 74. Physical activity by sleep duration moderated by most active hour

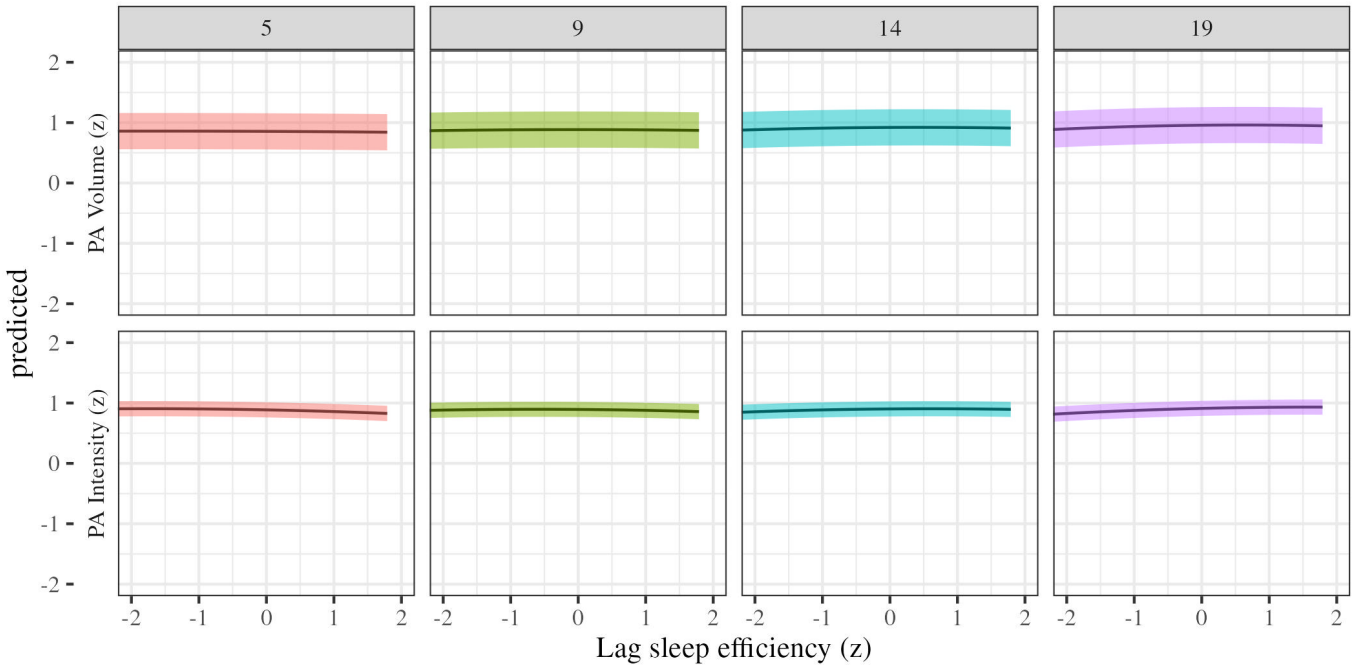

Figure 75. Physical activity by sleep efficiency moderated by most active hour

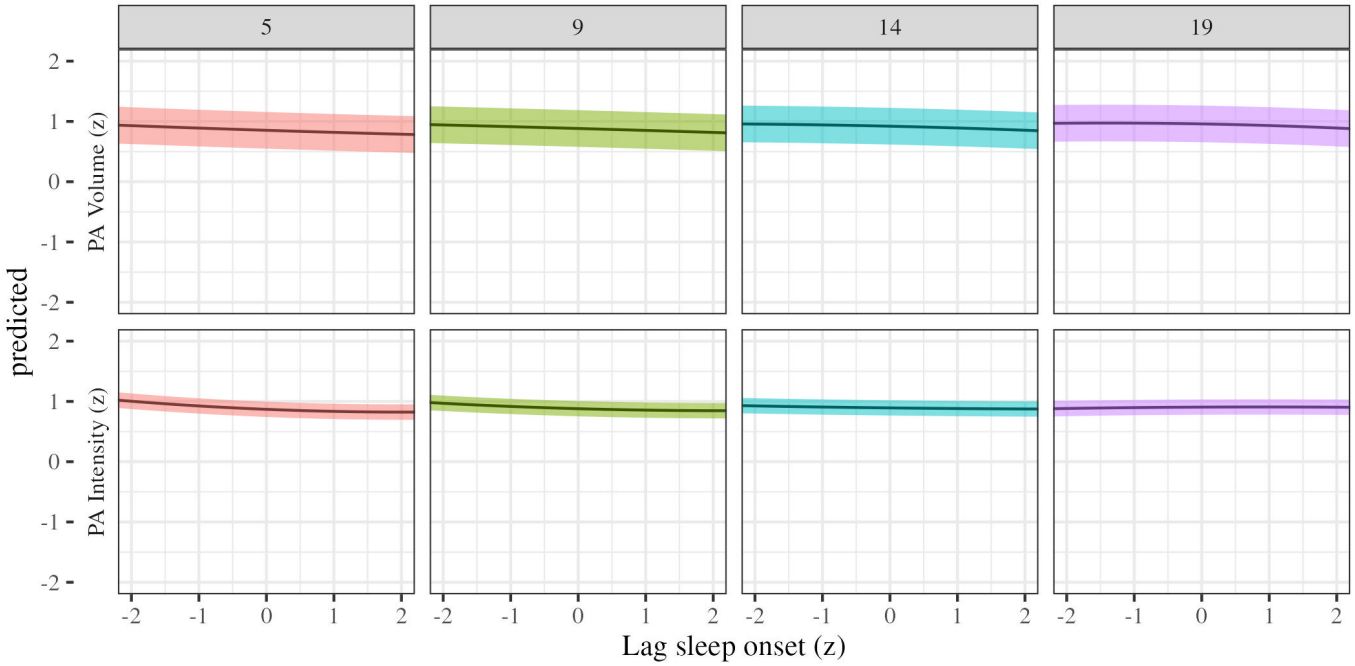

Figure 76. Physical activity by sleep onset moderated by most active hour

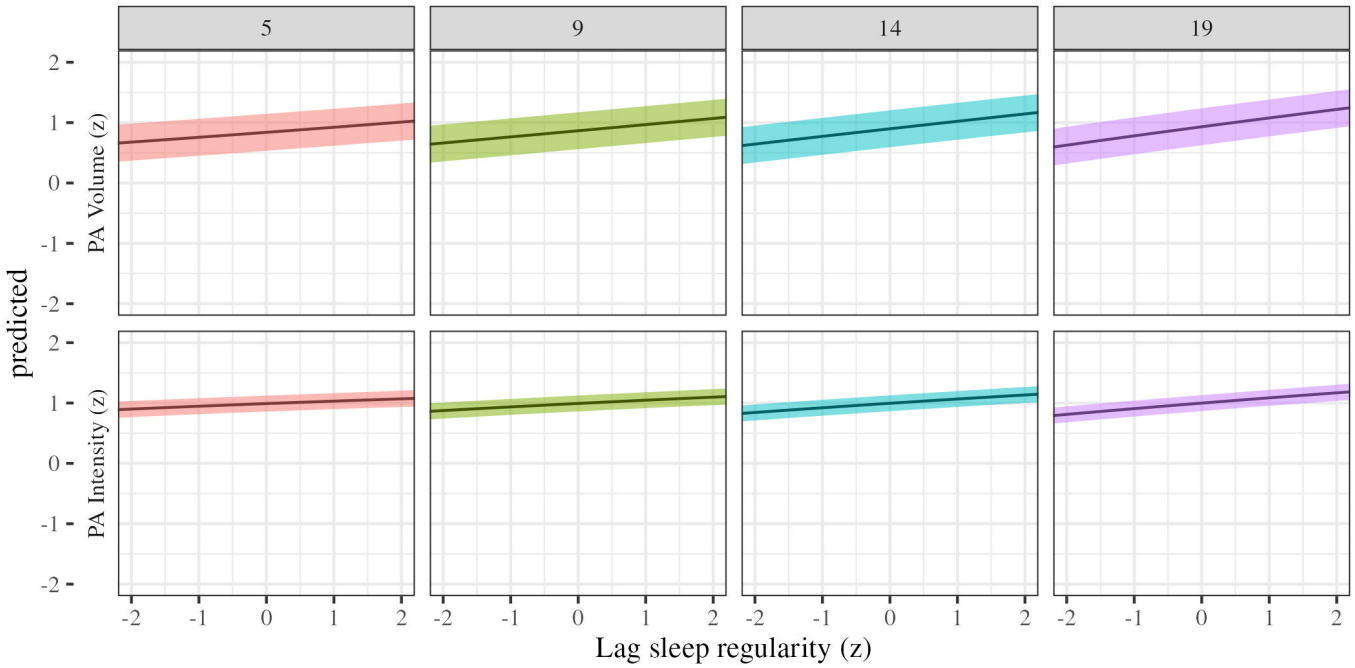

Figure 77. Physical activity by sleep regularity moderated by most active hour

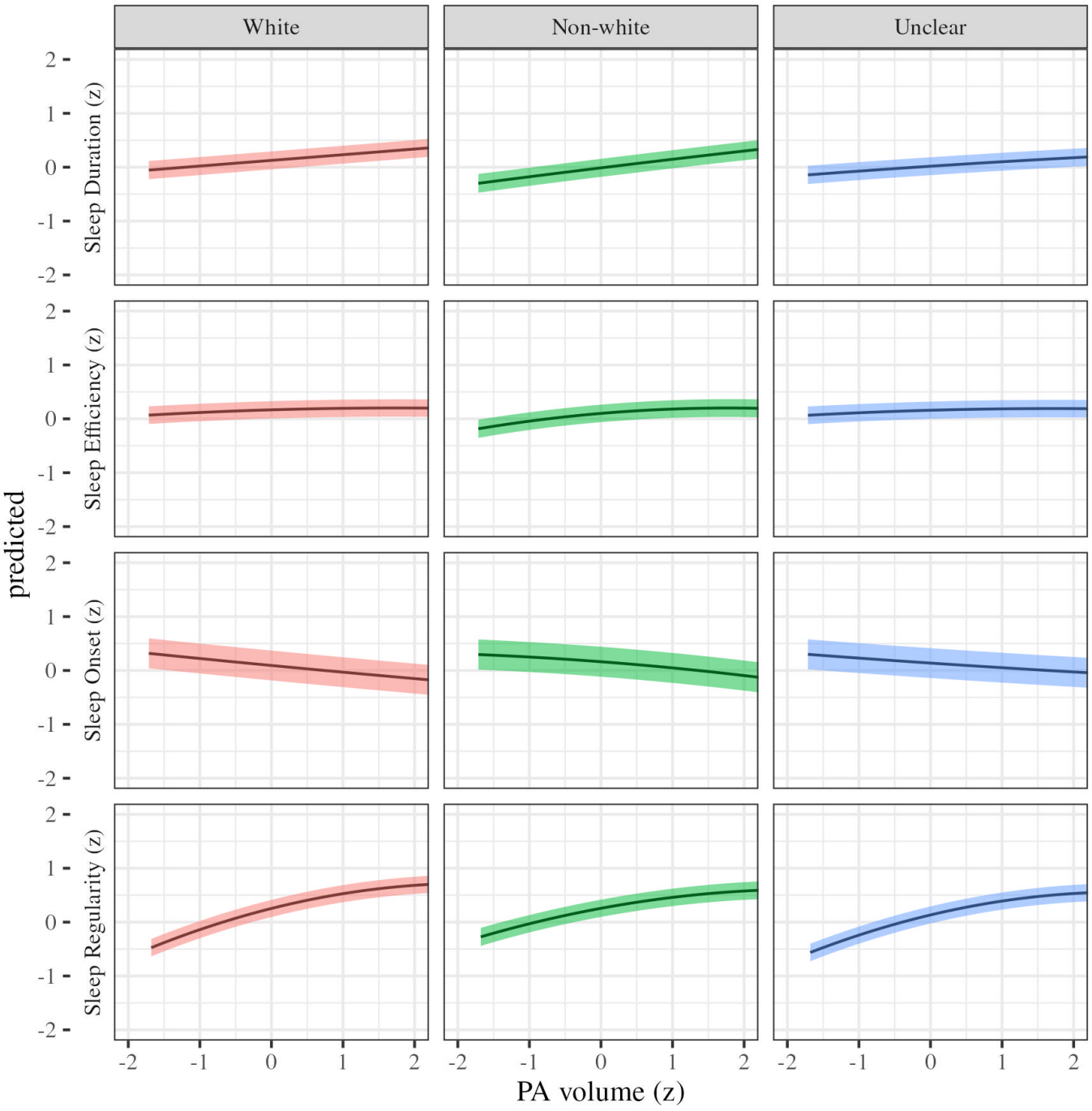

Figure 78. Sleep metrics on Physical activity volume by ethnicity

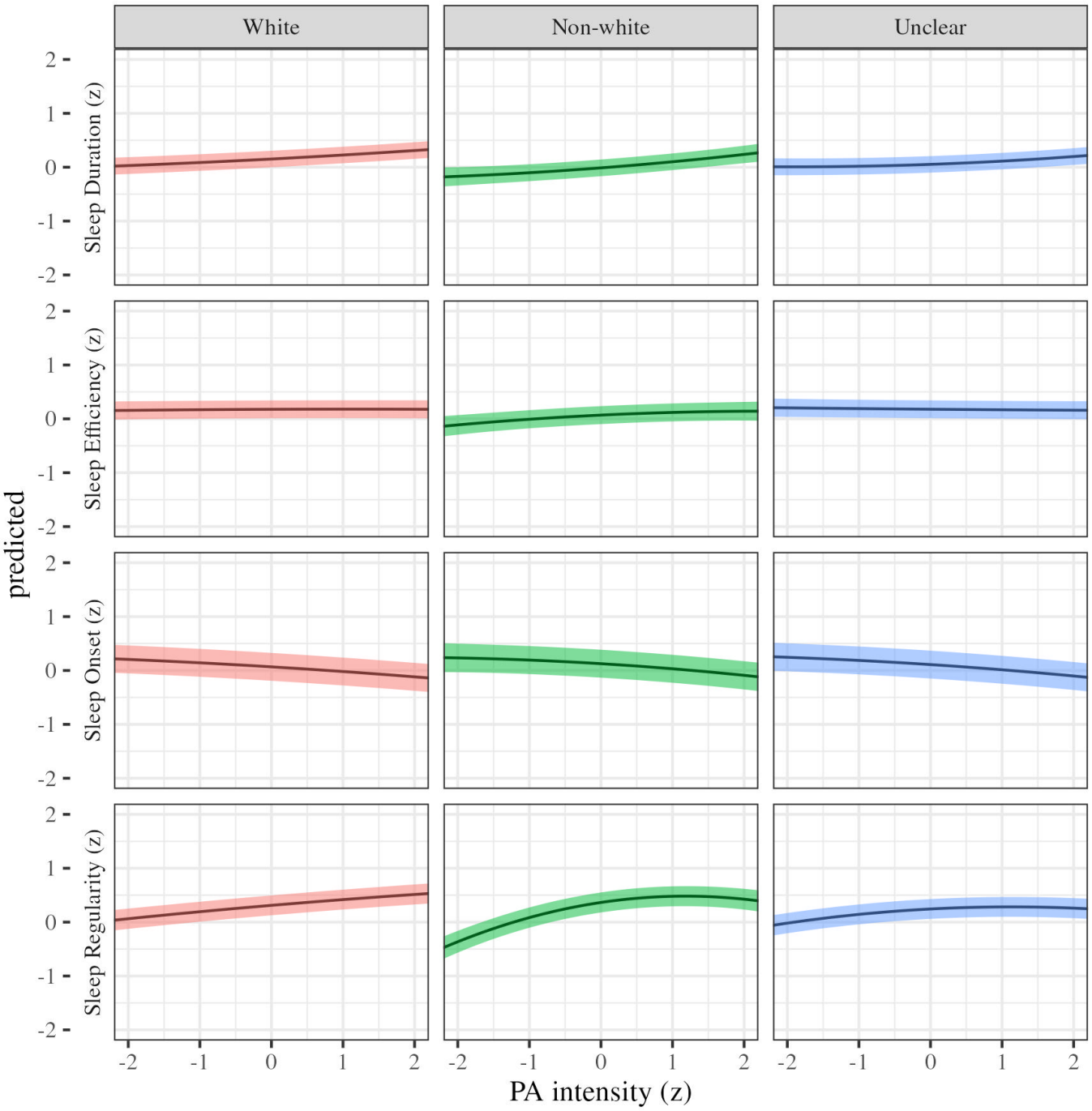

Figure 79. Sleep metrics on Physical activity intensity moderated by ethnicity

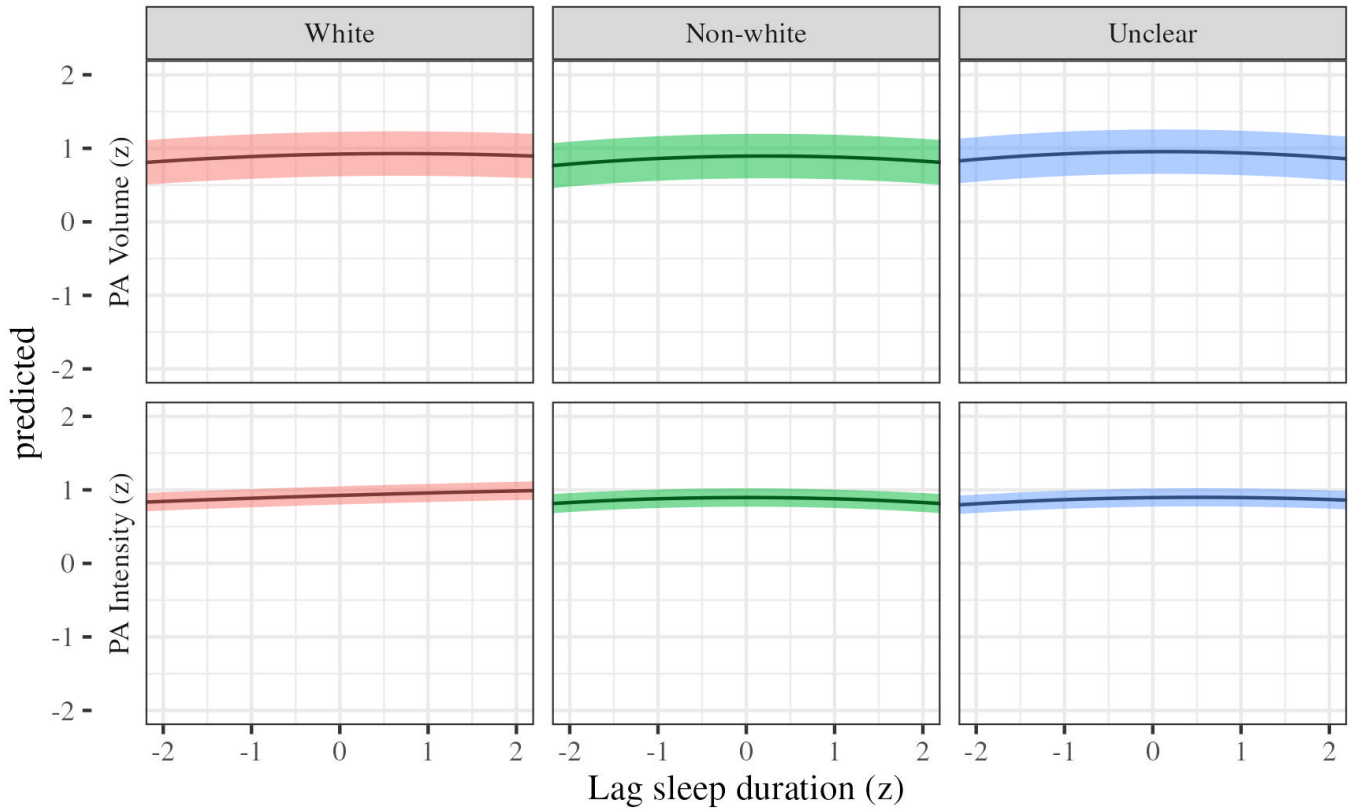

Figure 80. Physical activity by sleep duration moderated by ethnicity

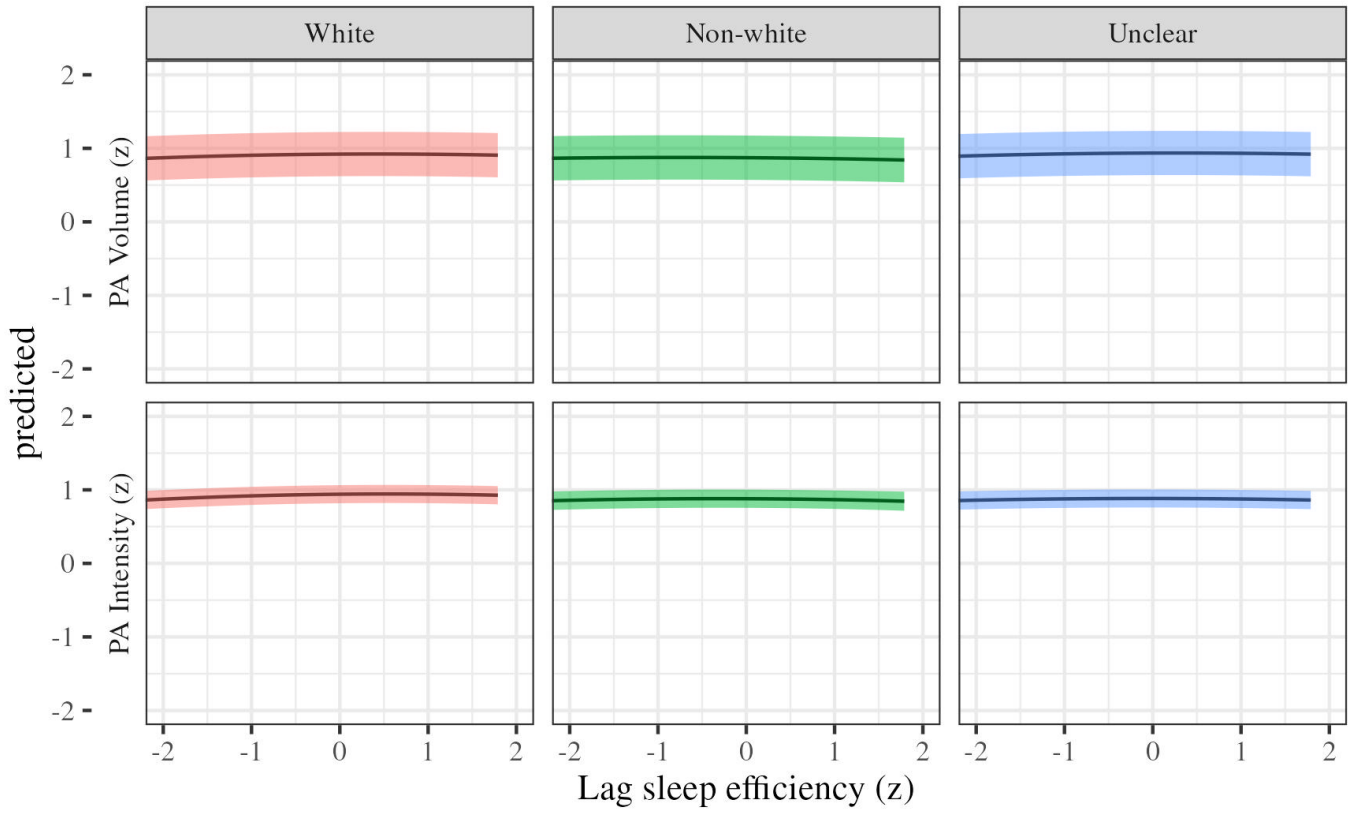

Figure 81. Physical activity by sleep efficiency moderated by ethnicity

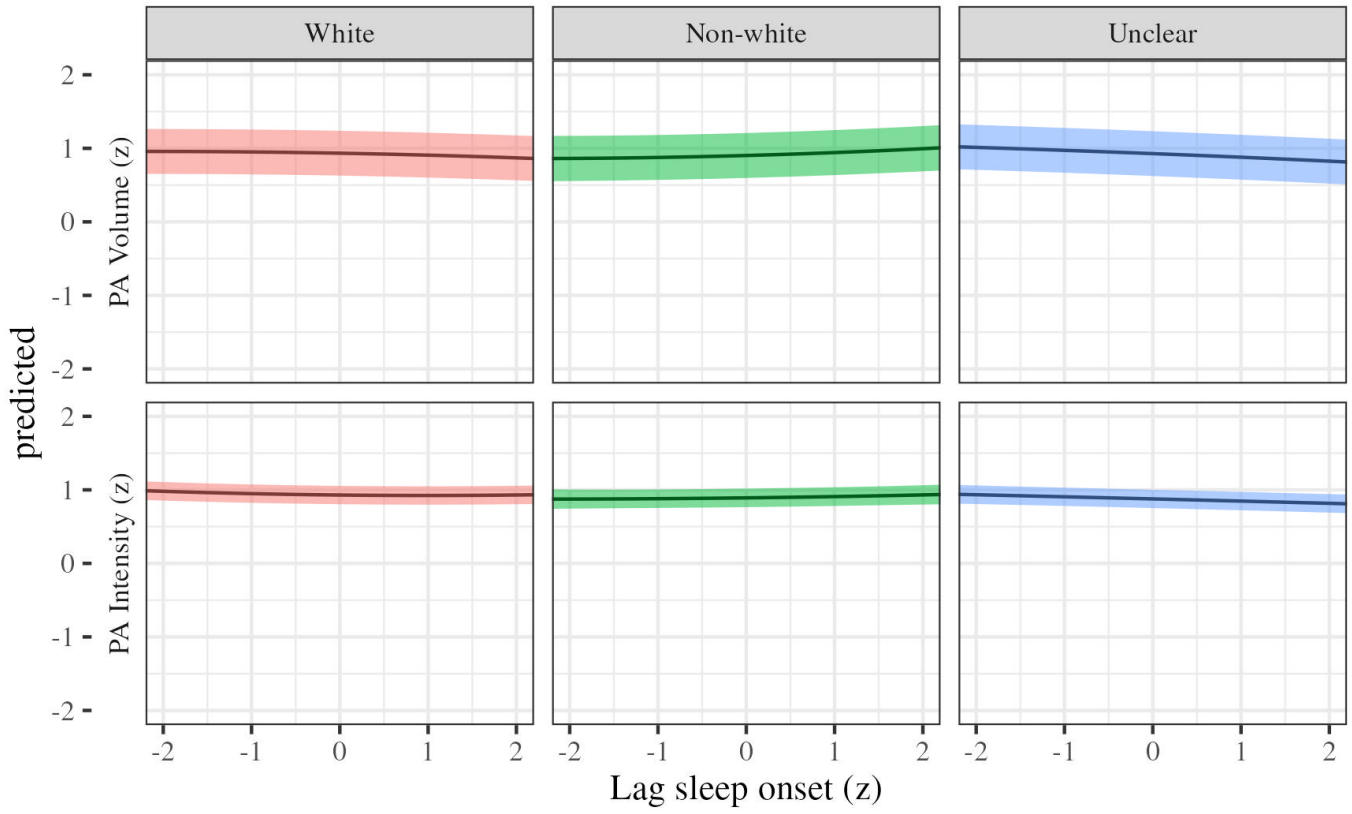

Figure 82. Physical activity by sleep onset moderated by ethnicity

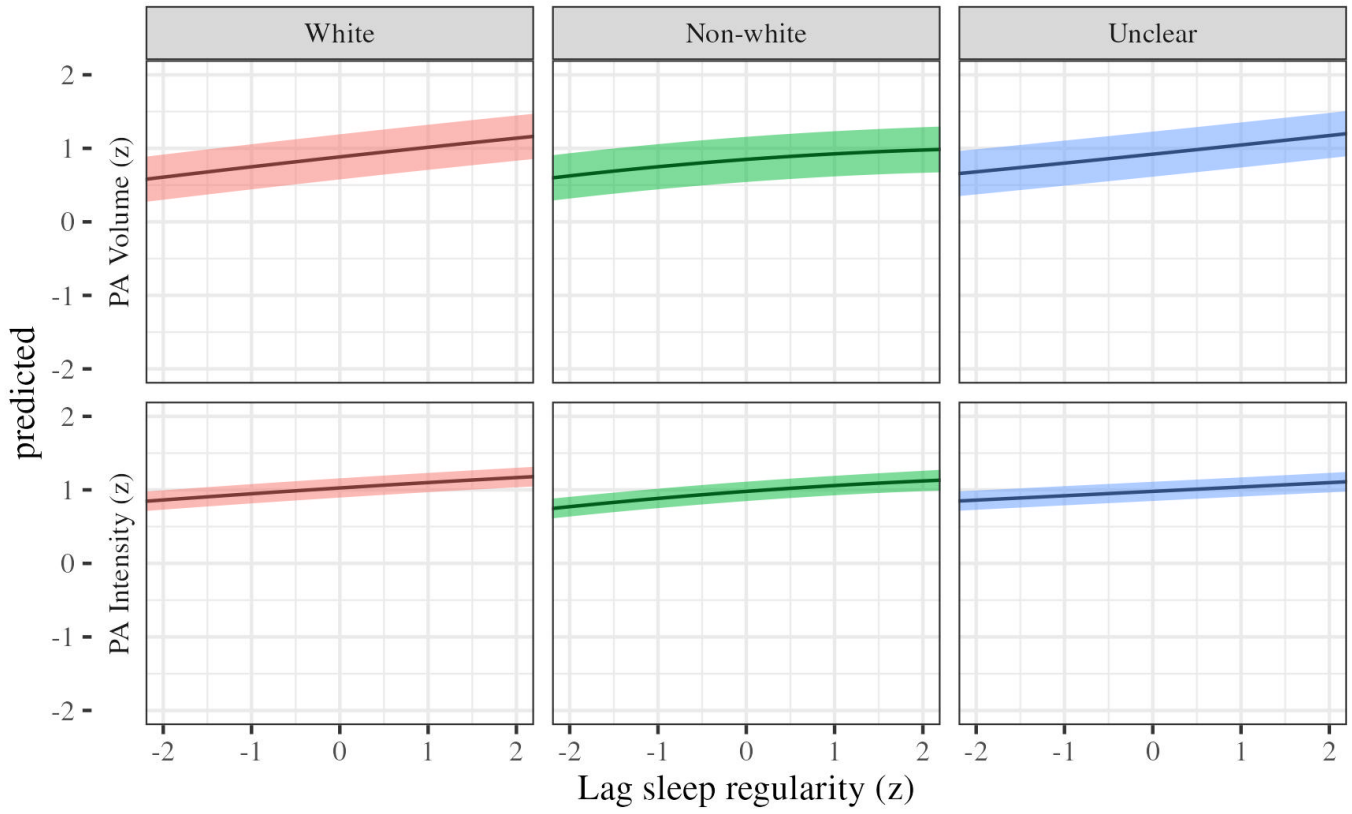

Figure 83. Physical activity by sleep regularity moderated by ethnicity